
INTRODUCTION TO ALGEBRAIC QUANTUM FIELD THEORY

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Chapter 1

Introduction

Entanglement is a characteristic feature of intrinsically quantum correlations, namely, correlations that have no classical counterpart. Although entanglement had already been recognized in the early days of quantum mechanics through the EPR thought experiment [1] and the work of Schrödinger [2], its significance in many-body quantum systems was not fully appreciated until the 1990s. Today, this concept plays a fundamental role in quantum information and quantum computation, condensed matter physics, and our understanding of quantum gravity.

As we know from Quantum Field Theory, local operators can be defined in a well-defined manner. Mathematically, let us consider a manifold M as the spacetime on which the quantum field theory is defined, with coordinates on this manifold denoted by x^i , where $i = 0, \dots, D$. Now suppose that we introduce a source $J(x)$ on M , defined within a region U . Let us also consider an arbitrary operator O . Then, locality can be expressed as

$$\langle \psi_J | O'(x') | \psi_J \rangle = \langle 0 | O(x) | 0 \rangle. \quad (1.1)$$

where

$$|\psi_J\rangle = \exp\left(-i \int dx, J(x)O(x)\right) |0\rangle. \quad (1.2)$$

Thus, in an ordinary quantum field theory, one can consistently define local operators.

Now, if we introduce gravity as a perturbation of the theory, diffeomorphism invariance prevents us from defining local operators in the same manner. In other words, as is well known in quantum gravity, local operators are not gauge-invariant quantities that can be defined a priori. Mathematically, if we consider the same configuration as before, under a diffeomorphism

$$x^\mu \rightarrow x'^\mu = x^\mu + \epsilon^\mu, \quad (1.3)$$

we have

$$\delta_\epsilon O(x) = -\epsilon^\mu \partial_\mu O(x). \quad (1.4)$$

As we can see, an arbitrary operator O evaluated at the two points x^μ and x'^μ is related by

a diffeomorphism; in other words, these two descriptions are gauge-equivalent.

Therefore, introducing gravity as a perturbation of a quantum field theory modifies, in an interesting way, notions such as locality and causality. We refer, in general, to a quantum field theory that includes gravity as quantum gravity. Furthermore, as is customary for any effective field theory, we begin by considering its action. At low energies, the action of the theory takes the form

$$S = \frac{1}{16\pi G_N} \int d^D x \sqrt{-g} (R - 2\Lambda + \mathcal{L}_{matter}). \quad (1.5)$$

As we know, since the metric is dimensionless in D spacetime dimensions while the Ricci scalar has mass dimension 2, the gravitational constant, unlike the dimensionless couplings usually encountered in quantum field theory, is dimensionful. It is given by

$$\frac{1}{G_N} \equiv (M_p)^{D-2}. \quad (1.6)$$

This interesting and longstanding feature gives rise to various difficulties in the theory, one proposed resolution of which is string theory [3]. We will not pursue this issue further here. Instead, we focus on the fact that the gravitational constant appearing in this effective field theory, or equivalently in the quantum theory of gravity, controls the degree of quantum fluctuations of the theory. For example, by taking the limit

$$G_N \rightarrow 0, \quad (1.7)$$

we move from a quantum theory to a semiclassical theory. This provides the starting point for the following question. In general, much of the work in this field over the past several years has focused on understanding how a quantum theory of gravity works and how classical geometry emerges from it. But what happens if we consider the reverse direction?

Question

Suppose that we have a quantum theory of gravity. In this case, due to metric fluctuations, a definite classical geometry cannot be assigned to the theory. How, then, by taking the limit $G_N \rightarrow 0$, do we recover a semiclassical theory, and how does a classical geometry emerge from a quantum state?

Furthermore, motivated by holography and, more generally, by the AdS/CFT correspondence, we can formulate the same question from another perspective. As we know from this duality [4, 5], the gravitational Newton constant G_N is related to the number of degrees of freedom, N , in the dual gauge theory through a relation of the form

$$G_N = \frac{1}{N^p}, \quad p > 0. \quad (1.8)$$

The question can therefore be reformulated as follows:

Question

How does a classical geometry emerge in a quantum theory of gravity as the entanglement between the degrees of freedom becomes infinite, namely in the limit $N \rightarrow \infty$?

The existence of this duality further strengthens the connections between quantum information theory and gravity. It is also worth emphasizing that, in order to conceptualize and investigate these questions, it is sufficient to adopt a common starting point: a quantum field theory which, in certain settings such as the aforementioned duality, possesses additional symmetries, such as conformal symmetry. This leads to the following question:

Question

If we are given only a conformal quantum field theory with N degrees of freedom, how does a classical geometry emerge in the limit $N \rightarrow \infty$?

The origin of this question is not limited to holographic duality. It can also be traced to related developments, such as those involving tensor networks and entanglement entropy [6, 7]. Thus, one may argue that the emergence of the bulk is not independent of what happens at the boundary. Up to this point, we characterize this phenomenon in terms of the divergence of entanglement among the degrees of freedom of the boundary theory.

Accordingly, in this thesis, we aim to investigate and revisit the connections between quantum information theory and gravity. This relatively recent perspective provides us with a small but potentially significant window into one of the major puzzles in quantum gravity [8].

Chapter 2

An Introduction to von Neumann Algebra

In this section, we study physical phenomena that, contrary to what we remember from quantum information, cannot be described easily.

2.1 Hilbert Space and Factorization

Entanglemen in quantum information theory can be described as follows. Consider a quantum system consisting of two arbitrary subsystems, such as L and R , where $R \cup L$ constitutes the entire system. The Hilbert space of this system can be written as

$$\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L \quad (2.1)$$

Thus, every quantum state of this system consists of two subsystems. If the quantum state of the system cannot be rewritten as the product of two separate states, then, by definition, it is an entangled state, and the two subsystems are entangled with each other. For example, suppose that the quantum state of a system is

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle|1\rangle + |1\rangle|0\rangle) \quad (2.2)$$

As can be seen, this quantum state cannot be expressed as a separate product of the two Hilbert spaces. Therefore, this state is an entangled state¹.

Also, if ψ is the quantum state of the system, a reduced density matrix can be defined such that it is associated only with one subsystem:

$$\rho_R \equiv \text{Tr}_L(|\psi\rangle\langle\psi|) \quad (2.3)$$

In this case, using this quantity, we can investigate and measure key information about that subsystem through the von Neumann entropy and the Rényi entropy, which are re-

¹This state is called a Bell state

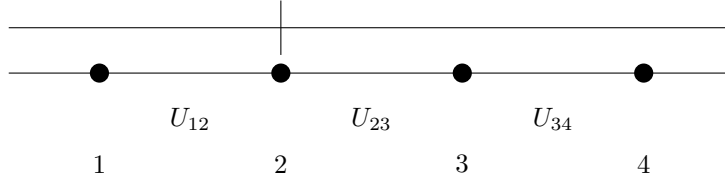
spectively given by

$$\begin{aligned} \text{von Neumann Entropy} &: S_R = -\text{Tr}_R \rho_R \log \rho_R \\ \text{Rényi Entropy} &: S_R^n = -\frac{1}{n-1} \log \text{Tr}_R \rho_R^n, \quad n = 2, 3, \dots \end{aligned} \quad (2.4)$$

However, in nature there are a limited number of systems for which the Hilbert space can be expressed separately, or, in other words, in a factorized form such as 2.1. The following examples are some of the systems for which a factorized Hilbert space cannot be considered,

1. Suppose that we have a lattice system in which the connection between every point of the lattice is described by a dynamical variable such as U_{ij} , which is not invariant under gauge transformations. Now, as shown in the figure below, if we consider this lattice to be one-dimensional and want to divide this one-dimensional lattice into two parts, point 2 of this lattice belongs jointly to both regions, or, more simply, it is regarded as the boundary between the two regions.

““



Now, given that the dynamics of U_{ij} is not invariant under gauge transformations, for the boundary of these two subsystems, which, as we can see, is point 2, the Hilbert space cannot simply be divided into two parts:

$$\mathcal{H} \neq \mathcal{H}_R \otimes \mathcal{H}_L \quad (2.5)$$

2. Suppose, as shown in the figure below, that we have a subsystem consisting of N pairs of particles, where each pair is in an entangled state. This system can be considered as two sets of qubits, with each pair interacting separately with one another. In this case, the quantum state of each pair can be written as

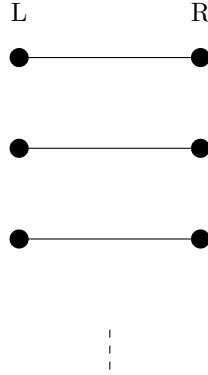
$$|\psi\rangle_\theta = \cos \theta |0\rangle |0\rangle + \sin \theta |1\rangle |1\rangle \quad (2.6)$$

And the quantum state of the entire system is

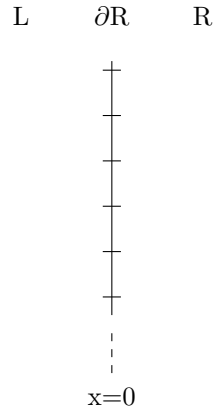
$$|\Psi\rangle_\theta = |\psi\rangle_{\theta_1} \otimes |\psi\rangle_{\theta_2} \otimes \dots \otimes |\psi\rangle_{\theta_N} \quad (2.7)$$

Now, if we take the number of these pairs of particles to be very large, such that $N \rightarrow \infty$, then an unlimited amount of energy is required to separate the two types of quantum states from expression 2.7, which is in fact impossible. Therefore,

$$\mathcal{H} \neq \mathcal{H}_R \otimes \mathcal{H}_L \quad (2.8)$$



3. Another example, which is related to the general continuation of this lecture, is found in quantum field theory. Suppose we are in Minkowski spacetime $\mathbb{R}^{1,d-1}$ and, by considering a time slice, we want to divide the spatial space into two regions R and L , which respectively correspond to $x \geq 0$ and $x \leq 0$. Now, as we know, in this



spacetime where quantum field theory is valid, there is a degree of freedom such as $\phi(x)$ at every point. As soon as a boundary is created, these degrees of freedom become infinite around the boundary, and consequently the entanglement entropy around the boundary can be obtained:

$$S_R = b \frac{\text{Area}(\partial R)}{\epsilon^{d-2}} + \dots \quad (2.9)$$

This entropy indicates that in ordinary quantum field theory, one cannot arbitrarily separate any region of spacetime and regard the total Hilbert space as factorized. Thus,

$$\mathcal{H} \neq \mathcal{H}_R \otimes \mathcal{H}_L \quad (2.10)$$

Therefore, as we observed in the examples above and as is the case in most systems in nature, the Hilbert space of the entire system cannot be expressed in a factorized form, and this limitation forces us to think of another way to investigate what occurs in nature.

2.2 What Is an Algebra?

Therefore, as a starting point and as an introduction to presenting this new approach, suppose first that we have a quantum system with a Hilbert space $\mathcal{H}$. Also, as we know from quantum mechanics, on every Hilbert space we have a set of bounded operators such as $\hat{x}$, $\hat{p}$, and so on, which act on the elements of this Hilbert space. We denote this set by $\mathcal{B}(\mathcal{H})$. Now suppose that we can express this Hilbert space in a factorized form. The question that arises is:

Question

Does the property of the Hilbert space being factorized or non-factorized affect the set of bounded operators acting on the Hilbert space or not?

The answer is no! Because, regardless of how we separate the Hilbert space, we still intrinsically have a type of operator set, or, more precisely, a type of algebra of operators. Therefore, another way to study quantum systems independently of the factorization property of the Hilbert space is to study the algebra of operators for each desired region. In other words, instead of assigning a Hilbert space to each subsystem, we assign an algebra to it, and from now on, for every region or subsystem, we study the algebra belonging to that region.

Now we examine this algebra and the emergence of this concept more precisely. First, suppose we have a quantum system with Hilbert space $\mathcal{H}$, and denote the set of all bounded operators acting on this Hilbert space by $\mathcal{B}(\mathcal{H})$, defined over the field $\mathbb{C}$. Then

Statement

We call a subset $\mathcal{A}$ of $\mathcal{B}(\mathcal{H})$ an algebra if

1. For every A and B in the subset $\mathcal{A}$, and also for every α and β in the field $\mathbb{C}$, we have $\alpha A + \beta B \in \mathcal{A}$, which expresses closure and the possession of the same structure as a vector space. Therefore, we can say that the subset $\mathcal{A}$ is a vector space over the field $\mathbb{C}$.
2. For every A and B in the subset $\mathcal{A}$, we have $AB \in \mathcal{A}$. This property likewise expresses closure under the binary operation in the structure of operators.

We can also state:

Statement

If, in an algebra such as $\mathcal{A}$, for every A in the algebra $\mathcal{A}$ we have $A^\dagger \in \mathcal{A}$, then $\mathcal{A}$ is called a $*$ -algebra.

Now that we have discussed the closure property of the subset $\mathcal{A}$, we need to address the concept of completeness, which gives rise to two types of algebras as follows:

1. Norm convergence: As we know, for bounded operators on the Hilbert space $\mathcal{H}$ represented by $\mathcal{B}(\mathcal{H})$, one can define a map called the norm as $|\cdot| : \mathcal{A} \times \mathcal{A} \rightarrow \mathbb{C}$. In this case, for an operator A , an operator sequence $A_1, A_2, \dots$ has convergence under the norm map, or norm convergence, if:

$$\lim_{n \rightarrow \infty} |A - A_n| = 0 \quad (2.11)$$

Therefore,

Statement

If norm convergence holds for every A in $\mathcal{A}$, then $\mathcal{A}$ is called a C^* -algebra.

2. Weak convergence: In this type of convergence, contrary to what was stated for norm convergence, we need the Hilbert space $\mathcal{H}$. If we have a sequence of operators A , $\{A_1, A_2, \dots\}$, this operator has weak convergence if, for every $|\eta\rangle, |\xi\rangle \in \mathcal{H}$, we have:

$$\lim_{n \rightarrow \infty} \langle \eta | A_n | \xi \rangle = \langle \eta | A | \xi \rangle \quad (2.12)$$

In this case,

Statement

If every A in the algebra $\mathcal{A}$ has weak convergence, then $\mathcal{A}$ is called a von Neumann algebra.

Therefore, the difference between a C^* -algebra and a von Neumann algebra is that a C^* -algebra can be defined abstractly and without using a Hilbert space, whereas defining a von Neumann algebra requires a Hilbert space.

1 Show that every von Neumann algebra is a C^* -algebra, but the converse is not true.

Now, given that in quantum mechanics and in the quantum systems we study in physics there is a specified Hilbert space, it is more appropriate to use weak convergence, and therefore von Neumann algebras are of greater importance to us. We can also define the commutant of an algebra such as $\mathcal{A}$ as

$$\mathcal{A}' = \{B \in \mathcal{B}(\mathcal{H}); BA = AB; \forall A \in \mathcal{A}\} \quad (2.13)$$

It can also be shown that if $\mathcal{A}$ is a von Neumann algebra, then

$$\mathcal{A} = \mathcal{A}'' \quad (2.14)$$

and if $\mathcal{M}$ is a $*$ -algebra, then

$$\mathcal{M}' = \mathcal{M}''' \quad (2.15)$$

and in this case we can state

Statement

The commutant of a $*$ -algebra is a von Neumann algebra.

On the other hand, if $\mathcal{M}_1$ and $\mathcal{M}_2$ are two von Neumann algebras, by definition the smallest algebra containing $\mathcal{M}_1$ and $\mathcal{M}_2$ is denoted by $\mathcal{M}_1 \vee \mathcal{M}_2$, and we define the largest common algebra as $\mathcal{M}_1 \wedge \mathcal{M}_2$. Thus,

$$\mathcal{M}_1 \vee \mathcal{M}_2 = (\mathcal{M}_1 \cup \mathcal{M}_2)'' \quad (2.16)$$

$$\mathcal{M}_1 \wedge \mathcal{M}_2 = \mathcal{M}_1 \cap \mathcal{M}_2 \quad (2.17)$$

and therefore

$$(\mathcal{M}_1 \vee \mathcal{M}_2)' = \mathcal{M}_1' \wedge \mathcal{M}_2' = \mathcal{M}_1' \cap \mathcal{M}_2' \quad (2.18)$$

We can also define the center of an algebra, denoted by $\mathcal{Z}(\mathcal{M})$, as

$$\mathcal{Z}(\mathcal{M}) = \mathcal{M} \cap \mathcal{M}' \quad (2.19)$$

Statement

If a von Neumann algebra has a trivial center, $\mathcal{Z}(\mathcal{M}) = \mathbb{C}I$, it is called a factor algebra.

Thus far, we have described the properties of von Neumann and C^* -algebras, and now it is time to advance our goals, to define a somewhat familiar map and in quantum mechanics represents the expectation value. It should be noted, however, that we are working within the mathematical description of algebras and have not yet discussed physics or quantum mechanics. In this case, we can define a linear map on the algebra, such as $\omega : \mathcal{M} \rightarrow \mathbb{C}$, with the following properties:

$$\omega(\alpha A + \beta B) = \alpha\omega(A) + \beta\omega(B) ; \quad \forall \alpha, \beta \in \mathbb{C} , \quad \forall A, B \in \mathcal{M} \quad (2.20)$$

$$\omega(A^\dagger) = (\omega(A))^* ; \quad \forall A \in \mathcal{M} \quad (2.21)$$

We also define a weight ω as a map that, in addition to satisfying 2.20 and 2.21, also satisfies

$$\omega(A^\dagger A) \geq 0 ; \quad \forall A \in \mathcal{M} \quad (2.22)$$

2 Show that, according to the definition of a weight, the Cauchy–Schwarz inequality holds:

$$|\omega(A^\dagger B)|^2 \leq \omega(A^\dagger A)\omega(B^\dagger B) \quad (2.23)$$

In this case, we can define

Statement

The weight map ω is a tr map if, for every $A, B \in \mathcal{M}$, we have $\omega(AB) = \omega(BA)$.

Note that, contrary to what we already know, the tr map can have an infinite value. For this reason, we call the weight ω a positive linear map if $\omega(I)$ has a finite value. In this case, we can introduce another new definition:

Statement

We call the weight map ω a state if

$$\omega(I) = 1 \quad (2.24)$$

and we define a normal state such that if A_i is a sequence of positive bounded operators in $\mathcal{M}$, then it must satisfy

$$\sup \omega(A_i) = \omega(\sup A_i) \quad (2.25)$$

It can also be shown that a state is normal if and only if there exists a density matrix ρ in the Hilbert space $\mathcal{H}$ such that

$$\omega(A) = \text{tr}(\rho A) \quad , \quad \forall A \in \mathcal{M} \quad (2.26)$$

Finally, we arrive at two final definitions for the weight map ω , which will be of particular importance to us later.

Statement

We call the state ω faithful if and only if

$$\omega(A^\dagger A) = 0 \Rightarrow A = 0 \quad (2.27)$$

Also, if we assume that ω_1 and ω_2 are states, then clearly, for $\lambda \in (0, 1)$,

$$\omega = \lambda \omega_1 + (1 - \lambda) \omega_2 \quad (2.28)$$

is a state. In this case, if a state cannot be expressed in the above form, we call it a pure state.

Thus far, we have examined the properties of algebras and defined the maps used with them. What is important for us from this viewpoint is that, contrary to what we had previously assumed, namely assigning a subsystem to a Hilbert space, as we observed, this approach fails in many examples in nature. Therefore, we changed our way of looking at the problem by assigning an algebra to each subsystem, which allows us to avoid the difficulties associated with the factorization of Hilbert space, and this is a positive step.

2.3 Projection Operator

As stated, defining a von Neumann algebra requires a Hilbert space. Therefore, this algebra can be considered by spanning projection operators, in which case every operator in the von Neumann algebra can be expressed in terms of projection operators. Hence, we need to examine the properties of projection operators².

First, suppose $\mathcal{H}$ is a Hilbert space and P is a projection operator that maps this Hilbert space onto a closed subspace of it, $P\mathcal{H} \subseteq \mathcal{H}$. It can also be stated that for every closed subspace of a Hilbert space, a corresponding projection operator can be defined. Therefore,

Statement

Suppose $\mathcal{A}$ is a von Neumann algebra. If $P \in \mathcal{A}$, then the closed subspace $P\mathcal{H}$ belongs to $\mathcal{A}$.

²It should be noted that up to this point we have not discussed quantum mechanics and have only examined its mathematics

As is clear, the largest projection operator is the identity operator I . But the question that arises is: what criterion can be used for the "largest" or "larger" relation between projection operators? To answer this question, we attempt to define a criterion for comparing two projection operators such as P and Q , as follows:

Statement

The projection operator P is smaller than the projection operator Q , $P < Q$, if

$$PQ = P \quad (2.29)$$

On this basis, projection operators can be divided into two categories:

1. Minimal projection operator: A projection operator P is a minimal projection operator if, other than the zero operator, there is no projection operator such as Q for which $Q < P$.
2. Finite projection operator: Suppose we have a quantum system whose Hilbert space can be factorized as $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$ and $|\psi\rangle \in \mathcal{H}$. In this case, we can define a projection operator such as $P = |\psi\rangle\langle\psi| \otimes I_B$ that generates the subspace $P\mathcal{H}_A$. Now, if we assume that the Hilbert space $\mathcal{H}_2$ is infinite-dimensional, then the projection-operator space on a subspace of this Hilbert space is infinite-dimensional. Therefore, the rank of the projection operator P on the Hilbert space $\mathcal{H}$ is infinite. However, the important point is that, from the algebraic viewpoint, the subalgebra containing the projection operator P is

$$\mathcal{M} = \mathcal{B}(\mathcal{H}_A) \otimes I_B \quad (2.30)$$

In this case, the projection operator P must have finite dimension. Therefore,

Statement

The finiteness of a projection operator does not depend on the entire Hilbert space but rather on the underlying algebra such as $\mathcal{M}$.

Then, to define the "finiteness" of an operator, we first need to define "equivalence":

Statement

Two projection operators $P, Q \in \mathcal{M}$ are equivalent, $P \sim Q$, if there exists an operator such as $V \in \mathcal{M}$ satisfying

$$P = V^\dagger V, \quad Q = VV^\dagger \quad (2.31)$$

The existence of such an operator means that a map can be applied through the operator V between the two subspaces $P\mathcal{H}$ and $Q\mathcal{H}$. Using equivalence, we can define finite and infinite projection operators. To this end,

Statement

A nonzero projection operator such as P is called finite if it is not equivalent to a projection operator smaller than itself. Also, if we have a nonzero projection operator such as P for which there exists a projection operator such as $Q \in \mathcal{M}$

satisfying $Q \sim P$ and/or $Q < P$, then it is called an infinite projection operator.

The important point now is that if the identity operator I , which is the largest projection operator, is finite, then for the von Neumann algebra $\mathcal{A}$ all projection operators in the algebra must be finite, and this algebra is called a finite von Neumann algebra. Also, if the identity operator I is infinite, then all projection operators in the algebra are infinite, and it is called an infinite von Neumann algebra. Therefore, as we have seen, the existence of projection operators is of great importance in classifying von Neumann algebras. For this purpose, we consider how the dimension of a projection operator is defined. Perhaps one answer is that by defining the map $\text{tr} : \mathcal{A} \rightarrow \mathbb{C}$, the dimension of projection operators can be represented. However, the point is that, as stated in the previous example, the projection operator P can be examined from two different viewpoints, and therefore the use of the tr map is not as well-defined as it should be. In this case, to define the dimension of a projection operator, we use the property of equivalence:

Statement

We consider the map $d : \mathcal{A} \rightarrow \mathbb{C}$ as the dimension of a projection operator, defined as

$$d(P) < d(Q) \text{ if } P < Q \qquad d(P) = d(Q) \text{ if } P \sim Q$$

In this case, the projection operator P has infinite dimension, $d(P) = \infty$, if there exists a projection operator such as $Q \in \mathcal{M}$ such that simultaneously

$$d(P) < d(Q) \text{ and } d(P) = d(Q) \tag{2.32}$$

As stated earlier, it is natural to use the map $\text{tr} : \mathcal{A} \rightarrow \mathbb{C}$ as the definition of the dimension of a projection operator. However, as observed in the above definition, this map is not well-defined for projection operators that have infinite dimension. Now, in the following, when defining the types of algebras, we will see that the definition of the map described above, just as we remember from quantum mechanics, is only valid for a part of the algebra, and in some cases this map cannot be defined, while in other cases its definition differs from what we know from before.

2.4 Types of von Neumann Algebras

As stated in the previous section, projection operators are an important element in identifying von Neumann algebras. In this section, we define and examine the properties of different types of von Neumann algebras. In general, according to the properties of their projection operators, von Neumann algebras are divided into three categories:

1. Type I von Neumann algebra: The algebra $\mathcal{A}$ has a nonzero minimal projection operator.

Suppose P_0 is the minimal projection operator. Then, by normalization, the dimension of this projection operator can be written as $d(P_0) = 1$, and for the largest projection operator, which is the identity operator, we have

$$d(I) = n \tag{2.33}$$

where n can take any value from the set of natural numbers. Now, if we define the map $\text{tr} : \mathcal{A} \rightarrow \mathbb{C}$, then

$$d(P) = \text{tr}(P) \quad , \quad P \in \mathcal{A} \quad (2.34)$$

We can now say that this map is the standard map defined on the Hilbert space. In other words, if $\mathcal{H}$ is the Hilbert space of a system that has a Type I von Neumann algebra, then $\text{tr} \equiv \text{tr}_{\mathcal{H}}$. In this case, the important point is that by defining this map as in quantum mechanics, we can give an interpretation to what is called "counting the states."

Therefore,

Statement

A Type I von Neumann algebra is the same algebra used in quantum mechanics, in which the dimension of projection operators, through the map tr , agrees with the dimension of the Hilbert space. It can also be stated that in this type of algebra, the factorization property of the Hilbert space exists.

It should be noted that, as we will discuss in the next chapter, a Type I algebra can also hold for non-factorizable Hilbert spaces. However, the important point is that the algebra used in quantum mechanics belongs to this type of algebra.

2. Type II von Neumann algebra: The algebra $\mathcal{A}$ contains finite projection operators, but there is no minimal projection operator.

In this case, because there is no minimal projection operator, the dimension map defined as $d : \mathcal{A} \rightarrow \mathbb{C}$ can take any arbitrary value, and not necessarily a value from the natural numbers. In other words, for every projection operator $P \in \mathcal{A}$, we can have a projection operator such as $Q \in \mathcal{A}$ such that the dimension of each of them can take any arbitrary value. For a more precise investigation, we need to pay attention to the largest projection operator of the algebra, namely the identity operator.

- (a) II_1 : The map $d(\text{I})$ has a lower bound, which by normalization can be written as

$$d(\text{I}) = \text{tr}(\text{I}) = 1 \quad (2.35)$$

In this case, the dimension of projection operators in this type of algebra lies in the interval $d(P) \in [0, 1]$, and it is called a Type II_1 von Neumann algebra.

- (b) II_∞ : The map $d(\text{I})$ is unbounded, and in this case

$$d(\text{I}) = \text{tr}(\text{I}) = \infty \quad (2.36)$$

Therefore, the dimension of projection operators in this type of algebra lies in the interval $d(P) \in [0, \infty)$, and it is called a Type II_∞ von Neumann algebra. Thus, the important point is that

Statement

The definition of the map $\text{tr} : \mathcal{A} \rightarrow \mathbb{C}$ in a Type II von Neumann algebra, unlike Type I, does not have the usual meaning from quantum mechanics and what we call "counting the states," and it has no physical interpretation.

3. Type III von Neumann algebra: All nonzero projection operators in the algebra $\mathcal{A}$ are infinite.

As stated, because all projection operators in this algebra are infinite, for every $P \in \mathcal{A}$ we have $d(P) = \infty$. Therefore, defining the map $\text{tr} : \mathcal{A} \rightarrow \mathbb{C}$ for the dimension of a projection operator is meaningless. In this type of algebra, because every projection operator such as P is infinite, according to the definition given above, for every projection operator P there exists a projection operator $Q \in \mathcal{A}$ such that simultaneously

$$d(Q) < d(P) \text{ and } d(Q) = d(P) \quad (2.37)$$

and for this reason we can write

$$d(Q) = d(P) \rightarrow Q \sim P \quad (2.38)$$

Therefore, we can claim that there always exists an operator such as $W \in \mathcal{A}$ satisfying

$$P = W^\dagger W, \quad Q = WW^\dagger \quad (2.39)$$

Also, because the largest projection operator I exists, we have

$$P = W^\dagger W, \quad I = WW^\dagger \quad (2.40)$$

Then we can state

Statement

In a Type III von Neumann algebra, for every projection operator such as $P \in \mathcal{A}$ there exists an isometry such as $W \in \mathcal{A}$ such that

$$P = W^\dagger W, \quad I = WW^\dagger \quad (2.41)$$

Also, by defining the map $\omega : \mathcal{A} \rightarrow \mathbb{C}$ as a "state", it can be shown that for every W there exists another state map such as ω_W such that

$$\omega_W(A) = \omega(W^\dagger A W), \quad A \in \mathcal{A} \quad (2.42)$$

where $\omega_W(P) = 1$.

2.5 Emergence of von Neumann Algebra and the GNS Structure

As stated earlier, the difference between a von Neumann algebra and a C^* -algebra is that defining a von Neumann algebra necessarily requires considering a Hilbert space. An inter-

esting discussion, which we will explain below, is that a Hilbert space and a von Neumann algebra can emerge from a C^* -algebra.

Therefore, first suppose $\mathcal{M}$ is a C^* -algebra and that there exists a map $\omega : \mathcal{M} \rightarrow \mathbb{C}$ on this algebra. In this case, the inner product can be defined as

$$\langle .|. \rangle : \mathcal{M} \times \mathcal{M} \rightarrow \mathbb{C} \quad \langle A|B \rangle = \omega(A^\dagger B) \quad , \quad \forall A, B \in \mathcal{M}$$

which has the following properties:

1. For every $A \in \mathcal{M}$, we have $\langle A|A \rangle = \omega(A^\dagger A) = 0$
2. If $A, B \in \mathcal{M}$, then

$$\langle A|B \rangle = \omega(A^\dagger B) = (\omega(B^\dagger A))^* = \langle B|A \rangle^*$$

3. (Cauchy–Schwarz inequality)

For every $A, B \in \mathcal{M}$, it can be shown that

$$|\omega(A^\dagger B)|^2 \leq \omega(A^\dagger A)\omega(B^\dagger B)$$

It should be noted that this property always accompanies the definition of the map ω (for a review, see expression 2.23).

Now, as we can see, this inner product is not the same as the one defined on a Hilbert space because, according to property 1, operators such as A can be found in the algebra $\mathcal{M}$ for which the map evaluated on the operator itself is zero. These are called null operators. In this case, suppose we denote the set of operators satisfying property 1 by $\mathcal{T}$:

$$\mathcal{T} = \{X \mid \omega(X^\dagger X) = 0\} \quad (2.43)$$

Then, using the Cauchy–Schwarz inequality, it can be shown that

$$\forall A \in \mathcal{M} \quad , \quad \forall X \in \mathcal{T} \mid \langle AX|AX \rangle = \omega((AX)^\dagger(AX)) = 0 \quad (2.44)$$

3 Prove the above expression.

In this case, according to the above expression, we can conclude that for every operator such as A from $\mathcal{M}$ and an operator such as X from $\mathcal{T}$, the operator AX is part of the algebra $\mathcal{M}$. It can also be shown that

$$\forall A, B \in \mathcal{M} \quad , \quad \forall X_1, X_2 \in \mathcal{T} \mid \langle A + X_1|B + X_2 \rangle = \omega(A^\dagger B) = \langle A|B \rangle \quad (2.45)$$

and therefore an equivalence relation can be defined as

$$\forall A, B \in \mathcal{M} \quad , \quad \forall X_1, X_2 \in \mathcal{T} \mid \langle A + X_1|B + X_2 \rangle = \langle A|B \rangle \rightarrow A \sim A + X \quad (2.46)$$

Then, due to the existence of an equivalence relation, by considering the quotient set $\mathcal{M}/\mathcal{T}$,

the elements of this set (algebra), such as $A \in \mathcal{M}$, can be expressed as equivalence classes in the form $[A]$. This also ensures that property 1 holds only for $A \sim 0$.

On the other hand, as stated earlier, given the properties of an algebra, the algebra $\mathcal{M}/\mathcal{T}$ can be regarded as a vector space over the underlying field. More precisely, the set $\mathcal{M}/\mathcal{T} = [A] | A \in \mathcal{M}/\mathcal{T}$ forms a vector space over the field $\mathbb{C}$. Now, to define objects called "vectors" in this Hilbert space, we first need to define a "representation".

Statement

A representation of an (group) algebra $\mathfrak{g}$, denoted by π , is a map from $\mathfrak{g}$ to a set of linear operators such as π that act on the vector space $\mathcal{V}$, $\pi : \mathfrak{g} \rightarrow \pi$, if

- If $a, b \in \mathfrak{g}$ and $\alpha, \beta \in \mathcal{F}$, then

$$\pi(\alpha a + \beta b) = \alpha \pi(a) + \beta \pi(b)$$

- Also, if $a, b \in \mathfrak{g}$, then

$$\pi(ab) = \pi(a)\pi(b)$$

where $\mathcal{F}$ is the field over which the vector space $\mathcal{V}$ is defined.

In this case, since $\mathcal{M}/\mathcal{T}$ is both an algebra and a vector space over the field $\mathbb{C}$ with equivalence classes $[A]$, we can define a map $\pi_\omega : \mathcal{M}/\mathcal{T} \rightarrow \mathcal{M}/\mathcal{T}$ and define the action of this map on the equivalence classes as

$$\pi_\omega(A)[B] = [AB] \quad , \quad \forall [A], [B] \in \mathcal{M}/\mathcal{T} \quad (2.47)$$

Then, by extending the equivalence class $[A]$ as a vector $|A\rangle$, we can state that the vector space $\mathcal{M}/\mathcal{T}$ contains vectors of the form $[A] \simeq |A\rangle$, and the representation of this algebra transforms these vectors into one another. Now, if the vector $|\Omega\rangle \equiv |I\rangle$ is regarded as an identity vector, then by applying the representation of any operator such as $A \in \mathcal{M}/\mathcal{T}$, we can generate its generalized vector:

$$\forall [A] \in \mathcal{M}/\mathcal{T} \quad , \quad \pi_\omega(A) |\Omega\rangle = \pi_\omega(A) |I\rangle = |A\rangle \quad (2.48)$$

In this case, we can write

$$\forall A \in \mathcal{M}/\mathcal{T} \quad | \quad \omega(A) = \langle \Omega | A \rangle = \langle \Omega | \pi_\omega(A) | \Omega \rangle \quad (2.49)$$

Therefore, we can conclude:

Statement

The set $\mathcal{M}/\mathcal{T}$ whose elements are defined by considering $|\Omega\rangle \equiv |I\rangle$ according to expression 2.48, and whose product between any two elements is given by expression 2.43, can be regarded as a Hilbert space, denoted by $\mathcal{H} * \omega$, constructed around $|\Omega\rangle \equiv |I\rangle$. This process is called the GNS^a construction.

^aGelfand-Naimark-Segal

As stated in the above definition, through the GNS construction, the Hilbert space $\mathcal{H} * \omega$ can be reconstructed around the vector $|\Omega\rangle$ by applying the representation $\pi_\omega(\mathcal{M})$ to the vector $|\Omega\rangle$:

$$\forall \psi \in \mathcal{H}_\omega, \quad \forall A_i \in \mathcal{M} \quad |\psi\rangle \simeq \sum_i \pi_\omega(A_i) |\Omega\rangle \quad (2.50)$$

In this case, we can define

Statement

The vector $|\Omega\rangle$ which, under the action of the representation $\pi_\omega(\mathcal{M})$, $\pi_\omega(A) |\Omega\rangle \mid A \in \mathcal{M}$, generates a dense subspace in the Hilbert space $\mathcal{H}_\omega$ is called a cyclic vector. The representation $\pi_\omega(\mathcal{M})$ that possesses a cyclic vector is called a cyclic representation. Thus, the structure $(\mathcal{H}_\omega, \pi_\omega(\mathcal{M}), |\Omega\rangle)$ gives rise to a cyclic representation of the algebra $\mathcal{M}$.

We can also state two propositions for this type of structure:

1. The representation $\pi_\omega(\mathcal{M})$ in the structure $(\mathcal{H}_\omega, \pi_\omega(\mathcal{M}), |\Omega\rangle)$ is irreducible if and only if the map ω is a pure state. In this case,

$$\mathcal{B}(\mathcal{H}_\omega) = (\pi_\omega(\mathcal{M}))'' \quad (2.51)$$

2. The vector $|\Omega\rangle$ is called a separating vector under the representation $\pi_\omega(\mathcal{M})$ if and only if the map ω is a faithful state.

4 Prove the two propositions above.

Now, according to proposition 1, we know that all representations in the algebra $\mathcal{M}$ are precisely the operators in the set of operators acting on the Hilbert space $\mathcal{H} * \omega$. In this case, we can state

Statement

If the map ω is a pure state, then

$$\mathcal{B}(\mathcal{H}_\omega) = (\pi_\omega(\mathcal{M}))''$$

Therefore, a von Neumann algebra, $\pi_\omega(\mathcal{M})$, can be generated from a C^* -algebra.

Now, to become more familiar with the GNS construction, we present an example of an ordinary system in quantum mechanics. Suppose $\mathcal{M}$ is the algebra of n -dimensional square matrices. Also, to define the map ω in the usual way in which it is defined on the algebra, we have

$$\forall A \in \mathcal{M} \quad \omega(A) = \text{tr}(\rho A) \quad (2.52)$$

where $\rho \in \mathcal{M}$ and satisfies $\text{tr}(\rho) = 1$. First, suppose that the operator ρ has rank 1. In this case, the map $\omega(A) = \text{tr}(\rho A)$ cannot be expressed in terms of two maps ω_1 and ω_2 defined

in the same way as 2.52, and therefore the map ω is a pure state. Now, given the algebra $\mathcal{M}$, which is a C^* -algebra, and considering the map ω as

$$\forall A, B \in \mathcal{M} \quad \langle A|B \rangle = \omega(A^\dagger B) = \text{tr}(\rho A^\dagger B) \quad (2.53)$$

we can verify that it possesses the properties corresponding to the map ω . In addition, since ρ has rank 1, we can state that ω is a faithful state. In this case, there is no need for the existence of a quotient set because $\mathcal{T}$ is trivial, and all operators in $\mathcal{M}$ can be regarded as equivalence classes. Furthermore, since the algebra $\mathcal{M}$ itself is a vector space over the field $\mathcal{F} = \mathbb{C}$, these equivalence classes can be regarded as vectors in this vector space. Now, according to the definition of an algebra representation, $\pi_\omega : \mathcal{M} \rightarrow \mathcal{M}$, the vectors can be transformed into one another according to expression 2.47. Therefore, by applying $\pi_\omega(\mathcal{M})$ to the identity vector $|\Omega\rangle = |I\rangle$, all vectors in the vector space $\mathcal{M}$ can be obtained, and we can also write

$$\forall A \in \mathcal{M} \quad , \quad \omega(A) = \text{tr}(\rho A) = \langle \Omega | \pi_\omega(A) | \Omega \rangle \quad (2.54)$$

In this case, the Hilbert space $\mathcal{H} * \omega$ can be reconstructed around the identity vector $|\Omega\rangle$. Also, because ρ has rank 1 and makes the state ω pure, we can conclude that the representation $\pi * \omega(\mathcal{M})$ of the structure $(\mathcal{H} * \omega, \pi * \omega(\mathcal{M}), |\Omega\rangle)$ is an irreducible representation, and we have

$$\mathcal{B}(\mathcal{H}_\omega) = (\pi_\omega(\mathcal{M}))'' \quad (2.55)$$

Since all bounded operators acting on the Hilbert space $\mathcal{H} * \omega$ are precisely the representation of the algebra $\mathcal{M}$, we can state that $\pi_\omega(\mathcal{M})$ is a von Neumann algebra. Now that we have understood that $\pi_\omega(\mathcal{M})$ is a von Neumann algebra, we can state $\mathcal{M} \simeq \pi_\omega(\mathcal{M})$. Therefore, according to expression 2.54, we have

$$\forall A \in \mathcal{M} \quad , \quad \omega(A) = \text{tr}(\rho A) = \langle \Omega | \pi_\omega(A) | \Omega \rangle = \text{tr}(|\Omega\rangle \langle \Omega | \pi_\omega(A)) = \text{tr}(|\Omega\rangle \langle \Omega | A) \quad (2.56)$$

In this case, since $|\Omega\rangle = |I\rangle$, we have $\rho = |\Omega\rangle \langle \Omega| = I_1$. Also, the Hilbert space has the form $\mathcal{H}_\omega = \mathcal{H}_n \times \mathcal{H}_1$. In this case, we can write

$$|\Omega\rangle = \sum_{i=1}^1 \rho_i^{\frac{1}{2}} |i\rangle_R |i\rangle_L \quad , \quad |A\rangle = \sum_{j=1}^n \sum_{i=1}^1 A_{ij} \rho_i^{\frac{1}{2}} |i\rangle_R |j\rangle_L \quad (2.57)$$

Then the representation $\pi_\omega(\mathcal{M})$ can be written as

$$\forall A \in \mathcal{M} \quad , \quad \pi_\omega(\mathcal{M}) = A \otimes I_1 \quad (2.58)$$

It should be noted that it can be shown that if $m = n$, then the state $|\Omega\rangle$ is also a separating vector.

5 Show that if $m = n$, then the state $|\Omega\rangle$ is simultaneously a cyclic and separating vector with respect to the representation $\pi_\omega(\mathcal{M})$.

Now we turn to a more practical example in physics, which we previously introduced

as an example of the non-factorization of Hilbert space and as motivation for studying the properties of algebras instead of Hilbert space. For review, suppose that, according to Figure 2.1, we have a subsystem consisting of N pairs of particles, each pair being in an entangled state. This system can be considered as two sets of qubits, with each pair interacting separately with one another. Therefore, the quantum state can be considered as a Bell state:

$$|\psi\rangle_\theta = \cos \theta |0\rangle |0\rangle + \sin \theta |1\rangle |1\rangle$$

And the quantum state of the entire system is

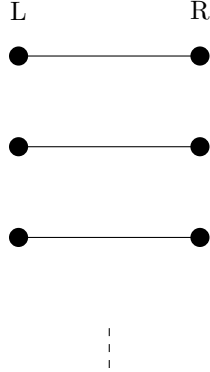


Figure 2.1: N pairs of entangled particles

$$|\Psi\rangle \equiv |\Psi\rangle_\theta = |\psi\rangle_{\theta_1} \otimes \cdots \otimes |\psi\rangle_{\theta_N}$$

Now, as we previously argued, if we let the number of these pairs of particles tend to infinity, $N \rightarrow \infty$, then, because an infinite amount of energy would be required, this state cannot be transformed into a pure state, and therefore Hilbert-space factorization will not hold. Now, if we study the algebra of operators of this system, we can obtain a great deal of information about it. In this case, suppose $\mathcal{M}$ is a C^* -algebra containing all finite-energy operators (that is, these operators only rotate a finite number of states). Therefore, if α_i is an operator that changes the i -th pair, these operators can be written as

$$\forall O \in \mathcal{M} \ , \ \text{For many } \alpha_i = \text{I}_2 \otimes \text{I}_2 \mid O = \alpha_1 \otimes \cdots \otimes \alpha_i \otimes \cdots \quad (2.59)$$

We can also define the map ω_θ on this algebra:

$$\forall O \in \mathcal{M} \mid \omega_\theta(O) = \langle \Psi | O | \Psi \rangle \quad (2.60)$$

Then, since the algebra $\mathcal{M}$ is a vector space, we define the inner-product map as

$$\forall O, P \in \mathcal{M} \mid \langle O | P \rangle = \omega_\theta(O^\dagger P) = \langle \Psi | O^\dagger P | \Psi \rangle \quad (2.61)$$

and it can be shown that this map is faithful. Therefore, $\mathcal{T}$ is trivial and all operators in $\mathcal{M}$ are vectors in this vector space. Then, as before, considering the representation of this algebra, $\pi_{\omega_\theta}(\mathcal{M})$, and the identity vector, which in this problem is the same as the initial

vector $|\Psi\rangle$, we consider

$$|\Psi\rangle = \alpha_i \otimes \cdots \otimes \alpha_i \otimes \cdots \quad (2.62)$$

and different vectors can be generated in the vector space. Therefore, by having the initial state $|\Omega\rangle$ and the representation $\pi_{\omega_\theta}(\mathcal{M})$, the Hilbert space $\mathcal{H}_{\Psi_\theta}$ can be generated. Now, given that the entanglement is very large, it is natural that 2.59 cannot be reconstructed in such a way that the map ω_θ becomes a pure state. Therefore,

$$\mathcal{B}(\mathcal{H}_\omega) \neq (\pi_\omega(\mathcal{M}))'' \quad (2.63)$$

However, the interesting point that leads us toward studying the algebra instead of the Hilbert space is that if we have access only to one subsystem of this problem—for example, all particles on the right-hand side of Figure 2.1—then the algebra available to us is $\mathcal{M}_R$, for which we know

$$\mathcal{B}(\mathcal{H}_\omega) = \mathcal{M}_R \vee \mathcal{M}_L \quad , \quad \mathcal{M}_L = \mathcal{M}'_R \quad (2.64)$$

Then, according to weak convergence in the Hilbert space $\mathcal{H}_{\Psi_\theta}$, it can be shown that the algebra $\mathcal{M}_R$ is a von Neumann algebra, and the algebra associated with the left side of the system, $\mathcal{M}_L$, can also be studied.

Therefore, as we observed in this example, if, as in the previous examples, we study Hilbert spaces assigned to subsystems in many examples from nature, we encounter a factorization problem that effectively makes the calculations inefficient. In this case, as we understood from the above example, considering algebras for a subsystem can provide valuable assistance in studying these subsystems, something that was not possible before. Furthermore, in most physical systems related to what is called holography and other areas of quantum gravity, the algebras corresponding to subsystems are von Neumann algebras, and this makes more information available and simplifies the analysis. In addition, the GNS construction, which gives rise to the Hilbert space and the von Neumann algebra, is very helpful in studying these physical phenomena, which we will address in the following sections.

Chapter 3

Entanglement Structure in von Neumann Algebras

As we have understood, by considering an algebra for a subsystem, it is possible to investigate its physical properties, something that could not be achieved merely by considering the Hilbert space. Therefore, in this new language, we attempt to study a physical subsystem by investigating its algebra. An important point is that, by considering an algebra for a subsystem and using the GNS construction, a von Neumann algebra as well as a Hilbert space emerge. This provides the necessary tools for a better investigation of the physical properties of the system. For this purpose, in this chapter we study different types of algebras in the physical system of pairs of spin particles.

3.1 Density Operators for Type I and II Algebras

To begin this section, we discuss the main difference between type I and type II von Neumann algebras and type III von Neumann algebras. As stated in the previous chapter, for the dimension of projection operators in these algebras, we can use the map tr ; however, in a type III von Neumann algebra, this is not possible. The existence of this map has an important effect on the properties and capabilities of the algebra and on its description of a physical system, which we discuss below.

Suppose that $|\Psi\rangle$ is the state of a system. If the von Neumann algebra associated with this system, which we denote by $\mathcal{M}$, admits a map $\text{tr} : \mathcal{M} \rightarrow \mathbb{C}$, then a density operator $\rho_{\mathcal{M}} \in \mathcal{M}$ corresponding to $\mathcal{M}$ can be defined for the state $|\Psi\rangle$. This operator must also satisfy the following condition:

$$\forall A \in \mathcal{M} \quad , \quad \text{tr}(\rho_{\mathcal{M}} A) = \langle \Psi | A | \Psi \rangle \quad (3.1)$$

An important point is that we can define the density operator $\rho_{\mathcal{M}} \in \mathcal{M}$ without assuming the factorization of a Hilbert space. This operator can also be referred to as the reduced density operator of the state $|\Psi\rangle$ belonging to $\mathcal{M}$, and can be used to calculate entropy and related quantum-information quantities. For example, the entanglement entropy $S_{\mathcal{M}}$ is

defined as

$$S_{\mathcal{M}} \equiv -\text{tr} \rho_{\mathcal{M}} \log \rho_{\mathcal{M}} \quad (3.2)$$

Interpreting $\rho_{\mathcal{M}}$ as a reduced density operator may give rise to a puzzle, since this definition is very different from what we know as the reduced density matrix, which is written as $\rho = |\Psi\rangle\langle\Psi|$. In particular, one may ask the following question:

Question

Suppose that we only have access to the algebra $\mathcal{M}$ and can define the reduced density operator $\rho_{\mathcal{M}} \in \mathcal{M}$ for it. In this case, how can access to an algebra that does not refer to entanglement reveal the entanglement in the system?

To answer this question, we need to classify von Neumann algebras. As we will show below, the differences between these classes give rise to different structures of entanglement.

3.2 Type I Algebra

As stated previously, if $\mathcal{M}$ is an algebra, the center of the algebra $\mathcal{Z}(\mathcal{M})$ is defined as

$$\mathcal{Z}(\mathcal{M}) = \mathcal{M} \cap \mathcal{M}'$$

Therefore, an algebra being a factor means that the center of the algebra is trivial. Now suppose that $\mathcal{M}$ is a type I factor algebra. As we know, the existence of a type I algebra makes the Hilbert space factorize. In this case, we can write

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B \quad (3.3)$$

We also know that in this type of algebra, the dimension of a projection operator can be defined by the map tr . In this case, the dimension of the Hilbert space can be taken to be equal to the dimension of the largest projection operator, which is the identity operator. In other words, if the dimension map $d : \mathcal{M} \rightarrow \mathbb{C}$ is considered for the identity operator, then, by defining the map $\text{tr} : \mathcal{M} \rightarrow \mathbb{C}$, we can write

$$\forall P \in \mathcal{M} \ , \ d(P) = \text{tr}(P) \quad (3.4)$$

Since this map in this type of algebra is the same as the map $\text{tr}_{\mathcal{H}_A}$, we therefore have

$$\text{tr} = \text{tr}_{\mathcal{H}_A} \quad (3.5)$$

Now, since we have a type I factor algebra, we can write

$$\mathcal{M} = \mathcal{B}(\mathcal{H}_A) \otimes \mathbb{I}_B \ , \ \mathcal{M}' = \mathbb{I}_A \otimes \mathcal{B}(\mathcal{H}_B) \quad (3.6)$$

In this case, as stated previously, by considering the algebra $\mathcal{M}$, we can express the reduced density operator $\rho_{\mathcal{M}}$ according to equation 3.1. Therefore, from the above equation

we have

$$\forall A \in \mathcal{M} \ , \ \text{tr}(\rho_{\mathcal{M}} A) = \langle \Psi | A | \Psi \rangle = \text{tr}_{\mathcal{H}_A}(\rho_A A) \quad (3.7)$$

Thus,

$$\rho_{\mathcal{M}} = \rho_A = |\Psi\rangle \langle \Psi| \quad (3.8)$$

Therefore, in this case we can conclude

Statement

In a type I factor algebra such as $\mathcal{M}$, the reduced density operator that satisfies only condition 3.1 is the same as the reduced density operator in quantum mechanics, and this is why this operator can contain information about the entanglement structure.

Two points concerning this type of algebra are as follows:

- First, we emphasize that these two approaches to obtaining the reduced density operator are different. In the quantum-mechanical approach, we need the entire system, part of which we do not have access to, and we also need the state of the entire system. Then, by considering a map on the Hilbert space corresponding to a subsystem, we obtain the density operator. In the algebraic approach, however, we only need the algebra corresponding to a subsystem, and then we can define the reduced density operator.
- As stated, if the map tr is the same map as the one that exists in quantum mechanics, then it is evident that this map, as well as the reduced density operator, has an interpretation in terms of state counting and entropy.

Now, suppose we study a system whose algebra is a type I von Neumann algebra but does not have a trivial center. How can all the previous statements be formulated in this case? For this purpose, suppose we denote this type of algebra by $\mathcal{M}$. In this case, this type of algebra can generally be associated with a Hilbert space as follows:

$$\mathcal{H} = \oplus_{\alpha} \mathcal{H}_{\alpha} \ , \ \mathcal{H}_{\alpha} = \mathcal{H}_{A_{\alpha}} \otimes \mathcal{H}_{B_{\alpha}} \quad (3.9)$$

This is obtained by putting together the Hilbert spaces associated with type I factor algebras. In a sense, one can say that the Hilbert space corresponding to a type I algebra can be expressed in terms of parts of the Hilbert spaces corresponding to type I factor algebras. Thus, using equation 3.5, the map tr is

$$\text{tr} = \sum_{\alpha} \text{tr}_{\mathcal{H}_{\alpha}} \quad (3.10)$$

Therefore, we have

$$\mathcal{M} = \oplus_{\alpha} (\mathcal{B}(\mathcal{H}_{A_{\alpha}}) \otimes \mathbf{I}_{B_{\alpha}}) \ , \ \mathcal{M}' = \oplus_{\alpha} (\mathbf{I}_{A_{\alpha}} \otimes \mathcal{B}(\mathcal{H}_{B_{\alpha}})) \quad (3.11)$$

In this case, the reduced density operator arising from the algebra $\mathcal{M}$ can be written as

$$\rho_{\mathcal{M}} = \sum_{\alpha} p_{\alpha} (\rho_{A_{\alpha}} \otimes \mathbf{I}_{B_{\alpha}}) \quad , \quad \rho_{A_{\alpha}} \in \mathcal{B}(\mathcal{H}_{A_{\alpha}}) \quad (3.12)$$

which must also satisfy

$$\forall A \in \mathcal{M} = \oplus_{\alpha} (\mathcal{B}(\mathcal{H}_{A_{\alpha}}) \otimes \mathbf{I}_{B_{\alpha}}) \quad , \quad \text{tr}(\rho_{\mathcal{M}} A) = \langle \Psi | A | \Psi \rangle \quad (3.13)$$

In this case, since $\text{tr}_{\mathcal{H}_{\alpha}} \rho_{A_{\beta}} = 1$, we must have

$$\text{tr} \rho_{\mathcal{M}} = \sum_{\alpha} \text{tr}_{\mathcal{H}_{\alpha}} \rho_{\mathcal{M}} = \sum_{\alpha} \text{tr}_{\mathcal{H}_{\alpha}} \sum_{\beta} p_{\beta} (\rho_{A_{\beta}} \otimes \mathbf{I}_{B_{\beta}}) = \sum_{\beta} p_{\beta} = 1 \quad , \quad p_{\beta} \in [0, 1] \quad (3.14)$$

Therefore, this reduced density operator is the same as the density operator in quantum mechanics, and hence

Statement

If $\mathcal{M}$ is a type I von Neumann algebra, then the reduced density operator in this algebra that is accessible to us is the same as the density operator in quantum mechanics, defined as

$$\rho_{\mathcal{M}} = \sum_{\alpha} p_{\alpha} (\rho_{A_{\alpha}} \otimes \mathbf{I}_{B_{\alpha}}) \quad , \quad \rho_{A_{\alpha}} \in \mathcal{B}(\mathcal{H}_{A_{\alpha}})$$

which, accordingly, contains information about the entanglement structure. In this case, the entanglement entropy for a subsystem whose algebra is $\mathcal{M}$ can be calculated:

$$\begin{aligned} S_{\mathcal{M}} &= -\text{tr} \rho_{\mathcal{M}} \log \rho_{\mathcal{M}} = -\text{tr} \sum_{\alpha} p_{\alpha} (\rho_{A_{\alpha}} \otimes \mathbf{I}_{B_{\alpha}}) \log \sum_{\beta} p_{\beta} (\rho_{A_{\beta}} \otimes \mathbf{I}_{B_{\beta}}) \\ &= -\sum_{\alpha} p_{\alpha} \log p_{\alpha} + \sum_{\alpha} p_{\alpha} S_{\alpha} \end{aligned} \quad (3.15)$$

where $S_{\alpha} = -\text{tr}_{\alpha} \rho_{A_{\alpha}} \log \rho_{A_{\alpha}}$.

Therefore, for physical systems possessing both factorized and non-factorized Hilbert spaces, there is a unified approach in the algebraic language to the structure of quantum information in a type I von Neumann algebra.

3.3 Type II Algebra

To investigate the structure of quantum information in a type II algebra, it is useful to review it by considering an example. Therefore, we consider the familiar problem of N pairs of spin particles. As we know, the quantum state of one pair is a Bell state given by

$$|\psi_{\theta}\rangle = \cos \theta |0\rangle |0\rangle + \sin \theta |1\rangle |1\rangle$$

and the state of the entire system is

$$|\Psi_{\theta}\rangle = |\psi_{\theta}\rangle_1 \otimes \cdots \otimes |\psi_{\theta}\rangle_N$$

Therefore, as stated, in the limit $N \rightarrow \infty$, due to the presence of infinite energy, it will not be possible to rotate the spin, and in this case the above quantum state cannot be regarded as a pure state or as a factorized Hilbert space. In this situation, by using the algebraic language and associating a part of the system (such as the right-hand side of the spin pairs) with an algebra, we were able, using the GNS construction, to generate a Hilbert space and obtain a von Neumann algebra. Now it is interesting to consider $\theta = \frac{\pi}{4}$ for the Bell state:

$$\begin{aligned} |\psi_{\theta=\frac{\pi}{4}}\rangle &= \frac{\sqrt{2}}{2}(|0\rangle|0\rangle + |1\rangle|1\rangle) \\ |\Psi_{\theta=\frac{\pi}{4}}\rangle &= |\psi_{\theta=\frac{\pi}{4}}\rangle_1 \otimes \cdots \otimes |\psi_{\theta=\frac{\pi}{4}}\rangle_N \end{aligned} \quad (3.16)$$

Let us also denote the algebra corresponding to the right-hand subsystem by $\mathcal{M}$, and let a be an operator acting on the right-hand side of each pair of particles. In this case, we can write

$$\frac{1}{2} \text{tr}_2(a) = \langle \psi_{\theta=\frac{\pi}{4}} | a | \psi_{\theta=\frac{\pi}{4}} \rangle \quad (3.17)$$

Then, in general, considering an operator acting on the right-hand side of k pairs, we have

$$\frac{1}{2^k} \text{tr}_2 a_1 \dots \text{tr}_2 a_k = \langle \Psi_{\theta=\frac{\pi}{4}} | A | \Psi_{\theta=\frac{\pi}{4}} \rangle \quad (3.18)$$

where $A \in \mathcal{M}$ is an operator that acts only on the right-hand side of the system. Using the above equation, one can also show that $\langle \Psi_{\theta=\frac{\pi}{4}} | AB | \Psi_{\theta=\frac{\pi}{4}} \rangle = \langle \Psi_{\theta=\frac{\pi}{4}} | BA | \Psi_{\theta=\frac{\pi}{4}} \rangle$. Therefore, the map tr can be defined as

$$\forall A \in \mathcal{M} \ , \ \text{tr}(A) = \langle \Psi_{\theta=\frac{\pi}{4}} | A | \Psi_{\theta=\frac{\pi}{4}} \rangle \quad (3.19)$$

We now examine the projection operators in this algebra for $\theta = \frac{\pi}{4}$ in order to determine the type of this algebra. First, we have the largest projection operator, which is the identity operator:

$$I = I_2 \otimes I_2 \otimes \dots \quad (3.20)$$

The projection operators can then be written as

$$P_1 = P_{\uparrow} \otimes I_2 \otimes I_2 \otimes \dots \ , \ P_2 = P_{\uparrow} \otimes P_{\uparrow} \otimes I_2 \otimes \dots \ , \ \dots \ , \ P_{\uparrow} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad (3.21)$$

where $P_{\uparrow}$ is the operator that rotates the spin of the right-hand particle in each pair. Since we are in the limit $N \rightarrow \infty$, we can therefore insert $P_{\uparrow}$ an arbitrary number of times and thereby construct a projection operator. Thus, this algebra contains no smallest projection operator, and we can conclude that the algebra $\mathcal{M}$ is a type II or III von Neumann algebra. To determine whether the projection operators have finite or infinite dimension, we take the map defined in 3.19 as the dimension map. In this case, for the largest projection operator

we have

$$d(I) = \text{tr}(I) = \langle \Psi_{\theta=\frac{\pi}{4}} | \Psi_{\theta=\frac{\pi}{4}} \rangle = 1 \quad (3.22)$$

Therefore, since a finite projection operator exists, $\mathcal{M}$ is a type II von Neumann algebra, and from the above equation we can identify this algebra as type II_1 . It is then natural that, upon calculating the dimension of each projection operator, its value lies in the interval $(0, 1]$. It should be noted that, as stated in the previous chapter, the map tr has a different meaning in type I and type II algebras. Now, to define the reduced density operator for the algebra $\mathcal{M}$, first suppose that $\Phi \in \mathcal{H}_{\theta=\frac{\pi}{4}}$ is an arbitrary state, where $\mathcal{H}_{\theta=\frac{\pi}{4}}$ is the Hilbert space. In this case, this state can be written as

$$|\Phi\rangle = |\eta_1\rangle \otimes |\eta_2\rangle \otimes \cdots \otimes |\eta_n\rangle \otimes \cdots \quad (3.23)$$

where, for k pairs, the $|\eta_i\rangle$ correspond to $|\psi_{\theta=\frac{\pi}{4}}\rangle$, while for the remaining pairs they are equal to the identity operator. Therefore, to define the reduced density operator for the state $|\Phi\rangle$, let us denote it by $\rho_{\mathcal{M}} \in \mathcal{M}$, satisfying

$$\forall A \in \mathcal{M} \quad , \quad \text{tr}(A\rho_{\mathcal{M}}(\Phi)) = \langle \Phi | A | \Phi \rangle \quad (3.24)$$

In this case, one can show that

$$\rho_{\mathcal{M}}(\Phi) = 2\rho_1 \otimes 2\rho_2 \otimes \cdots \otimes 2\rho_n \otimes \cdots \quad (3.25)$$

where, according to equation 3.17, we know that a factor of 2 is required for normalization, and this factor therefore appears in the above expression. Thus, from equation 3.25, we obtain

$$\rho_{\mathcal{M}}(\Psi_{\theta=\frac{\pi}{4}}) = I \quad (3.26)$$

Therefore, up to this point, the reduced density operator is very different from what we know from quantum mechanics. This is not surprising, however, because, as stated previously, the algebra corresponding to what exists in quantum mechanics is a type I von Neumann algebra. Thus, for a type II von Neumann algebra, the reduced density operator is not the same as the one in quantum mechanics. Nevertheless, it is interesting to investigate what the entanglement entropy looks like when calculated using this operator. For this purpose, we have

$$\begin{aligned} S_{\mathcal{M}}(\Phi) &= -\text{tr}\rho_{\mathcal{M}} \log \rho_{\mathcal{M}} = - \sum_{i=1}^{N=\infty} \text{tr}\rho_i \log \rho_i - k \log 2 \\ &= \sum_{i=1}^{N=\infty} [S_2(\rho_i) - S_2(\frac{I}{2})] = \lim_{N \rightarrow \infty} [S_R^N(\Phi) - S_R^N(\Psi_{\theta=\frac{\pi}{4}})] \end{aligned} \quad (3.27)$$

where S_R^N is the entropy corresponding to the system of N pairs of spin particles, for which we consider the right-hand subsystem. As we can see from the above equation, in the limit $N \rightarrow \infty$, the entropy in this type of algebra is interpreted as the difference between an arbitrary state and the maximally entangled state. In this case, we can define

Statement

The relative entropy for an arbitrary state Φ in a type II von Neumann algebra is defined as

$$S_{\mathcal{M}}(\Phi) = -S_{\mathcal{M}}(\Phi || \Psi_{\theta=\frac{\pi}{4}}) = -\text{tr}[\rho_{\mathcal{M}}(\Phi)(\log \rho_{\mathcal{M}}(\Phi)) - \log \rho_{\mathcal{M}}(\Psi_{\theta=\frac{\pi}{4}})] \quad (3.28)$$

Thus, as we stated previously, the map tr in a type II algebra, unlike in type I, does not mean state counting in the sense familiar from statistical mechanics. Therefore, it is natural that the entanglement entropy represents the distance of an arbitrary state $|\Phi\rangle$ from the maximally entangled state rather than the entropy known from statistical mechanics. It should also be noted that the map tr in a type II_{∞} von Neumann algebra is no longer normalized as in 3.26.

On the other hand, as we know from this example, the algebra corresponding to the entire system is given by 2.64, in which case the algebra described in this example is a type II factor algebra. If we want to consider a general type II algebra, then, as in the previous section, it is necessary to construct a general algebra of this type from a sum of type II factor algebras. One can then show that the reduced density operator and the entanglement entropy have the same interpretation as that assigned to them for the corresponding factor algebra.

6 Derive the reduced density operator and the entanglement entropy for a general type II algebra.

Therefore, in a type II von Neumann algebra, as stated, the factorization property of the Hilbert space does not exist. This means that every arbitrary state in the Hilbert space corresponding to the entire system is entangled between the subsystems with corresponding algebras $\mathcal{M}$ and $\mathcal{M}'$. In this case, by defining the map ω on the algebra $\mathcal{M}$, it cannot be regarded as a pure state, which, according to the above discussion, is natural. This is in contrast to what occurs in a type I algebra. We then defined the reduced density operator and, considering that the map tr in this type of algebra has a mathematical definition rather than a physical interpretation, calculated the entanglement entropy and found that, in a type II algebra, this definition represents the distance from the maximally entangled state.

3.4 Type III Algebra

We still need to analyze the type III von Neumann algebra. However, because the reduced density operator cannot be defined for this type of algebra, it is not possible to study the entanglement structure using these operators in the same way as for the other types. But does this mean that, in practice, the entanglement structure for a type III algebra cannot be studied? As we know, most physical systems, such as statistical systems and quantum field theories, possess type III algebras. If we are unable to understand and obtain the entanglement structure through the algebraic framework, then essentially this framework provides us with no help in studying quantum information in these systems!

In the next chapter, by introducing a new structure, we will show that it is still possible to discuss entanglement and quantum information even for type III algebras, and that the existence or absence of reduced density operators is not a problem.

Chapter 4

Modular Theory

4.1 Density Operator for a Type III Algebra

We now turn to the study of type III von Neumann algebras, which are the most important algebras for the physics of interest to us. As we discussed earlier, similar to type II von Neumann algebras and unlike type I algebras, these algebras do not possess a factorized Hilbert space. However, unlike both of the aforementioned algebras, since all projection operators in this type of algebra are infinite, it is not possible to define the map tr . For this reason, one cannot define a well-defined reduced density operator, and consequently, investigating the entanglement structure for this type of algebra is particularly intriguing for us. In this section, we attempt to provide an overview of what exists in this algebra, classify this algebra using mathematical definitions, and present an example. We will then introduce, in the next chapter, the concept relevant to this algebra in physics, known as the Tomita-Takesaki theorem.

Our motivation for describing the entanglement structure of a type III von Neumann algebra requires us first to return to type I algebras, where the concept of entanglement is properly formulated in quantum mechanics. As we stated, for the quantum state of the entire system, $A \cup B$, which we denote by $|\Psi\rangle$, we can express the density operator for subsystem A as

$$\rho_R = |\Psi\rangle\langle\Psi| \equiv \exp(-K_A) \quad , \quad K_A = -\log \rho_A \quad (4.1)$$

Since ρ_A contains quantum information, K_A , which is expressed in the above equation in analogy with the Hamiltonian of subsystem A , also contains this information. Therefore, it is appropriate to consider $K_A \geq 0$.

Now, if this quantity acts like a Hamiltonian, then by considering the Heisenberg picture, we can generate all operators of $\mathcal{M}$ using this quantity¹ (usually, in the algebraic literature, the Heisenberg picture is referred to as a flow):

$$\forall A \in \mathcal{M} \quad , \quad A(s) = \exp(iK_A s) A \exp(-iK_A s) \in \mathcal{M} \quad (4.2)$$

¹It should be noted that because of the existence of a type I algebra, what we state for the algebra corresponding to the subsystem can also be considered for the algebra corresponding to the entire system. However, in this text we prefer to consider the subsystem.

An interesting point is that this operator flow, for any real value of s , still belongs to the algebra $\mathcal{M}$, and in a sense, one can say that this flow traverses the entire algebra. Thus,

Statement

By considering a type I von Neumann algebra for a subsystem, which we denote by $\mathcal{M}$, all operators within the algebra $\mathcal{M}$ can be expressed in the form 4.2 using a quantity such as K_A , which is defined according to 4.1. Therefore, the quantity s can emerge as a "time-like" quantity or "time" in the entanglement structure.

Similarly, for the other subsystem, one can define K_B and ρ_B , in which case we have

Statement

In fact, by combining these two subsystems, thereby forming the total system, one can define an operator such as Δ_Ψ , which is called the modular operator^a:

$$\Delta_\Psi \equiv \rho_A \otimes \rho_B^{-1} \quad , \quad -\log \Delta_\Psi = K_A - K_B \quad (4.3)$$

An interesting property of this operator is that it leaves the state $|\Psi\rangle$ invariant:

$$\Delta_\Psi |\Psi\rangle = \Delta_\Psi^{-1} |\Psi\rangle = |\Psi\rangle \quad , \quad (K_A - K_B) |\Psi\rangle = 0 \quad (4.4)$$

^aModular Operator

On the other hand, by considering the Heisenberg picture, or equivalently the flow, for operators, one can define

Statement

The modular flow for the algebra $\mathcal{M}$, denoted by $\sigma_s(\mathcal{M})$, can be defined as

$$\forall A \in \mathcal{M} \quad , \quad s \in \mathfrak{R} \quad | \quad \sigma_s(A) \equiv \Delta_\Psi^{-is} A \Delta_\Psi^{+is} \in \mathcal{M} \quad (4.5)$$

where s is called time, or more precisely, modular time.

As we stated, one can also define the modular conjugation operator J_Ψ , which satisfies $J_\Psi^2 = 1$. It should be noted that the Hamiltonians corresponding to subsystems A and B , denoted by K_A and K_B , respectively, as well as the modular operator Δ_Ψ , are well-defined quantities if the density operators ρ_A and ρ_B have full rank, or equivalently, are invertible. This statement is clear in the definition of the modular operator 4.3, since it explicitly uses the inverse of ρ_B . This statement requires the Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, which are factorized in this algebra, to have equal dimensions. It also guarantees the existence of a correspondence between states and operators in the two Hilbert spaces. In this case, we turn to the definition of another important operator that exchanges the two subsystems.

Statement

Assume that $|\Psi\rangle$ is the quantum state of the entire system and is represented in the Schmidt decomposition as $|\Psi\rangle = \sum_n \sqrt{\lambda_n} |n\rangle_A |n\rangle_B$. In this case, based on $\{|n\rangle\}_{i=1}^D$,

an arbitrary state in this Hilbert space can be written as

$$|\psi\rangle = \sum_{n,m} \psi_{nm} |n\rangle_A |m\rangle_B \quad (4.6)$$

Therefore, the modular conjugation operator J_Ψ can be defined as

$$J_\Psi |\psi\rangle = \sum_{n,m} \psi_{nm}^* |m\rangle_A |n\rangle_B \quad (4.7)$$

where J_Ψ is an anti-unitary operator satisfying $J_\Psi^2 = 1$.

The operator Δ_Ψ , the modular flows expressed in 4.5, and the existence of the modular conjugation operator together provide a powerful framework for studying and classifying entanglement structures.

Up to this point, what we have stated was a repetition of part of the previous chapter, and we attempted to provide an overview of what follows. Therefore, the existence of modular operators is, in a sense, dependent on the definition and existence of reduced density operators. On the other hand, these operators by themselves are rich in information about the algebra (the subsystem under consideration), and one can study a collection of such operators independently of the background from which they originated. Thus, this framework can be regarded as a modular structure that does not require an explicit definition of reduced density operators. This perspective is particularly useful for type III von Neumann algebras, for which we cannot define density operators. However, a question arises in this regard:

Question

How can one define well-defined modular operators for a type III von Neumann algebra when the condition for their being well-defined is based on the full-rank property of the reduced density operators?

To answer this question, we attempt to use another perspective based on the Hilbert space of the GNS construction. Suppose we have a subsystem with a C^* -algebra, which we denote by $\mathcal{M}$. Then, as we stated, it is sufficient to define an inner product by considering the map ω :

$$\forall A, B \in \mathcal{M}; \langle A|B \rangle = \omega(A^\dagger B) \quad (4.8)$$

Then, by removing the null operators from the algebra $\mathcal{M}$, equivalence classes can be regarded as vectors in the vector space $\mathcal{M}/\mathcal{T}$. Furthermore, by considering a vector such as $|\Psi\rangle$ and applying all equivalence classes to it, the Hilbert space $\mathcal{H}_\omega$ can be generated. In this case, the vector $|\Psi\rangle$ is called a cyclic vector. As we stated, there are two important statements concerning it. The first states that if a representation of the algebra $\mathcal{M}$, denoted by $\pi_\omega(\mathcal{M})$, is irreducible, then one can state:

$$\mathcal{B}(\mathcal{H} * \omega) = (\pi_\omega(\mathcal{M}))'' \quad (4.9)$$

which means the emergence of a von Neumann algebra. The other statement is that if the map ω is faithful, then the vector $|\Psi\rangle$ is called a separating vector with respect to the

algebra $\mathcal{M}$.

As we mentioned in the previous chapter, if the dimensions of the Hilbert spaces are equal, or equivalently, if the reduced density operators have full rank, then one can state that $|\Psi\rangle$ is a cyclic and separating vector with respect to the algebra $\mathcal{M}$, with no restriction on the type of algebra². We also found that the full-rank property of the aforementioned operators means that the modular operators are well-defined. Therefore, we can conclude that the existence and well-definedness of the modular structure are related to the existence of a cyclic and separating vector such as $|\Psi\rangle$ with respect to the algebra $\mathcal{M}$. Therefore,

Statement

As we stated, the modular structure contains the richest quantum information associated with the subsystem. Initially, the existence of this structure was based on the existence of reduced density operators, but we found that this structure can also be considered for type III algebras, for which reduced density operators cannot be defined. Accordingly, the existence of this structure in general depends only on the existence of a cyclic and separating vector with respect to the algebra under consideration.

In this way, a vast class of physical phenomena whose subsystems are described by type III algebras can be studied. For this purpose, we need to introduce a theory that discusses more completely and precisely the relation between the modular structure and a cyclic and separating vector with respect to an algebra.

4.2 Tomita-Takesaki Theory

As we stated in the previous section, the existence of $|\Psi\rangle$ as a cyclic and separating vector with respect to the algebra $\mathcal{M}$ is equivalent to the modular operator Δ_Ψ , the modular flow $\sigma_s(\mathcal{M})$, and the modular conjugation operator J_Ψ being well-defined. Since the importance of the modular structure rests on the existence of a cyclic and separating vector, all these statements can be formulated as a theory called the Tomita-Takesaki theory:

Statement

This theory states that for every algebra $\mathcal{M}$ possessing a cyclic and separating vector such as $|\Psi\rangle$, we have:

1. There exists a positive modular operator Δ_Ψ that leaves the vector $|\Psi\rangle$ invariant,

$$\Delta_\Psi |\Psi\rangle = |\Psi\rangle \quad (4.10)$$

and $K_\Psi = -\log \Delta_\Psi$ is a generator of the modular flow,

$$\begin{aligned} \forall A \in \mathcal{M} \ , \ \sigma_s(A) &\equiv \exp(+iK_\Psi s) A \exp(-iK_\Psi s) = \Delta_\Psi^{-is} A \Delta_\Psi^{+is} \in \mathcal{M} \\ \forall A' \in \mathcal{M} \ , \ \sigma_s(A') &\equiv \exp(+iK_\Psi s) A' \exp(-iK_\Psi s) = \Delta_\Psi^{-is} A' \Delta_\Psi^{+is} \in \mathcal{M}' \end{aligned} \quad (4.11)$$

2. There exists an anti-unitary operator called the modular conjugation operator

²That is, it may be a type I, II, or III algebra.

J_Ψ , which has the following properties:

$$\begin{aligned} J_\Psi |\Psi\rangle &= |\Psi\rangle, \quad J_\Psi = J_\Psi^{-1} = J_\Psi^\dagger, \quad J_\Psi \Delta_\Psi J_\Psi = \Delta_\Psi^{-1} \\ J_\Psi \mathcal{M} J_\Psi &= \mathcal{M}', \quad J_\Psi \mathcal{M}' J_\Psi = \mathcal{M} \end{aligned} \quad (4.12)$$

3. Moreover, the vector

$$\forall A \in \mathcal{M}, \quad \Delta_\Psi^{-is} A |\Psi\rangle \quad (4.13)$$

can be analytically continued along the line $\text{Im}; s \in (0, \frac{1}{2})$, in which case we have

$$\forall A \in \mathcal{M}, \quad t \in \mathbb{R}, \quad \Delta_\Psi^{-i(t+\frac{1}{2})} A |\Psi\rangle = \Delta_\Psi^{-it} J_\Psi A^\dagger |\Psi\rangle \quad (4.14)$$

4. On the other hand, if we define the correlation function for the modular flow operator as

$$\forall A, B \in \mathcal{M}, \quad f_{AB}(s) \equiv \langle \Psi | \sigma_s(A) | \Psi \rangle \quad (4.15)$$

then this correlation function can be analytically continued along the line $\text{Im}; s \in (-1, 0)$ and satisfies the KMS condition, $f_{AB}(s) = f_{BA}(-s - i)$.

5. There exists a Tomita operator S_Ψ [9], such that

$$\begin{aligned} \forall A \in \mathcal{M}, \quad \forall A' \in \mathcal{M}', \quad S_\Psi A |\Psi\rangle &= A^\dagger |\Psi\rangle, \quad S_\Psi^\dagger A' |\Psi\rangle = A'^\dagger |\Psi\rangle \\ S_\Psi^2 &= I, \quad S_\Psi |\Psi\rangle = |\Psi\rangle \end{aligned} \quad (4.16)$$

In this case, according to the definitions of the modular operator and modular conjugation, we have

$$S_\Psi = J_\Psi \Delta_\Psi^{\frac{1}{2}} \quad (4.17)$$

Therefore, as we observed in the Tomita-Takesaki theory, $\mathcal{M}$ is a general algebra, and it doesn't need to belong to any particular class of algebras, especially whether or not it is a type III algebra. This theory also expresses several properties:

1. This theory states that if $|\Psi\rangle$ is a cyclic and separating vector with respect to the algebra $\mathcal{M}$, then a type of time, s , called modular time (as we stated in the previous chapter), emerges in $\mathcal{M}$ and leaves $|\Psi\rangle$ invariant.
2. From a physical point of view, the existence of this modular time is interpreted as follows: if an observer has access to a subsystem whose algebra is $\mathcal{M}$, then the state $|\Psi\rangle$ is perceived as a thermal state with temperature $\beta = 1$.
3. The cyclic and separating condition for the vector $|\Psi\rangle$ in type I and II von Neumann algebras is somewhat restrictive because it requires the dimensions of the Hilbert spaces of the two subsystems to be equal. However, in type III von Neumann algebras, which in physics include examples from quantum field theory and statistical systems, this condition is naturally satisfied. This property allows us to study many physical systems solely by considering the Tomita-Takesaki theory.

4. As we stated, in this theory there exists an operator called the Tomita operator. It can be shown that, using this operator alone, one can define all modular operators and the modular structure.
5. It can also be shown that if the modular flow operator, σ_s , exists for an algebra and satisfies the KMS condition, then the modular operator must necessarily exist.

It should be noted that in some physical systems, such as quantum field theory, it is not straightforward to demonstrate that a vector is separating with respect to the algebra corresponding to a subsystem. There is a theorem, which we leave as an exercise, that allows the separating property to be examined more easily.

7 Show that the vector $|\Psi\rangle$ is separating with respect to the algebra $\mathcal{M}$ if and only if it is cyclic with respect to the algebra $\mathcal{M}'$.

Now that we have been able to access all types of von Neumann algebras using the Tomita-Takesaki theory, we can classify these algebras using the modular structure governing them. This is the subject of the next section, which discusses the important implications of this theory for a better understanding of the different types of algebras and for the study of entanglement.

4.3 Classification of Algebras

One of the most important features of the Tomita-Takesaki theory is that it provides a powerful tool enabling us to give an appropriate classification of von Neumann algebras. For this purpose, suppose $\mathcal{M}$ is a von Neumann algebra for which the vector $|\Psi\rangle$ is cyclic and separating. In this case, we can conclude that the modular operators for this algebra are well-defined. However, for the classification of algebras, another definition is also required:

Statement

An inner automorphism is called a flow if there exists a unitary operator such as $U_s \in \mathcal{M}$ such that

$$\forall A \in \mathcal{M} \quad , \quad A_s = U_s^\dagger A U_s \in \mathcal{M} \quad (4.18)$$

Now, as can be seen from the above definition, by comparing it with the definition of modular flow, we can arrive at a statement for classifying von Neumann algebras.

Statement

The modular flow $\sigma_s(\mathcal{M})$ is an inner automorphism of the algebra $\mathcal{M}$ for all $s \in \mathbb{R}$ if and only if $\mathcal{M}$ is a type I or II von Neumann algebra.

Therefore, the above statement says that for a type III von Neumann algebra, not all modular flows of the algebra $\mathcal{M}$ are inner automorphisms. It should be noted that because the modular structure exists, there are cyclic and separating vector(s) for the algebra, and the operators belonging to this structure can therefore be associated with these vectors. In this case, if we suppose that $|\Omega\rangle$ is another cyclic and separating vector with respect to the

algebra $\mathcal{M}$, then all operators belonging to the modular structure can be indexed according to these vectors.

Now suppose $\sigma_s^\Psi(\mathcal{M})$ and $\sigma_s^\Omega(\mathcal{M})$ are the modular flows corresponding to the cyclic and separating vectors $|\Psi\rangle$ and $|\Omega\rangle$:

$$\begin{aligned}\sigma_s^\Psi(\mathcal{M}) &= \Delta_\Psi^{-is} A \Delta_\Psi^{+is} \\ \sigma_s^\Omega(\mathcal{M}) &= \Delta_\Omega^{-is} A \Delta_\Omega^{+is}\end{aligned}\tag{4.19}$$

In this case, we have

$$\begin{aligned}\sigma_s^\Psi(\mathcal{M}) &= \Delta_\Psi^{-is} A \Delta_\Psi^{+is} = \Delta_\Psi^{-is} \Delta_\Omega^{-is} \Delta_\Omega^{+is} A \Delta_\Omega^{-is} \Delta_\Omega^{+is} \Delta_\Psi^{+is} \\ &= (\Delta_\Psi \Delta_\Omega)^{-is} \sigma_s^\Omega(\mathcal{M}) (\Delta_\Psi \Delta_\Omega)^{+is} = u_{\Psi\Omega}^\dagger(s) \sigma_s^\Omega(\mathcal{M}) u_{\Psi\Omega}(s)\end{aligned}\tag{4.20}$$

where $u_{\Psi\Omega}(s)$ is a unitary operator. Therefore, one can conclude that the modular flow corresponding to each of the two cyclic vectors is an inner automorphism of the algebra $\mathcal{M}$.

8 Show that if $\mathcal{M}$ is a von Neumann algebra and the vectors $|\Psi\rangle$, $|\Omega\rangle$, and $|\Phi\rangle$ are cyclic and separating vectors with respect to this algebra, then

$$u_{\Psi\Omega}(s) u_{\Omega\Phi}(s) = u_{\Psi\Phi}(s)\tag{4.21}$$

Another property of this unitary operator is

$$u_{\Psi\Omega}(s_1 + s_2) = u_{\Psi\Omega}(s_1) [\Delta_\Omega^{-is_1} u_{\Psi\Omega}(s_2) \Delta_\Omega^{+is_1}] = u_{\Psi\Omega}(s_1) \sigma_{s_1}^\Omega(u_{\Psi\Omega}(s_2))\tag{4.22}$$

where $\sigma_{s_1}^\Omega(u_{\Psi\Omega}(s_2)) = \Delta_\Omega^{-is_1} u_{\Psi\Omega}(s_2) \Delta_\Omega^{+is_1}$. According to the above equation, one can also state

$$\forall A \in \mathcal{M} \ , \ \sigma_{s_1+s_2}^{\Psi\Omega}(A) = \sigma_{s_1}^{\Psi\Omega}(\sigma_{s_2}^{\Psi\Omega}(A))\tag{4.23}$$

Thus, one can conclude

Statement

If an operator such as $u_{\Psi\Omega}(s_1) \in \mathcal{M}$ exists that satisfies 4.22, then there exists a cyclic and separating vector such as $|\Psi\rangle$ with respect to the algebra $\mathcal{M}$.

Therefore, considering the property that a modular flow operator can be defined corresponding to every cyclic and separating vector, we proceed to the classification of algebras, particularly type III algebras. As we stated, for this type of algebra we have only found that the modular flow is not itself an inner automorphism. Therefore, it is necessary to examine it more carefully. The first property for classification is as follows:

Statement

The set $T(\mathcal{M})$ is a collection of modular flow parameters corresponding to a cyclic and separating vector $|\Psi\rangle$ for which the modular flow is an inner automorphism of $\mathcal{M}$, and

is defined as

$$T(\mathcal{M}) = \{s \mid \sigma_s^\Psi(\mathcal{M}) = U_s^\dagger A U_s, \forall A \in \mathcal{M}, \exists U_s \in \mathcal{M}\} \quad (4.24)$$

It should be noted that this set has no dependence on the cyclic and separating vector. In other words, for every vector with these properties in a given algebra, this set is well-defined and invariant.

For this purpose, as we stated, for type I and II von Neumann algebras, since for all $s \in \mathbb{R}$ the modular flow is an inner automorphism, $T(\mathcal{M}) = \mathbb{R}$. Therefore, for a type III algebra, we know that $T(\mathcal{M})$ is a subset of $\mathbb{R}$. We can also define the second property as follows:

Statement

The set $S(\mathcal{M})$ is the intersection of the spectra of the modular flow corresponding to all cyclic and separating vectors with respect to the algebra $\mathcal{M}$:

$$S(\mathcal{M}) = \cap_\Psi \lambda(\Delta_\Psi) \subset \mathbb{R}_+ \equiv [0, \infty) \quad (4.25)$$

As is clear, if $\mathcal{M}$ is a type I or II von Neumann algebra, because it includes $\Delta_\Psi = I$, we have $S(\mathcal{M}) = 1$. Also, for a type III algebra, one can state:

$$\begin{aligned} \text{Type; III}_0 &: S(\mathcal{M}) = 0, 1 \\ \text{Type; III}_\alpha &: S(\mathcal{M}) = 0 \cup \alpha^n \mid n \in \mathbb{N} \\ \text{Type; III}_1 &: S(\mathcal{M}) = \mathbb{R}_+ \end{aligned} \quad (4.26)$$

Therefore, merely having a cyclic and separating vector (or vectors) with respect to the algebra, which guarantees the existence of the modular structure, allows one to classify von Neumann algebras appropriately. We can also state:

Statement

The sets $T(\mathcal{M})$ and $S(\mathcal{M})$ of $\mathcal{M}$ are two invariant properties of this algebra.

Therefore, classifying type III von Neumann algebras can provide a classification of the entanglement structure for systems possessing these algebras. To better understand what was discussed in this section, we attempt, as with the type II algebra, to present an example involving pairs of spin particles.

As we stated previously, in a system of N pairs of spin particles, where the state of each pair is given by 2.6, the state of the entire system can be written as 2.7. Then, by taking $N \rightarrow \infty$, the total state of the system cannot remain pure, and therefore factorization of the Hilbert space will not be possible. In this case, studying this system through the Hilbert space becomes difficult. Thus, by adopting the algebraic perspective for the subsystems, we were able to study the entanglement structure more conveniently. We then showed that by considering the maximally entangled state, which occurs for $\theta = \frac{\pi}{4}$, the algebra corresponding to the subsystems in this problem is of type II, and we investigated the details of that system.

Now suppose that for this system $\theta \in (0, \frac{\pi}{4})$ and that the algebra corresponding to the right subsystem is denoted by $\mathcal{M}$. Also suppose that the initial state of the system is an

arbitrary state such as 3.23. Then, to ensure the existence of the modular structure, we first consider the problem at finite N [10]. In this case, as we know, the reduced density operators are

$$\rho_A(\Phi) = \rho_{1,A} \otimes \rho_{2,A} \otimes \cdots \otimes \rho_{N,A} \quad , \quad \rho_B(\Phi) = \rho_{1,B} \otimes \rho_{2,B} \otimes \cdots \otimes \rho_{N,B} \quad (4.27)$$

where $\rho_{i,A/B} = \begin{pmatrix} \cos^2 \theta & 0 & 0 & \sin^2 \theta \end{pmatrix}$. Therefore, the density operator is invertible, which implies that the modular structure exists for this system³. In this case, we consider $|\Phi\rangle$ to be a cyclic and separating vector with respect to the algebra $\mathcal{M}$ and obtain the modular operator:

$$\Delta_\Phi = \rho_A(\Phi) \otimes \rho_B^{-1}(\Phi) = \rho_{1,A} \otimes \rho_{1,B}^{-1} \otimes \cdots \otimes \rho_{N,A} \otimes \rho_{N,B}^{-1} \quad (4.28)$$

Now, to calculate $T(\mathcal{M})$, it is necessary to determine whether the above expression is an inner automorphism or not. For this purpose, since N is finite, let us examine the modular flow operator for one pair of this system:

$$\begin{aligned} \forall A \in \mathcal{M} \quad , \quad \sigma_s^\Phi(\mathcal{M}) &= \Delta_\Phi^{-is} A \Delta_\Phi^{+is} = [\rho_A(\Phi) \otimes \rho_B^{-1}(\Phi)]^{-is} A [\rho_A(\Phi) \otimes \rho_B^{-1}(\Phi)]^{+is} \\ &= [\rho_{1,A} \otimes \rho_{1,B}^{-1}]^{-is} A [\rho_{1,A} \otimes \rho_{1,B}^{-1}]^{+is} = U_s^\dagger A U_s \end{aligned} \quad (4.29)$$

which is an inner automorphism, and therefore $T(\mathcal{M}) = \mathbb{R}$. Consequently, as we know, the algebra corresponding to the subsystems is a type I von Neumann algebra. To calculate the second property, $S(\mathcal{M})$, we need to obtain the eigenvalues of the modular operator. As we know from 4.28, we have

$$\begin{aligned} \lambda(\rho_{i,A} \otimes \rho_{i,B}^{-1}) &= \lambda \left(\begin{pmatrix} \cos^4 \theta & 0 & 0 & 0 \\ 0 & \cos^2 \theta \sin^2 \theta & 0 & 0 \\ 0 & 0 & \sin^2 \theta \cos^2 \theta & 0 \\ 0 & 0 & 0 & \sin^4 \theta \end{pmatrix} \right) \\ &= \begin{pmatrix} \cos^4 \theta - \alpha & 0 & 0 & 0 \\ 0 & \cos^2 \theta \sin^2 \theta - \alpha & 0 & 0 \\ 0 & 0 & \sin^2 \theta \cos^2 \theta - \alpha & 0 \\ 0 & 0 & 0 & \sin^4 \theta - \alpha \end{pmatrix} \\ &= (\cos^4 \theta - \alpha)(\sin^4 \theta - \alpha)(\cos^2 \theta \sin^2 \theta - \alpha)^2 = 0 \end{aligned} \quad (4.30)$$

The eigenvalues are of the form $\alpha = 1, \tan^2 \theta, \tan^{-2} \theta$. Therefore,

$$S(\mathcal{M}) = \cap_\Phi \lambda(\Delta_\Phi) = \prod_{i=1}^N \lambda^{t_i} = 1 \quad , \quad \lambda = \tan^2 \theta \quad , \quad t_i = -1, 0, 1 \quad (4.31)$$

which indicates the existence of a type I or II algebra. Therefore, based on the properties of $T(\mathcal{M})$ and $S(\mathcal{M})$, we find that the algebra corresponding to the right subsystem in this problem is a type I von Neumann algebra. Naturally, because N is finite, this result is somewhat obvious. But if we assume $N \rightarrow \infty$, how do these invariant properties characterize the algebra corresponding to the system? For this purpose, if we consider the problem in

³This is obvious because in the finite- N limit, the algebra corresponding to the subsystem is of type I.

this limit, we know from the above expression that $\Delta^{is} * \Phi = I$, and therefore $\lambda^{is} = 1$. We can then state

$$T(\mathcal{M}) = s \mid \lambda^{is} = 1, s = \frac{2\pi n}{\log \lambda}, n \in \mathbb{Z} \subset \mathbb{R}_+ \quad (4.32)$$

which, considering $\lambda = \tan^2 \theta$, reduces to the same type II von Neumann algebra that we studied in the previous section when the maximally entangled state $\theta = \frac{\pi}{4}$ is considered.

Also, for calculating $S(\mathcal{M})$ in this limit, we have

$$S(\mathcal{M}) = \cap_{\Phi} \lambda(\Delta_{\Phi}) = \lim_{N \rightarrow \infty} \prod_{i=1}^N \lambda^{t_i} = 0 \cup \lambda^n \mid n \in \mathbb{Z} \quad (4.33)$$

Thus, we can conclude that in the limit $N \rightarrow \infty$, with $\theta \in (0, \frac{\pi}{4})$, the algebra corresponding to each of the right and left subsystems is a type III_{λ} von Neumann algebra.

9 For systems of spin-1 particles, investigate what type of algebra exists for a subsystem of this problem.

Therefore, in this section, we introduced one of the advantages of the existence of the modular structure and observed several examples of algebra classification. However, this structure and the Tomita-Takesaki theory founded upon it contain even more interesting material. In the next section, we discuss a concept called "the emergence of time" from the algebraic perspective, which may also be fruitful from a physical point of view.

4.4 Emergence of Time

As we stated, the existence of the modular structure has a profound effect on our perspective toward the algebraic literature. One of the features that this structure gives rise to is the "emergence of time." The existence of this feature in a type III algebra is particularly interesting, but in this section we attempt to address this issue for an arbitrary algebra by relying on the modular structure.

Suppose we have an algebra $\mathcal{M}$ with $|\Omega\rangle$ as a cyclic and separating vector with respect to $\mathcal{M}$. Therefore, according to the Tomita-Takesaki theory, the modular structure can be considered for it. We thus consider the modular operator $\Delta_{\mathcal{M}} = S_{\mathcal{M}}^{\dagger} S_{\mathcal{M}} = \exp(-K_{\mathcal{M}})$, where $S_{\mathcal{M}}$ is the Tomita operator. Now suppose there exists a subalgebra $\mathcal{N}$ of $\mathcal{M}$, $\mathcal{N} \subset \mathcal{M}$, for which the vector $|\Omega\rangle$ is also cyclic and separating. Therefore, a modular structure exists for the subalgebra $\mathcal{N}$. In this case, according to a theorem in [11], one can show that

$$\Delta_{\mathcal{N}} \geq \Delta_{\mathcal{M}} \quad (4.34)$$

where $\Delta_{\mathcal{N}} = \exp(-K_{\mathcal{N}})$. Accordingly,

$$\begin{aligned} \Delta_{\mathcal{M}} |\Omega\rangle &= \Delta_{\mathcal{N}} |\Omega\rangle = |\Omega\rangle \\ (\Delta_{\mathcal{M}} - \Delta_{\mathcal{N}}) |\Omega\rangle &= (K_{\mathcal{M}} - K_{\mathcal{N}}) |\Omega\rangle = 0 \end{aligned} \quad (4.35)$$

As a result, one can define an operator such as $G \equiv \frac{1}{2\pi}(K_{\mathcal{M}} - K_{\mathcal{N}})$,

$$G|\Omega\rangle = 0, \quad G \geq 0 \quad (4.36)$$

As is clear, G behaves like a Hamiltonian, and we can consider $\exp(iGs)$ as a flow for generating time, as usual. This phenomenon becomes relevant when there exists a subalgebra such as $\mathcal{N}$ for which $|\Omega\rangle$ is cyclic and separating, and in this case it gives rise to the emergence of time, s .

Statement

If $\mathcal{M}$ has a cyclic and separating vector such as $|\Omega\rangle$ and there also exists a subalgebra such as $\mathcal{N}$ for which $|\Omega\rangle$ remains cyclic and separating, then one can consider a Hamiltonian such as G satisfying $G|\Omega\rangle = 0$. In this case, the flow $\exp(iGs)$ generates the evolution of the system. Thus, the existence of a subalgebra gives rise to the emergence of time, s .

The emergence of time through a subalgebra gives rise to interesting properties. For this purpose, as we stated, the algebras $\mathcal{M}$ and $\mathcal{N}$ possess modular structures. Therefore, the modular conjugation operators $J_{\mathcal{M}}$ and $J_{\mathcal{N}}$ exist. We then define a new operator as follows [12, 13]:

$$V = J_{\mathcal{M}}J_{\mathcal{N}} \quad (4.37)$$

Now, as we know, $J_{\mathcal{M}}\mathcal{M}J_{\mathcal{M}} = \mathcal{M}'$ and $J_{\mathcal{N}}\mathcal{N}J_{\mathcal{N}} = \mathcal{N}'$. Therefore,

$$V^\dagger \mathcal{M} V = J_{\mathcal{N}} J_{\mathcal{M}} \mathcal{M} J_{\mathcal{M}} J_{\mathcal{N}} = J_{\mathcal{N}} \mathcal{M}' J_{\mathcal{N}} \equiv J_{\mathcal{N}} (\mathcal{N} \subset \mathcal{M})' J_{\mathcal{N}} \subset \mathcal{N} \quad (4.38)$$

In this case, using the operator V , we can generate various subalgebras of $\mathcal{N}$:

$$\begin{aligned} \hat{\mathcal{N}}_1 &\equiv V^\dagger \mathcal{M} V \subset \mathcal{N} \\ \hat{\mathcal{N}}_2 &\equiv V^\dagger \mathcal{N} V \subset \hat{\mathcal{N}}_1 \\ \hat{\mathcal{N}}_3 &\equiv V^\dagger \hat{\mathcal{N}}_1 V \subset \hat{\mathcal{N}} * 2 \\ &\vdots \end{aligned} \quad (4.39)$$

Therefore,

Statement

By repeatedly applying $V^\dagger \cdot V$ to the algebra $\mathcal{M}$, we can generate subalgebras of the form $\hat{\mathcal{N}}_{2n-1}$, where $n = 1, 2, \dots$. Similarly, by applying $V^\dagger \cdot V$ to the subalgebra $\mathcal{N}$, we can construct the subalgebras $\hat{\mathcal{N}}_{2n}$, with $n = 1, 2, \dots$. These subalgebras have the following property:

$$\hat{\mathcal{N}}_a \subset \hat{\mathcal{N}}_b, \quad a < b \in \mathbb{N} \quad (4.40)$$

Thus, due to the existence of a subalgebra, a specific time can be defined for each of them.

One can also define a new operator in terms of the modular operators of the two algebras

$\mathcal{M}$ and $\mathcal{N}$:

$$D(t) \equiv \Delta_{\mathcal{M}}^{-it} \Delta_{\mathcal{N}}^{+it} \quad (4.41)$$

For very small t , we have

$$\begin{aligned} D(t) &= \Delta_{\mathcal{M}}^{-it} \Delta_{\mathcal{N}}^{+it} = \exp[-it(K_{\mathcal{N}} - K_{\mathcal{M}})] \simeq 1 - it(K_{\mathcal{N}} - K_{\mathcal{M}}) \\ &= 1 + 2\pi itG \end{aligned} \quad (4.42)$$

On the other hand, because $\mathcal{N} \subset \mathcal{M}$ and both algebras possess modular structures, modular flow operators also exist for them. As we know, this operator implies

$$\forall A \in \mathcal{M} \mid \Delta_{\mathcal{M}}^{-it} A \Delta_{\mathcal{M}}^{+it} \in \mathcal{M} \quad (4.43)$$

Then, a subalgebra can be defined using the operator $D(t)$:

$$\mathcal{N}_t \equiv D(t) \mathcal{N} D^\dagger(t) = \Delta_{\mathcal{M}}^{-it} \Delta_{\mathcal{N}}^{+it} \mathcal{N} \Delta_{\mathcal{N}}^{-it} \Delta_{\mathcal{M}}^{+it} = \Delta_{\mathcal{M}}^{-it} \mathcal{N} \Delta_{\mathcal{M}}^{+it} \subseteq \mathcal{M} \quad (4.44)$$

Therefore,

Statement

If we have $\mathcal{N} \subset \mathcal{M}$, where both algebras possess a common cyclic and separating vector such as $|\Omega\rangle$, then by defining the operator

$$D(t) = \Delta_{\mathcal{M}}^{-it} \Delta_{\mathcal{N}}^{+it}$$

one can generate multiple subalgebras of the form

$$\mathcal{N}_t \equiv D(t) \mathcal{N} D^\dagger(t) \subset \mathcal{M} \quad (4.45)$$

In this way, we examined various possibilities for generating subalgebras and, consequently, the emergence of time. Finally, we can examine the properties of the operator $D(t)$, which will be used in the next chapter.

1. The operator $D(t)$ is continuous and unitary in t , and $D(t=0) = \mathbf{I}$. As we know from 4.41,

$$D^\dagger(t) D(t) = \Delta_{\mathcal{N}}^{-it} \Delta_{\mathcal{M}}^{+it} \Delta_{\mathcal{M}}^{-it} \Delta_{\mathcal{N}}^{+it} = \mathbf{I} \quad (4.46)$$

and at $t=0$,

$$D(t=0) = \Delta_{\mathcal{M}}^{-it} \Delta_{\mathcal{N}}^{+it} |_{t=0} = \mathbf{I} \quad (4.47)$$

2. For every $t \in \mathbb{R}$, we have $D(t) |\Omega\rangle = |\Omega\rangle$.

As we know, the cyclic and separating vector $|\Omega\rangle$ is common to both algebras $\mathcal{M}$ and $\mathcal{N}$. Therefore,

$$\Delta_{\mathcal{M}}^{-it} |\Omega\rangle = \Delta_{\mathcal{N}}^{-it} |\Omega\rangle = |\Omega\rangle \quad (4.48)$$

Thus,

$$D(t) |\Omega\rangle = \Delta_{\mathcal{M}}^{-it} \Delta_{\mathcal{N}}^{+it} |\Omega\rangle = |\Omega\rangle, \quad \forall t \in \mathbb{R} \quad (4.49)$$

3. The operator $D(t)$ satisfies $D(t+s) = \Delta_{\mathcal{M}}^{-it} D(s) \Delta_{\mathcal{M}}^{+it} D(t)$. To prove the above expression, we start from the right-hand side and obtain

$$\begin{aligned} \Delta_{\mathcal{M}}^{-it} D(s) \Delta_{\mathcal{M}}^{+it} D(t) &= \Delta_{\mathcal{M}}^{-it} \Delta_{\mathcal{M}}^{-is} \Delta_{\mathcal{N}}^{+is} \Delta_{\mathcal{M}}^{+it} \Delta_{\mathcal{M}}^{-it} \Delta_{\mathcal{N}}^{+it} = \Delta_{\mathcal{M}}^{-it} \Delta_{\mathcal{M}}^{-is} \Delta_{\mathcal{N}}^{+it} \Delta_{\mathcal{N}}^{+is} \\ &= \Delta_{\mathcal{M}}^{-i(t+s)} \Delta_{\mathcal{N}}^{+i(t+s)} = D(t+s) \end{aligned} \quad (4.50)$$

We can now state a proposition that we leave as an exercise:

10 Show that the operator $D(t)$ is generated by the operator G if the subalgebra $\mathcal{N}$ satisfies the following condition:

$$\mathcal{N}_t \equiv \Delta_{\mathcal{M}}^{-it} \mathcal{N} \Delta_{\mathcal{M}}^{+it} = D(t) \mathcal{N} D^\dagger(t) \subset \mathcal{M}, \quad \forall t \leq 0 \quad (4.51)$$

This statement says that the subalgebras $\mathcal{N}$ generated under the modular flow operator remain invariant in the half-plane $t \leq 0$. This structure is called half-sided modular inclusion. This statement leads to three very interesting properties [14, 15, 16, 17]:

Statement

1. The operators $K_{\mathcal{M}}$ and $K_{\mathcal{N}}$ satisfy the algebra

$$\begin{aligned} [K_{\mathcal{M}}, K_{\mathcal{N}}] &= -2\pi i (K_{\mathcal{M}} - K_{\mathcal{N}}) = -4\pi^2 i G \\ J_{\mathcal{M}} \exp(iGs) J_{\mathcal{M}} &= \exp(-iGs), \quad J_{\mathcal{N}} \exp(iGs) J_{\mathcal{N}} = \exp(-iGs) \end{aligned} \quad (4.52)$$

2. The stated proposition actually causes the generator of the algebra $\mathcal{M}$ to generate itself within $\mathcal{M}$ in the half-plane of the emergent time s :

$$\exp(iGs) \mathcal{M} \exp(-iGs) \subset \mathcal{M}, \quad \forall s \leq 0 \quad (4.53)$$

In this case, we have

$$\mathcal{N} = \exp(-iG) \mathcal{M} \exp(iG) \quad (4.54)$$

3. $\mathcal{M}$ must be a type III₁ von Neumann algebra.

The interesting point is the third property, which states that this proposition and its associated properties hold in a type III₁ algebra. Thus, as we know, this encourages us to study quantum field theories, which possess this type of algebra. In the following chapters, we will examine this theory from the foundations, as it constitutes the main framework connecting this mathematical perspective to physics. We will also continue part of the discussion from this section in a portion of the next chapter in order to benefit from the physical perspective. At the end of this section, we need to introduce a lemma that will be highly influential in the following chapters:

Statement

Suppose we have two von Neumann algebras $\mathcal{M}$ and $\mathcal{N}$ with a common cyclic and separating vector $|\Psi\rangle$ satisfying $\mathcal{N} \subset \mathcal{M}$. Then the modular structure exists. If we consider the modular flow operator for the algebra $\mathcal{M}$, then by applying it to the subalgebra $\mathcal{N}$, the entire algebra $\mathcal{M}$ can be generated.

As can be seen from this lemma, in problems such as the subregion-subalgebra duality discussed in Chapter 7, it can be used to reconstruct the regions of interest to us.

4.5 Relative Entropy

We now turn to a definition that is highly useful not only in algebraic mathematics but also in physics and, in particular, in high-energy physics, known as relative entropy. This discussion begins with the observation that the Tomita-Takesaki theory can be generalized to two cyclic and separating vectors such as $|\Omega\rangle$ and $|\Psi\rangle$ with respect to the algebra $\mathcal{M}$. From the physical perspective, this is analogous to what exists between two quantum states in quantum information theory. For this purpose, by considering the generalized theory, we have

Statement

The Tomita-Takesaki theory can be generalized to two cyclic and separating vectors with respect to the algebra $\mathcal{M}$, which we denote by $|\Omega\rangle$ and $|\Psi\rangle$. In this case, we denote the generalized Tomita operator between these two vectors by $S_{\Psi\Omega}$, which has the following property:

$$\forall A \in \mathcal{M}, \forall A' \in \mathcal{M}', S_{\Psi\Omega} A |\Omega\rangle = A^\dagger |\Psi\rangle, S_{\Psi\Omega}^\dagger A' |\Omega\rangle = A'^\dagger |\Psi\rangle \quad (4.55)$$

Also, because this theory exists, the modular structure corresponding to this theory is present. Therefore, the modular operator and modular conjugation operator are defined as

$$S_{\Psi\Omega} = J_{\Psi\Omega} \Delta_{\Psi\Omega}^{\frac{1}{2}} \quad (4.56)$$

$$\Delta_{\Psi\Omega} = S_{\Psi\Omega}^\dagger S_{\Psi\Omega} \quad (4.57)$$

$\Delta_{\Psi\Omega}$ is called the relative modular operator, and $J_{\Psi\Omega}$ is called the relative modular conjugation operator.

On the other hand, by defining the generalized Tomita operator, we have the following property:

$$\begin{aligned} S_{\Psi\Omega} S_{\Omega\Psi} A |\Psi\rangle &= S_{\Psi\Omega} A^\dagger |\Omega\rangle = A |\Psi\rangle \\ S_{\Psi\Omega} S_{\Omega\Psi} &= I \end{aligned} \quad (4.58)$$

In this case, using the relative modular operator and relative modular conjugation operator,

one can show

$$\begin{aligned} S_{\Psi\Omega} S_{\Omega\Psi} &= J_{\Psi\Omega} \Delta_{\Psi\Omega}^{\frac{1}{2}} J_{\Omega\Psi} \Delta_{\Omega\Psi}^{\frac{1}{2}} = I \\ J_{\Psi\Omega} \Delta_{\Psi\Omega}^{\frac{1}{2}} J_{\Omega\Psi} &= \Delta_{\Omega\Psi}^{-\frac{1}{2}} \end{aligned} \quad (4.59)$$

Also, by squaring the above expression, we obtain

$$(J_{\Psi\Omega} \Delta_{\Psi\Omega}^{\frac{1}{2}} J_{\Omega\Psi})^2 = J_{\Psi\Omega} \Delta_{\Psi\Omega}^{\frac{1}{2}} J_{\Omega\Psi} J_{\Psi\Omega} \Delta_{\Psi\Omega}^{\frac{1}{2}} J_{\Omega\Psi} = J_{\Psi\Omega} \Delta_{\Psi\Omega} J_{\Omega\Psi} = \Delta_{\Omega\Psi}^{-1} \quad (4.60)$$

One of the very interesting properties of the existence of the relative modular structure is its relation to the KMS condition. To demonstrate this condition in this structure, first, as before, suppose that $|\Omega\rangle$ and $|\Psi\rangle$ are cyclic and separating vectors with respect to the algebra $\mathcal{M}$. Then, using 4.55, we have

$$\begin{aligned} \forall A, B \in \mathcal{M}, \langle \Psi | AB | \Psi \rangle &= \langle A^\dagger \Psi | B \Psi \rangle = \langle S_{\Psi\Omega} A \Omega | S_{\Psi\Omega} B^\dagger \Omega \rangle \\ &= \langle J_{\Psi\Omega} \Delta_{\Psi\Omega}^{\frac{1}{2}} A \Omega | J_{\Psi\Omega} \Delta_{\Psi\Omega}^{\frac{1}{2}} B^\dagger \Omega \rangle = \langle A \Omega | J_{\Psi\Omega} \Delta_{\Psi\Omega}^{\frac{1}{2}} J_{\Psi\Omega} \Delta_{\Psi\Omega}^{\frac{1}{2}} | B^\dagger \Omega \rangle \\ &= \langle A \Omega | \Delta_{\Psi\Omega} | B^\dagger \Omega \rangle = \langle \Omega | B \Delta_{\Psi\Omega} A | \Omega \rangle \end{aligned} \quad (4.61)$$

Therefore, in this relative modular structure, the KMS condition still exists:

$$\forall A, B \in \mathcal{M}, \langle \Psi | AB | \Psi \rangle = \langle \Omega | B \Delta_{\Psi\Omega} A | \Omega \rangle \quad (4.62)$$

Consequently, in the algebraic literature, if we have access to a subsystem with corresponding algebra $\mathcal{M}$ that possesses cyclic and separating vectors $|\Omega\rangle$ and $|\Psi\rangle$ with respect to this algebra, then one can establish a relation between the correlation functions of these two vectors. This result is highly useful in physics, since correlation functions play a significant role in calculations as well as in our approach to the problem under consideration. The strength of this discussion becomes apparent in the following definition.

Statement

Suppose we have a subsystem with algebra $\mathcal{M}$ possessing cyclic and separating vectors $|\Omega\rangle$ and $|\Psi\rangle$. The relative entropy for this algebra is defined as

$$S_{\mathcal{M}}(\Psi || \Omega) \equiv - \langle \Psi | \log \Delta_{\Omega\Psi} | \Psi \rangle \quad (4.63)$$

This definition is very useful and consistent with what we defined for entropy of an algebra. In other words, suppose $\mathcal{M}$ is a type II von Neumann algebra. As we stated previously in 3.28, the entropy for this algebra is given by the difference between an arbitrary state and the maximally entangled state. In this way, this definition agrees with the type II algebra and, trivially, with the type I algebra. The positive aspect of this definition concerns type III algebras. As we know, most physical systems possess this type of algebra, but a property of this algebra prevents the definition of the map tr and consequently prevents the definition of entropy. The definition of relative entropy allows us to define a quantity analogous to entropy for this algebra as well. This enables us to use this definition in physical systems and to study these systems and theories such as quantum field theory more easily and more effectively.

It should be noted that relative entropy is a non-negative quantity. To demonstrate this property, we have

$$-\langle \Psi | \log \Delta_{\Omega\Psi} | \Psi \rangle \geq \langle \Psi | 1 - \Delta_{\Omega\Psi} | \Psi \rangle = \langle \Psi | \Psi \rangle - \langle \Psi | \Delta_{\Omega\Psi} | \Psi \rangle \quad (4.64)$$

Then, using the KMS condition, equation 4.62, we have

$$S_{\mathcal{M}}(\Psi || \Omega) = -\langle \Psi | \log \Delta_{\Omega\Psi} | \Psi \rangle \geq \langle \Psi | \Psi \rangle - \langle \Psi | \Delta_{\Omega\Psi} | \Psi \rangle = \langle \Psi | \Psi \rangle - \langle \Omega | \Omega \rangle = 0 \quad (4.65)$$

Therefore, relative entropy is always a non-negative quantity.

Up to this point, we have presented the mathematical preliminaries of the algebraic approach. In the next chapter, we gradually move toward the physical context and examine the implications of this literature in physics. As we stated earlier, our focus is on type III von Neumann algebras, which occur in quantum field theory as well as gauge field theories. For this purpose, we will first turn to quantum field theory and, after employing the algebraic mathematics, investigate the duality between gravity and gauge field theories. It should be noted that one of the important parts of this discussion, known as the crossed product algebra, will be introduced in the next chapter because it forms the basis and the main strength of the construction for type III algebras. For this reason, we have decided to introduce it in the next chapter.

Chapter 5

Quantum Field Theory

As stated in the previous chapter, we were able to employ the modular structure for all von Neumann algebras without the restriction of the existence of reduced density operators, simply by considering a cyclic and separating vector with respect to the algebra under study. This allows us to discuss the entanglement structure and quantum information of even subsystems whose von Neumann algebra is of type III. In this way, we investigate these systems using Tomita–Takesaki theory.

5.1 Algebra of Local Operators

Quantum field theory is an important example of what we need to study from the algebraic point of view, and we also wish to observe the effects of the existence of Tomita–Takesaki theory in advancing our understanding of this physical theory. First, suppose that we denote the Hilbert space of the entire spacetime by $\mathcal{H}$ and the vacuum state in this theory by $|\Omega\rangle$. In addition, suppose that we have the spacetime $\mathbb{R}^{1,d-1}$ on which the quantum field theory is defined. Now consider a time slice of this spacetime and divide it into two spatial regions R and L , with their boundary denoted by ∂ . Then the algebras $\mathcal{M}_R$ and $\mathcal{M}_L$ are, respectively, the algebras of local operators in the subsystems R and L . In this case, due to the existence of causality in this theory, we can write

$$\forall A \in \mathcal{M}_R, \forall B \in \mathcal{M}_L, [A, B] = 0 \quad (5.1)$$

which is another expression of the existence of the commutant of the algebra $\mathcal{M}_R$. Thus, $\mathcal{M}'_R = \mathcal{M}_L$.

Therefore, as we stated, on a time slice we were able to study the algebra corresponding to each region of the space in which the quantum field theory is defined. However, it should be noted that this theory also contains the time parameter. We are now interested in how the local operators contained in $\mathcal{M}_R$ change with time in quantum field theory, and whether they remain in this algebra. Accordingly, as time evolves, the Rindler regions for R and L , shown in Figure 5.1, can be defined as follows,

$$R_{\text{Rindler}} \equiv \hat{R} = \{(t, \vec{x}) \mid x > |t|\} , \quad L_{\text{Rindler}} \equiv \hat{L} = \{(t, \vec{x}) \mid -x > |t|\} \quad (5.2)$$

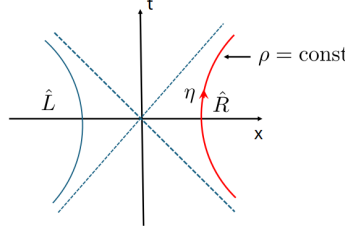


Figure 5.1: The Rindler regions are considered in $1 + 1$ -dimensional Minkowski spacetime, where the metric can be written as $ds^2 = -dt^2 + dx^2 = -\rho^2 d\eta^2 + d\rho^2$. The worldline of a Rindler observer with constant ρ is given by a hyperbola parametrized by Rindler time η . Therefore, the proper time of a Rindler observer is given by $d\tau = \rho, d\eta$. A boost transformation induces a translation in the coordinate η and maps the right Rindler region $\hat{R}$ to itself.

where K denotes the Hamiltonian. Then, for every operator in $\mathcal{M}_R$ and its commutant, using [18], we have

$$\begin{aligned} \forall A \in \mathcal{M}_R, \quad A(s) &= \exp(+iKs)A(0)\exp(-iKs) \in \mathcal{M}_R \\ \forall A' \in \mathcal{M}_L, \quad A'(s) &= \exp(+iKs)A'(0)\exp(-iKs) \in \mathcal{M}_L \end{aligned} \quad (5.3)$$

An important point to mention is the closure of the algebras corresponding to the right and left regions. In other words, due to causality, we expect that if at $t = 0$ an operator belongs to the algebra $\mathcal{M}_R(\mathcal{M}_L)$, it should remain in this algebra as time evolves; otherwise, in view of the violation of Eq. 5.1, we would also obtain a contradiction with the principle of causality. Therefore,

$$\mathcal{M}_R \cong \mathcal{M}_{\hat{R}}, \quad \mathcal{M}_L \cong \mathcal{M}_{\hat{L}} \quad (5.4)$$

In this case, considering a Hilbert space for the entire system, denoted by $\mathcal{H}$, and using weak convergence, we can state that the algebra $\mathcal{M}_R$ is a von Neumann algebra. At this point, it is necessary to introduce the Reeh–Schlieder theorem

Statement

This theorem states that in a quantum field theory, by applying all local operators in an open region such as O to the vacuum state, $|\Omega\rangle$, one can reconstruct a Hilbert space.

This theorem leads us to a very important point:

Statement

Suppose that we have a quantum field theory on a spacetime possessing a vacuum state $|\Omega\rangle$, and that we have access only to a region such as R and denote the algebra of this region by $\mathcal{M}_R$. Then

- According to the Reeh–Schlieder theorem, by applying all operators inside the algebra $\mathcal{M}_R$ to the vacuum state $|\Omega\rangle$, one can construct a Hilbert space for that subsystem, in which case $|\Omega\rangle$ is a cyclic vector.

- In view of the principle of causality, by applying all operators inside the algebra $\mathcal{M}_R$ to $|\Omega\rangle$, one can construct a Hilbert space. Therefore, $|\Omega\rangle$ is still a cyclic vector with respect to the commutant of $\mathcal{M}_R$ and a separating vector with respect to the algebra $\mathcal{M}_R$.

In this case, we can conclude that the vacuum state $|\Omega\rangle$ is cyclic and separating with respect to the algebra $\mathcal{M}_R$, and therefore the modular structure can be considered for it.

Now, if we consider the region R of the entire space, then, due to the existence of the modular structure, we can claim that an operator such as K_Ω exists such that

$$\Delta_\Omega |\Omega\rangle = |\Omega\rangle \quad , \quad K_\Omega \equiv -\log \Delta_\Omega \quad (5.5)$$

where Δ_Ω is called the modular operator. Also, as we have realized, due to causality, the algebra $\mathcal{M}_R$ is closed under time evolution. In this case, we have

$$\forall A \in \mathcal{M}_R \quad , \quad A(s) = \exp(+iK_\Omega s)A(0)\exp(-iK_\Omega s) \in \mathcal{M}_R \quad (5.6)$$

Then the modular flow can be defined as above, $\sigma_s(A) \equiv A(s)$. Also, using Tomita–Takesaki theory, one can show that the modular conjugation operator J_Ω also exists, with an interesting property that is left as an exercise.

11 Show that the modular conjugation operator J_Ω can be written as

$$J_\Omega = CRT \quad (5.7)$$

where C is the charge conjugation operator, R is the reflection operator, and T is the time-reversal operator, all of which have important meanings in quantum field theory.

As stated previously, K_Ω generates the evolution in the same way as the Hamiltonian, and is therefore called the modular Hamiltonian. Accordingly, if we consider this in this way, we can state that the energy eigenvalues for this subsystem lie in the interval $(-\infty, +\infty)$. At this point, a question should be raised:

Question

Considering an arbitrary region of the manifold on which the quantum field theory is defined, what type of algebra is the algebra of operators corresponding to this region?

We have already stated the answer to this question, but now we want to answer it using the powerful modular tools and the classification of algebras presented in the previous chapter. As we know, given the allowed range of energy eigenvalues, the modular flow operator, and the intersection over all cyclic and separating vectors with respect to the algebra $\mathcal{M}_R$, it is a subset of the set of positive real numbers. More precisely, one can show that

$$S(\mathcal{M}_R) = \cap_\Omega \sigma(\Delta_\Omega) \subset \mathbb{R}_+ \quad (5.8)$$

Therefore, according to Eq. 4.26, we conclude

Statement

The algebra of every arbitrary region of the manifold on which the quantum field theory is defined is a type III₁ von Neumann algebra.

Another point is that the existence of the principle of causality in quantum field theory leads to a type III₁ algebra for every region of the manifold containing the theory. Accordingly, in nonrelativistic quantum field theory¹, local operators do not necessarily have to obey the principle of causality, and therefore the algebra of an arbitrary region is no longer a type III₁ von Neumann algebra. On the other hand, so far we have only discussed

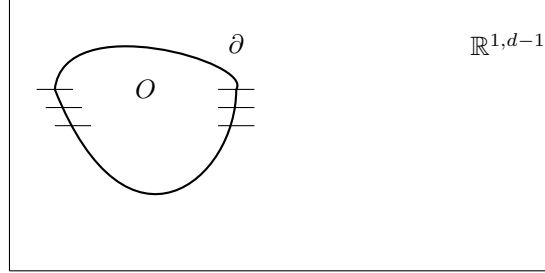


Figure 5.2: Quantum field theory is defined on the manifold $\mathbb{R}^{1,d-1}$. An arbitrary region O on this manifold, with boundary ∂ , contains short-range entanglement near the boundary, represented by straight lines.

the algebra of local operators belonging to a region and ultimately found that it is a type III₁ algebra. In other words, suppose that on the manifold $\mathbb{R}^{1,d-1}$, where the quantum field theory is defined, we consider an arbitrary region on a time slice and denote it by O . Then, according to Figure 5.2, one can observe short-range entanglement between degrees of freedom near the boundary, which causes the Hilbert space to be non-factorizable and also leads to the existence of a type III₁ algebra. Therefore, one can give the following interpretation of what happens in the algebraic literature, which may be related to a part of physics referred to as quantum gravity:

Statement

In quantum field theory (where the principle of causality is assumed by default), considering a boundary causes or gives rise to a type III₁ von Neumann algebra.

This interpretation is interesting in a sense, because without considering a boundary we cannot obtain any information about the algebra of the theory, and as soon as a boundary is added, the algebra belonging to the separated region emerges. Consequently, the role of the boundary in physics is very interesting and important, and therefore we are interested in studying the properties associated with the boundary and the emergence of different algebras.

5.2 Boundary and Emergence of the Algebra

To introduce this section, we prefer to begin by discussing our starting point regarding the boundary in quantum field theory. Suppose that we have a $(D+1)$ -dimensional manifold M

¹We refer to relativistic quantum field theory simply as quantum field theory.

on which a quantum field theory is defined. Naturally, there are very many, and uncountably many, operators on this manifold. However, up to this point we cannot make a statement about the algebra of operators unless, using a boundary, we restrict our attention to a region. In other words, suppose that we place a D -dimensional manifold N around the entire manifold M . Then, by considering the regions near the boundary with the manifold M lying on one side, the situation acts like a Rindler region. Therefore, according to the Reeh–Schlieder theorem, the modular structure can be considered for it.

Accordingly, by considering only the modular Hamiltonian, one can state that the energy eigenvalues lie in the interval $(-\infty, +\infty)$, and in this case, by classifying the algebras, one can regard the algebra of local operators on the manifold M as a type III_1 algebra. Also, by considering any other D -dimensional manifold, different regions in M can be studied, for which the algebra of local operators belonging to that region is of type III_1 . Therefore,

Statement

Considering a boundary causes a type III_1 algebra to emerge.

But does this interpretation physically express holography? In quantum gravity, and especially in the duality between gravity and gauge theories such as AdS/CFT, we can precisely envision the algebraic structure, which we will discuss in later chapters. Also, in part of the research literature, the emergence of spacetime geometry through a boundary is a very important question. Therefore, this interpretation (in my opinion) does not provide us with a new finding from the physical point of view, but one can say that the existence of this feature in the algebraic literature allows us to observe similarities with what we are seeking in physics in this approach. In other words, as we said, in physics the role of the boundary as a tool or phenomenon that shapes spacetime is highly intriguing. Now, from the algebraic point of view, where we do not require much information about the physics, we still observe that the role of the boundary causes the algebra associated with the region under consideration (or, in physical language, the desired volume) to emerge.

We now wish to take one step further and pose the following problem:

Question

Suppose that M is a $(D + 1)$ -dimensional manifold on which a quantum field theory exists. Considering a manifold N of dimension D , the algebra of local operators belonging to every region with boundary N is a type III_1 algebra. At this point, can we ask what type of algebra the local operators belonging to the boundary manifold N form?

As we stated, in the general case there is no condition imposed on the algebra of local operators on the boundary manifold, and it depends only on the theory defined on it. In other words, suppose that an ordinary quantum theory is defined on the boundary manifold N . Then the algebra of local operators of the boundary theory is a type I algebra, while the algebra belonging to the interior region is a III_1 algebra. Therefore, there is essentially no contradiction or prerequisite. However, if we enter the discussion of holography, this question cannot be simply ignored. As we explain below, one of the conditions for the existence of a duality between two theories is the equality of their correlation functions. On the other hand, these functions depend on the operators existing in the two theories. Therefore, the sets of operators of the two theories affect the correlation functions and consequently the

principles of holography. For example, in the AdS/CFT duality, the correlation functions of the two theories must essentially be equal, and in this case the algebras of operators corresponding to the two theories must be identical.

Therefore, the existence of a boundary plays a significant role in defining the algebra, especially in quantum field theory. Also, the non-factorizability of the Hilbert space in this theory arises from the existence of this algebra, which represents extensive entanglement around regions close to the boundary. However, using the split property, the role of this near-boundary entanglement can in a sense be modified.

5.3 Split Property

Suppose that on the manifold M , by considering a time slice, we divide the spatial regions into two parts R and L . Also, each region has its own Hilbert space; thus, for region R we consider the Hilbert space $\mathcal{H}_R$ and for region L the Hilbert space $\mathcal{H}_L$. As stated in the previous sections, because the algebra of local operators on the manifold M is of type III₁, the total Hilbert space is not factorizable:

$$\mathcal{H} \neq \mathcal{H}_R \otimes \mathcal{H}_L \quad (5.9)$$

and, in other words, there is entanglement between the two regions. We also denote the set of all local operators for each region by $\mathcal{M}_R$ and $\mathcal{M}_L$.

Now suppose that a gap ϵ , as shown in Figure 5.3, lies between the two regions. In this case the total Hilbert space becomes factorizable, because the entanglement in the regions near the boundary is removed, and we have

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \quad (5.10)$$

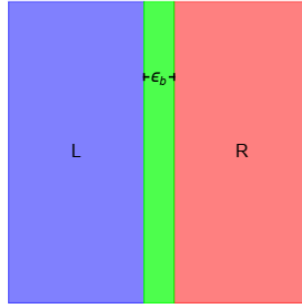


Figure 5.3: The regions R and L on a spatial slice are slightly separated from each other. We denote the green-colored region by I_ϵ . $R_\epsilon = R \cup I_\epsilon$ denotes the union of R and the green region, while $L_\epsilon = L \cup I_\epsilon$ denotes the union of L and the green region.

In this case, the algebras of local operators belonging to the new regions, denoted by the

indices 1 and 2, can be written as $\mathcal{B}(\mathcal{H}_1)$ and $\mathcal{B}(\mathcal{H}_2)$. Therefore,

$$\begin{aligned}\mathcal{M}_R &\subset \mathcal{B}(\mathcal{H}_1) \otimes \mathcal{I}_{\mathcal{H}_2} \subset \mathcal{M}'_L \equiv \mathcal{M}_{R_\epsilon} \\ \mathcal{M}_L &\subset \mathcal{I}_{\mathcal{H}_1} \otimes \mathcal{B}(\mathcal{H}_2) \subset \mathcal{M}'_R \equiv \mathcal{M}_{L_\epsilon}\end{aligned}\tag{5.11}$$

where $R_\epsilon \equiv R \cup I_\epsilon$ and $L_\epsilon \equiv L \cup I_\epsilon$. Now, as we observe in Eq. 5.11, the type I algebras, namely $\mathcal{B}(\mathcal{H}_1) \otimes \mathcal{I}_{\mathcal{H}_2}$ and $\mathcal{I}_{\mathcal{H}_1} \otimes \mathcal{B}(\mathcal{H}_2)$, lie between the type III₁ algebras, and in a sense one can make type I algebras emerge by using a gap. Consequently, we can state,

Statement

In quantum field theory, every region separated by a boundary has a type III₁ von Neumann algebra, representing the entanglement around the boundary region. Now, if this boundary is regarded as a gap with a specified finite thickness such as ϵ , then a type I algebra can be produced, and consequently the entanglement is removed.

Therefore, the existence of the split property [19, 20, 21, 22, 23] causes a type I von Neumann algebra to emerge. More generally, one can show

Statement

If we have two regions in the form $O_2 \subset O_1$, then by considering a thickness for the boundary of these two regions, one can produce a type I algebra with factor N ,

$$\mathcal{M}_{O_1} \subset N \subset \mathcal{M}_{O_2}\tag{5.12}$$

On the other hand, the existence of the split property gives rise to other interesting features, one of the most important of which is Strong Local Preparability [24]. To introduce this property, we return to the previous example. As we know, the algebra of local operators on the manifold M can be considered as $\mathcal{M}_R \vee \mathcal{M}_L$, and accordingly we can define the map $\omega : \mathcal{M}_R \vee \mathcal{M}_L \rightarrow \mathbb{C}$ as follows,

$$\begin{aligned}\omega(\alpha A + \beta B) &= \alpha\omega(A) + \beta\omega(B) \ ; \ \forall \alpha, \beta \in \mathbb{C} \ , \ \forall A, B \in \mathcal{M}_R \vee \mathcal{M}_L \\ \omega(A^\dagger) &= (\omega(A))^* \ ; \ \forall A \in \mathcal{M}_R \vee \mathcal{M}_L\end{aligned}\tag{5.13}$$

Then, due to the existence of the type III₁ algebra and the entanglement between regions R and L , we have

$$\omega(AB) \neq \omega(A)\omega(B) \ ; \ \forall A \in \mathcal{M}_R \ , \ \forall B \in \mathcal{M}_L\tag{5.14}$$

Now, if we consider a split with thickness ϵ , then the Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$ can be represented as $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$. Also, due to the existence of the type I algebra, according to Eq. 5.11, one can show that a vector such as $|\zeta\rangle$ exists in the Hilbert space $\mathcal{H}_1$ such that

$$\forall A \in \mathcal{M}_R \subset \mathcal{B}(\mathcal{H}_1) \otimes \mathcal{I}_{\mathcal{H}_2} \mid \phi(A) = \langle \zeta | A | \zeta \rangle\tag{5.15}$$

One can also define the projection operator corresponding to this vector as $P_\zeta = |\zeta\rangle \langle \zeta| \otimes \mathcal{I}_{\mathcal{H}_2} \subset \mathcal{B}(\mathcal{H}_1) \otimes \mathcal{I}_{\mathcal{H}_2} \subset \mathcal{M}_{R_\epsilon}$. Then, since $\mathcal{M}_{R_\epsilon}$ is a type III₁ algebra and according to the theorems of Chapter 2, all projection operators are equivalent to one another and are also

all equivalent to the largest projection operator, I . Accordingly, there exists an operator such as W in the algebra $\mathcal{M}_{R_\epsilon}$ such that

$$\exists W \in \mathcal{M}_{R_\epsilon}, \forall |\zeta\rangle \in \mathcal{H}_1 \mid P_\zeta = W^\dagger W, I = WW^\dagger \quad (5.16)$$

Accordingly, we define the map associated with the operator W as $\omega_W : \mathcal{M}_L \rightarrow \mathbb{C}$, which has the following property,

$$\forall B \in \mathcal{M}_L \mid \omega_W(B) = \omega(W^\dagger BW) = \omega(BWW^\dagger) = \omega(B) \quad (5.17)$$

Also, from Eq. 5.15, we have

$$\forall A \in \mathcal{M}_R \mid P_\zeta A P_\zeta = (|\zeta\rangle \langle \zeta| \otimes I_{\mathcal{H}_2}) A (|\zeta\rangle \langle \zeta| \otimes I_{\mathcal{H}_2}) = \phi(A) P_\zeta \quad (5.18)$$

and, using the operator W , the above expression can be written as

$$\begin{aligned} P_\zeta A P_\zeta &= W^\dagger W A W^\dagger W = \phi(A) W^\dagger W \\ W A W^\dagger &= \phi(A) \end{aligned} \quad (5.19)$$

Consequently, using Eqs. 5.17 and 5.19, we obtain

$$\forall A \in \mathcal{M}_R, \forall B \in \mathcal{M}_L \mid \omega_W(AB) = \omega(W^\dagger A W B) = \omega(\phi(A) B) = \phi(A) \omega(B) \quad (5.20)$$

In this case, by considering a split between regions that were entangled, one can define a state in which there is not only no entanglement or correlation between the two regions, but the defined state depends only on one region. In other words, one can state,

Statement

Physical states can be generated solely through local operators, without the need to modify the external regions. This statement expresses the concept of locality as well as causality.

This statement is very similar to the Connes–Stromer theorem²,

Statement

If $\mathcal{M}$ is a type III₁ algebra, then for every $\epsilon > 0$, there exist states ϕ and ω such that for a unitary operator $W \in \mathcal{M}$ we have:

$$\|\phi - \omega_W\| < \epsilon \quad (5.21)$$

5.4 Haag Duality and Additivity

We remain interested in quantum field theory and wish to obtain a better and deeper understanding of it from the algebraic point of view. As we have realized, every arbitrary region of the manifold on which this theory is defined possesses a type III₁ algebra. In the mathematical literature, the algebra belonging to an arbitrary region such as O is denoted

²Connes-Stromer Theorem

by $\mathcal{M}(O)$. Also, by considering all collections or families of all algebras of different regions, $\{\mathcal{M}(O)\}$, we can obtain a complete description of the theory. Accordingly, this theory should still possess concepts such as locality and causality. Therefore, we begin by stating the concept of locality in the algebraic literature.

Statement

If the regions O_1 and O_2 satisfy $O_2 \subseteq O_1$, then, due to locality, we have

$$\mathcal{M}(O_2) \subseteq \mathcal{M}(O_1) \quad (5.22)$$

This definition is so intuitive and simple that the concept of locality can easily be studied from the algebraic point of view. However, for the concept of causality, two different statements and consequences can be considered.

Statement

Suppose that we have a region O . Now suppose that, due to the existence of the modular Hamiltonian, a time evolution can be envisioned for all operators belonging to the algebra of this region, $\mathcal{M}(O)$. Also, according to the time evolution, a deformation of the region can be considered, which we denote by $\hat{O}$. The concept of causality states that

$$\mathcal{M}(O) = \mathcal{M}(\hat{O}) \quad (5.23)$$

However, the causal nature of spacetime can also be discussed from the algebraic point of view in another way. Accordingly, rather than regarding causality as a property specific to spacetime and its regions, we consider it as a property of the algebra and the operators contained in the algebra under study.

Statement

Suppose that O is an open region in spacetime and denote its algebra by $\mathcal{M}(O)$. Also, from the definition of the commutant algebra we have

$$(\mathcal{M}(O))' = \{A | [A, B] = 0, B \in \mathcal{M}(O)\} \quad (5.24)$$

In this case, one can state that causality means that there exists an operator that belongs to the commutant of the algebra $\mathcal{M}(O)$ but does not belong to the algebra of the spacetime complement, $\mathcal{M}(O')$. Consequently,

$$\mathcal{M}(O') \subseteq (\mathcal{M}(O))' \quad (5.25)$$

Therefore, using the algebraic point of view, we can formulate a framework for the concepts of locality and causality. Under special conditions, these two concepts can be aligned with one another. In other words, we focus our attention on Eq. 5.25 when equality holds. In this case, we arrive at a stronger property than locality.

Statement

If, from the concept of causality, we regard Eq. 5.25 as an equality,

$$\mathcal{M}(O') = (\mathcal{M}(O))' \quad (5.26)$$

then the above relation is called Haag's duality.

As we stated, this property is stronger than locality. To prove this, suppose that on a manifold M on which a quantum field theory exists, we consider an arbitrary open region such as O . Accordingly, we can write $M = O \cup O'$. Now locality states that every operator contained in the algebra belonging to region O must lie in the commutant of the algebra of region O' . Then, using the relations in both directions, one can show that Eq. 5.26 holds.

It should be noted that the validity of this duality is not generally correct and is not a universal property of every theory. Also, Haag duality is easily observable and provable for regions with trivial topology, but this does not mean that this duality is true only for trivial topologies. For example, in quantum field theory, because there are regions with trivial topology, this duality always holds. The existence of regions with trivial topology is not characterized only by Haag duality; another important property called additivity also holds for these topologies. To this end, suppose that on the manifold M we have regions O_1 and O_2 . We also denote the algebra belonging to each region by $\mathcal{M}(O_1)$ and $\mathcal{M}(O_2)$. Then we know that $\mathcal{M}(O_1) \wedge \mathcal{M}(O_2) \equiv (\mathcal{M}(O_1) \cup \mathcal{M}(O_2))''$. It can also be shown that

$$\mathcal{M}(O_1) \wedge \mathcal{M}(O_2) \equiv (\mathcal{M}(O_1) \cup \mathcal{M}(O_2))'' \subseteq \mathcal{M}(O_1 \cup O_2) \quad (5.27)$$

In this case, similarly to Haag duality, one can state

Statement

If we consider the above relation in the case of equality,

$$\mathcal{M}(O_1) \vee \mathcal{M}(O_2) = \mathcal{M}(O_1 \cup O_2) \quad (5.28)$$

we express the fact that there are no additional operators on either side of the equality for the two arbitrary regions O_1 and O_2 . This property is called additivity.

As stated, this property also holds for regions with trivial topology. Thus, in quantum field theory, because every arbitrary region of the manifold on which this theory is defined has trivial topology, the properties of Haag duality and additivity hold for it.

These two properties lead to interesting results in theories such as quantum field theory. Accordingly, we use the additivity property and take the commutant to obtain

$$\mathcal{M}(O_1) \wedge \mathcal{M}(O_2) = \mathcal{M}(O_1 \cup O_2) \rightarrow (\mathcal{M}(O_1) \wedge \mathcal{M}(O_2))' = (\mathcal{M}(O_1 \cup O_2))' \quad (5.29)$$

Now, considering the right-hand side of the above equality, only by using Haag duality we can write

$$(\mathcal{M}(O_1 \cup O_2))' = (\mathcal{M}(O_1 \cup O_2))' = \mathcal{M}(O_1' \cap O_2') \quad (5.30)$$

Also, from the left-hand side of Eq. 5.29, according to the first chapters, we have

$$(\mathcal{M}(O_1) \vee \mathcal{M}(O_2))' = (\mathcal{M}(O_1))' \cap (\mathcal{M}(O_2))' = \mathcal{M}(O_1)' \cap \mathcal{M}(O_2)' \quad (5.31)$$

Consequently, by equating Eqs. 5.30 and 5.31, we obtain

$$\mathcal{M}(O_1)' \cap \mathcal{M}(O_2)' = \mathcal{M}(O_1' \cap O_2') \rightarrow \mathcal{M}(O_1) \cap \mathcal{M}(O_2) = \mathcal{M}(O_1 \cap O_2) \quad (5.32)$$

This property is, in a sense, very interesting and important. For example, in what we know about the duality between gravity and gauge field theories, the Ryu–Takayanagi entropy plays a significant role. In that discussion, as we will address later, the common algebra of different regions on the gauge field theory is a subset of the algebra of the common regions. For this reason, the existence of equality rather than inclusion in Eq. 5.32 in quantum field theory is very interesting. In later chapters, we will discuss whether, in gauge field theories, the properties of additivity and Haag duality fail because of nontrivial topologies or for another reason. Finally, using the stated properties, we can claim

Statement

Theories that possess the properties of Haag duality and additivity for all regions (regardless of whether the topology is trivial) are called complete theories [25].

In the next section, we turn to one of the most important structures in the algebraic literature. This structure has significant effects and applications in physics, particularly in dS spacetimes.

5.5 Crossed Product Algebra

Up to this point, we have studied the reason for using the algebraic point of view, become familiar with the types of von Neumann algebras, and then discussed the modular structure and Tomita–Takesaki theory. Now, what becomes important and raises a question for us is the following:

Question

Can the algebraic structure be changed without considering a boundary?

As stated previously, in a type III algebra, by considering the split property, one can transform the algebra into a type I algebra and remove the entanglement between regions near the boundary. But is the existence of a boundary with non-negligible thickness the only mechanism that changes the algebra? The answer to this question is based on what we know as the crossed product algebra [26, 9, 27].

To define this algebra, first suppose that $\mathcal{M}$ is a von Neumann algebra whose Hilbert space is denoted by $\mathcal{H}$. On the other hand, as we know, many physical systems possess symmetries acting on their algebra. This means that there exists a group such as G containing continuous unitary actions α of this group acting on the algebra $\mathcal{M}$,

$$\forall A \in \mathcal{M}, g \in G, U_g \in \mathcal{B}(\mathcal{H}), \alpha_g(A) = U_g A U_g^\dagger \quad (5.33)$$

For simplicity, we take G to be a locally compact group. In this case, given the triple $(\mathcal{M}, G, \alpha)$, one can reconstruct a von Neumann algebra called the crossed product algebra of $\mathcal{M}$ by the action α of the group G , and denote it by $\widehat{\mathcal{M}} = \mathcal{M} \rtimes_\alpha G$. It should also be noted that, because of the additional algebraic structure, the Hilbert space changes as well, and we have $\widehat{\mathcal{H}} = \mathcal{H} \otimes L^2(G)$.

Now, to describe the crossed product algebra $\widehat{\mathcal{M}}$ more precisely according to what we have in mind, suppose that we add a one-dimensional quantum system to the original system, whose corresponding algebra we called $\mathcal{M}$, whose Hilbert space is denoted by $\mathcal{H}$, and where K is the Hamiltonian operator. Therefore, first we can write the Hilbert space of the new system as

$$\mathcal{H} \rightarrow \widehat{\mathcal{H}} = \mathcal{H} \otimes L^2(\mathbb{R}) \quad (5.34)$$

Also, for this one-dimensional quantum system, $\hat{q}$ and $\hat{p}$ are respectively the position and momentum operators acting on the Hilbert space $L^2(\mathbb{R})$, and we have $[\hat{q}, \hat{p}] = i$. Therefore, the crossed product algebra $\widehat{\mathcal{M}}$ can be regarded as a subalgebra of $\mathcal{M} \otimes L^2(\mathbb{R})$ containing elements that remain invariant under the unitary transformation generated by

$$C \equiv K + q \quad (5.35)$$

³. In this case, because q commutes with C , an element of this algebra can be taken to be $\exp(-iqs)$, $s \in \mathbb{R}$. Also, for the other elements of this algebra, we have

$$\forall A \in \mathcal{M}, [\exp(+iKp)A \exp(-iKp), C] = 0 \quad (5.36)$$

Consequently,

Statement
The crossed product algebra $\widehat{\mathcal{M}}$ can therefore be written as
$\widehat{\mathcal{M}} = \{\exp(+iKp)A \exp(-iKp), \exp(-iqs) \mid A \in \mathcal{M}, s \in \mathbb{R}\}'' \quad (5.37)$
and, to represent an arbitrary element in this algebra, by defining $A(\hat{p}, s) = \exp(+iKp)A(s) \exp(-iKp)$, we have
$\forall \hat{A} \in \widehat{\mathcal{M}}, A \in \mathcal{M}, \hat{A} = \int ds A(\hat{p}; s) \exp(-iks), \quad (5.38)$

On the other hand, since every element of $\mathcal{M}$ commutes with every element of $\mathcal{M}'$, the crossed product condition also holds for the commutant algebra, and we have

$$\widehat{\mathcal{M}}' = \{A', \exp(iCs) \mid A' \in \mathcal{M}', s \in \mathbb{R}\}'' \quad (5.39)$$

Another point is the transformation of the definition of the algebra using unitary operators. In other words, suppose that U is a unitary operator. Accordingly, if $\widehat{\mathcal{M}}$ is the crossed

³This can be stated as

$$\widehat{\mathcal{M}} = \{a \mid [a, C] = 0\}$$

product algebra and C is the sum of the generators of each of the algebras, then we have

$$\widehat{\mathcal{M}} \rightarrow U\widehat{\mathcal{M}}U^\dagger, C \rightarrow UCU^\dagger, \widehat{\mathcal{M}}' \rightarrow U\widehat{\mathcal{M}}'U^\dagger \quad (5.40)$$

In this case, considering Eq. 5.37 and the fact that $U = \exp(+iKp)$ is a unitary operator, another form of the crossed product algebra can be obtained,

$$\begin{aligned} \widehat{\mathcal{M}} &= \{\exp(+iKp)A\exp(-iKp), \exp(-iqs) \mid A \in \mathcal{M}, s \in \mathbb{R}\}'' \equiv U\widehat{\mathcal{M}}U^\dagger \\ \widehat{\mathcal{M}} &= \{A, \exp(-iqs) \mid A \in \mathcal{M}, s \in \mathbb{R}\}'' , C = q \end{aligned} \quad (5.41)$$

It should be noted that with this transformation, the sum of the generators simultaneously changes and becomes $C = q$.

Now that we have defined the crossed product algebra, we want to investigate what type of von Neumann algebra it is. Suppose that the state $|\Psi\rangle$ is a cyclic and separating vector with respect to the algebra $\mathcal{M}$ and that the state $|p=0\rangle$ is an arbitrary vector in the Hilbert space $L^2(\mathbb{R})$. Then

$$|\hat{\Psi}\rangle = |\Psi\rangle \otimes |p=0\rangle \quad (5.42)$$

is a cyclic and separating vector with respect to the crossed product algebra $\widehat{\mathcal{M}}$. Also, using Tomita–Takesaki theory, we have

$$\hat{A}, \hat{B} \in \widehat{\mathcal{M}}, \langle \hat{\Psi} | \hat{A}\hat{B} | \hat{\Psi} \rangle = \langle \hat{\Psi} | \hat{B}\hat{\Delta}\hat{A} | \hat{\Psi} \rangle \quad (5.43)$$

where $\hat{\Delta}$ is the modular operator for the crossed product algebra $\widehat{\mathcal{M}}$, and one can show that $\hat{\Delta} = \Delta_\Psi$. Since the modular structure exists for this crossed product algebra, its modular flow is given by

$$\begin{aligned} \forall \hat{A} \in \widehat{\mathcal{M}}, \sigma_s(\hat{A}) &= \hat{\Delta}^{-is} \hat{A} \hat{\Delta}^{+is} = \hat{\Delta}_\Psi^{-is} \hat{A} \hat{\Delta}_\Psi^{+is} = \exp(+iKs) \hat{A} \exp(-iKs) \\ &= \exp(+iqs) \exp[+i(K-q)s] \hat{A} \exp[-i(K-q)s] \exp(-iqs) \end{aligned} \quad (5.44)$$

If $s \in \mathbb{R}$, then $\exp(-iqs)$ is recognized as an element of $\widehat{\mathcal{M}}$, and in this case the modular flow can be an inner automorphism for $s \in \mathbb{R}$. Now, according to the classification of the types of algebras, because for every $s \in \mathbb{R}$ the modular flow is an inner automorphism, and also because the Hilbert space is non-factorizable, we can conclude that $\widehat{\mathcal{M}}$ is a type II von Neumann algebra [28]. Therefore,

Statement

If $\mathcal{M}$ is a type III von Neumann algebra, then by adding a one-dimensional quantum system and imposing the invariance condition that gives rise to the crossed product algebra $\widehat{\mathcal{M}}$, one can produce a type II von Neumann algebra.

Now, since the crossed product algebra is a type II von Neumann algebra, we can define the map tr for this algebra. Therefore, considering $|\hat{\Psi}\rangle$ as a cyclic and separating vector with respect to $\widehat{\mathcal{M}}$ and comparing with Eq. 5.42, we have

$$\forall \hat{A} \in \widehat{\mathcal{M}}, \text{tr}(\hat{A}) = \langle \hat{\Psi} | \hat{A} \exp(-q) | \hat{\Psi} \rangle = \int_{-\infty}^{+\infty} dq \exp(-q) \langle \Psi | \hat{A} | \Psi \rangle \quad (5.45)$$

Also, according to the previous sections concerning the types of von Neumann algebras, we understand that by considering the dimension of the largest projection operator, namely the identity operator $\hat{I}$, the precise type of this algebra can be determined (with the point that in a type II von Neumann algebra we regard the dimension map as the map tr). Therefore,

$$\text{tr}(\hat{I}) = \langle \hat{\Psi} | \hat{I} \exp(-q) | \hat{\Psi} \rangle = \int_{-\infty}^{+\infty} dq \exp(-q) \langle \Psi | \hat{I} | \Psi \rangle = \int_{-\infty}^{+\infty} dq \exp(-q) = +\infty \quad (5.46)$$

In this case, the crossed product algebra $\widehat{\mathcal{M}}$ is a type II_∞ von Neumann algebra. On the other hand, if we impose the constraint $q \geq 0$, then the above expression becomes

$$\text{tr}(\hat{I}) = \langle \hat{\Psi} | \hat{I} \exp(-q) | \hat{\Psi} \rangle = \int_0^{+\infty} dq \exp(-q) \langle \Psi | \hat{I} | \Psi \rangle = \int_0^{+\infty} dq \exp(-q) = 1 \quad (5.47)$$

which indicates a type II_1 von Neumann algebra.

Now that we have realized that the crossed product is a type II algebra, we expect, as before, to investigate and define the reduced density operator for it. To this end, consider an arbitrary state in the new Hilbert space,

$$|\hat{\Phi}\rangle = |\Phi\rangle \otimes |g\rangle \equiv |\Phi, g\rangle, \quad |\Phi\rangle \in \mathcal{H}, |g\rangle \in L^2(\mathbb{R}) \quad (5.48)$$

In this case, we define the reduced density operator $\rho_{\hat{\Phi}}(\widehat{\mathcal{M}})$ as follows,

$$\forall \hat{A} \in \widehat{\mathcal{M}} \mid \text{tr}(\rho_{\hat{\Phi}} \hat{A}) = \langle \hat{\Phi} | \hat{A} | \hat{\Phi} \rangle \quad (5.49)$$

On the other hand, using Eq. 5.45, we can write

$$\text{tr}(\rho_{\hat{\Phi}} \hat{A}) = \langle \hat{\Psi} | \rho_{\hat{\Psi}} \hat{A} \exp(-q) | \hat{\Psi} \rangle = \langle \hat{\Phi} | \hat{A} | \hat{\Phi} \rangle \quad (5.50)$$

Then, using the unitary transformation introduced previously and Eq. 5.41, we can rewrite the above expression as

$$\langle \hat{\Psi} | \rho_{\hat{\Psi}} A \exp[i(K-q)s] \exp(-q) | \hat{\Psi} \rangle = \langle \hat{\Phi} | e^{+iKp} A \exp[i(K-q)s] e^{-iKp} | \hat{\Phi} \rangle \quad (5.51)$$

Now, from the right-hand side of the above equality, we have

$$\begin{aligned} \langle \hat{\Phi} | e^{+iKp} A e^{i(K-q)s} e^{-iKp} | \hat{\Phi} \rangle &= \langle \Phi, g | e^{+iKp} A e^{i(K-q)s} e^{-iKp} | \Phi, g \rangle \\ &= \langle \Psi, g | \Delta_{\Phi\Psi} e^{+iKp} A e^{i(K-q)s} e^{-iKp} | \Psi, g \rangle \\ &= \int dq \langle \Psi, g | \Delta_{\Phi\Psi} e^{+iKp} A e^{i(K-q)s} e^{-iKp} | \Psi, g \rangle \langle q | q \rangle \\ &= \int dq \langle \Psi, g | \Delta_{\Phi\Psi} e^{+iKp} A e^{i(K-q)s} | q \rangle \langle q | e^{-iKp} | \Psi, g \rangle \\ &= \int dq \langle \Psi, g | \Delta_{\Phi\Psi} e^{+iKp} | q \rangle A e^{i(K-q)s} \langle q | e^{-iKp} | \Psi, g \rangle \\ &= \int dq \langle \Psi | \Delta_{\Phi\Psi} \langle g | e^{+iKp} | q \rangle A e^{i(K-q)s} \langle q | e^{-iKp} | g \rangle | \Psi \rangle \end{aligned} \quad (5.52)$$

Then, by defining $\langle g|e^{+iKp}|q\rangle \equiv g(q-K)$, the above expression can be written as

$$\langle \hat{\Phi}|e^{+iKp}Ae^{i(K-q)s}e^{-iKp}|\hat{\Phi}\rangle = \int dq \langle \Psi|\Delta_{\Phi\Psi}g^*(q-K)Ae^{i(K-q)s}g(q-K)|\Psi\rangle \quad (5.53)$$

From the left-hand side of Eq. 5.51, we have

$$\begin{aligned} \langle \hat{\Psi}|\rho_{\hat{\Psi}}Ae^{i(K-q)s}e^{-q}|\hat{\Psi}\rangle &= \int dq dq' \langle \hat{\Psi}|\rho_{\hat{\Psi}}Ae^{i(K-q)s}e^{-q} \langle q|q\rangle \langle q'|q'\rangle |\hat{\Psi}\rangle \\ &= \frac{1}{2\pi} \int dq dq' e^{-q} \langle \Psi, q'|\rho_{\hat{\Psi}}Ae^{i(K-q)s}|\Psi, q\rangle \end{aligned} \quad (5.54)$$

Now, by equating Eqs. 5.53 and 5.54, we can write

$$\frac{1}{2\pi} \int dq dq' e^{-q} \langle \Psi, q'|\rho_{\hat{\Psi}}Ae^{i(K-q)s}|\Psi, q\rangle = \int dq \langle \Psi|\Delta_{\Phi\Psi}g^*(q-K)Ae^{i(K-q)s}g(q-K)|\Psi\rangle \quad (5.55)$$

Then the reduced density operator corresponding to the crossed product algebra $\widehat{\mathcal{M}}$ can be obtained,

$$\rho_{\hat{\Psi}} = 2\pi g^*(q-K) \exp(+q) \Delta_{\Phi\Psi} g(q-K) \quad (5.56)$$

Therefore, the existence of this reduced density operator allows us to calculate the entropy for the crossed product algebra. Accordingly, we have

$$S_{\widehat{\mathcal{M}}} \equiv -\text{tr}(\rho_{\hat{\Phi}} \log \rho_{\hat{\Phi}}) \quad (5.57)$$

Then, using the definition of the map tr , the above expression can be written as

$$S_{\widehat{\mathcal{M}}} = -\text{tr}(\rho_{\hat{\Phi}} \log \rho_{\hat{\Phi}}) = -\langle \hat{\Phi}|e^{+iKp} \log \rho_{\hat{\Phi}} e^{-iKp}|\hat{\Phi}\rangle \quad (5.58)$$

Now, using Eq. 5.56, we have

$$\begin{aligned} S_{\widehat{\mathcal{M}}} &= -\langle \hat{\Phi}|e^{+iKp} \log \rho_{\hat{\Phi}} e^{-iKp}|\hat{\Phi}\rangle \\ &= -\langle \hat{\Phi}|e^{+iKp} \log[2\pi g^*(q-K) \exp(+q) \Delta_{\Phi\Psi} g(q-K)] e^{-iKp}|\hat{\Phi}\rangle \\ &= \langle \hat{\Phi}|e^{+iKp} [\log(2\pi) - q - \log \Delta_{\Phi\Psi} - \log |g(q-K)|^2] e^{-iKp}|\hat{\Phi}\rangle \\ &\simeq \langle \hat{\Phi}|e^{+iKp} [-q - \log \Delta_{\Phi\Psi} - \log |g(q-K)|^2] e^{-iKp}|\hat{\Phi}\rangle \\ &\simeq -\langle \hat{\Phi}|q|\hat{\Phi}\rangle - \langle \hat{\Phi}|\log |g(q-K)|^2|\hat{\Phi}\rangle - \langle \hat{\Phi}|\log \Delta_{\Phi\Psi}|\hat{\Phi}\rangle \\ &\simeq -\int dq |g(q)|^2 q - \int dq \log |g(q)|^2 |g(q-K)|^2 - S_{\mathcal{M}}(\Phi||\Psi) \end{aligned} \quad (5.59)$$

where we have used the definition of relative entropy in Eq. 4.63, and $g(q)$ represents the probability distribution associated with the added one-dimensional system. In this case,

Statement

In the crossed product algebra $\widehat{\mathcal{M}}$, which is a type II von Neumann algebra, the entropy

is given by

$$S_{\widehat{\mathcal{M}}} = - \int dq |g(q)|^2 q - \int dq \log |g(q)|^2 |g(q - K)|^2 - S_{\mathcal{M}}(\Phi || \Psi) \quad (5.60)$$

As we observe, the above entropy is a negative quantity because of the minus sign in front of the relative entropy. The existence of this minus sign, similar to what we investigated for a type II algebra in the example of pairs of qubits, is a sign of the relative nature of the entropy and is not the same as the definition used in statistical mechanics. It should of course be noted that in some areas of high-energy physics, this quantity can be considered in the same way as in statistical mechanics; however, as we show below, the existence of this entropy in some branches of physics means the entropy between an arbitrary state and the state with the greatest entanglement (which usually has zero entropy), and for this reason the minus sign arises correctly.

Chapter 6

Gauge–Gravity Duality

Up to this point, we have become acquainted with the algebraic viewpoint and introduced theories such as Tomita–Takesaki theory. We then became interested in the connection between the algebraic viewpoint and quantum information. For this reason, we developed an algebraic approach to obtaining and computing the reduced density operator, and consequently the entropy, for arbitrary algebras. To this end, we attempted to study quantum field theories, which are more familiar to us, from the algebraic viewpoint. We now turn to quantum gravity, at the heart of which lies the gravity–gauge duality. Based on this duality, we aim to study it within the algebraic framework in the limit $N \rightarrow \infty$ [29, 30, 31].

6.1 Introduction to AdS/CFT

To begin this discussion, let us step back and approach this duality from the perspective of superstring theory. As we know, in this theory we find that Type IIB superstring theory on $AdS_5 \times S_5$ is equivalent to $\mathcal{N} = 4$ super Yang–Mills theory with gauge group $SU(N)$, defined on the geometry $\mathbb{R} \times S_3$. Accordingly, physicists proposed a duality between the bulk, which possesses an AdS_5 geometry, and a boundary on which a theory known as a conformal field theory is defined.

What gives this duality its power is the dictionary relating the two theories, namely the gravitational theory and the conformal field theory. In other words, one of the most important features of this duality is the equality between the gravitational partition function and the correlation functions of the conformal field theory,

$$Z_{\text{Gravity}} \cong Z_{\text{CFT}}. \quad (6.1)$$

Accordingly, the following table can be regarded as a dictionary between the two theories.

Bulk	Boundary
$\mathcal{H}_{\text{AdS}}$	$\mathcal{H}_{\text{CFT}}$
$G_N \rightarrow 0$	$N \rightarrow \infty$
$\alpha' \rightarrow 0$	$\lambda \rightarrow \infty$
Fundamental fields	Single-trace operators
Classical geometry	Semiclassical state $ \Psi\rangle$

Furthermore, in this duality, one can establish a relation between the fundamental fields, $\phi(X)$, in the bulk and operators located at infinity on the boundary, $O(x)$, of the form

$$O(x) = \lim_{r \rightarrow \infty} r^{-\Delta} \phi(X) \quad (6.2)$$

where $X = (r, x)$, with r denoting the radial coordinate in the bulk and x denoting a point on the boundary. The quantity Δ is the conformal dimension, which plays a crucial role in conformal field theory. On the other hand, in $\mathcal{N} = 4$ super Yang-Mills theory, arbitrary operators such as $O(x)$ are defined schematically as

$$O(x) = \text{tr}(\dots). \quad (6.3)$$

Thus, one can establish a correspondence, analogous to the dictionary above, between the fundamental fields in the bulk and single-trace operators,

$$O(x) = \lim_{r \rightarrow \infty} r^{-\Delta} \phi(X) = \text{tr}(\dots) \quad (6.4)$$

Our goal in this duality, as stated in Chapter 1, is to investigate whether spacetime geometry emerges as the entanglement between degrees of freedom becomes infinite. This duality provides a suitable setting for examining this question because, as can be seen from the dictionary, as the entanglement between the degrees of freedom in the conformal field theory becomes arbitrarily large, corresponding to $N \rightarrow \infty$, quantum gravitational effects in the bulk become negligible, and the bulk approaches a classical geometry, $G_N \rightarrow 0$. This regime is referred to as the semiclassical regime. Therefore, studying the above question in this setup is particularly useful.

We begin by placing ourselves in the gauge theory and considering an arbitrary state $|\Psi_N\rangle$. If our analysis is to be performed in the large- N limit, then we need to choose a state that remains well-defined in this regime. Thus,

$$|\Psi\rangle = \lim_{N \rightarrow \infty} |\Psi_N\rangle. \quad (6.5)$$

Since we are in the semiclassical regime, we also refer to this state as a semiclassical state. The existence of a well-defined semiclassical state in the large- N regime, however, is not sufficient, since there are operators that do not remain well-defined in this limit. Therefore,

Statement

In the conformal field theory, given a semiclassical state $|\Psi\rangle$ in the limit $N \rightarrow \infty$, there exist operators such as A for which a sequence of operators, denoted by $\{A_N\}$, satisfies

$$\lim_{N \rightarrow \infty} \langle \Psi_N | A_N | \Psi_N \rangle < \infty. \quad (6.6)$$

Here, $|\Psi\rangle = \lim_{N \rightarrow \infty} |\Psi_N\rangle$ [32, 33].

We can therefore define the set of all operators that remain well-defined in the large- N limit as an algebra, which we denote by $\mathcal{A}_\Psi$ and which naturally depends on the semiclassical state $|\Psi\rangle$. As stated earlier, the operators of interest to us are single-trace operators, which form a subalgebra of $\mathcal{A}_\Psi$. Thus, the algebra available to us, denoted by $\mathcal{S}$, consists of

single-trace operators that remain well-defined in the limit $N \rightarrow \infty$,

$$\mathcal{S} \subseteq \mathcal{A}_\Psi. \quad (6.7)$$

At this point, we have a semiclassical state $|\Psi\rangle$ and a C^* -algebra, denoted by $\mathcal{S}$. This situation naturally reminds us of the GNS construction, through which, starting from a C^* -algebra, we were able to construct a von Neumann algebra. To this end, we define the map $\omega : \mathcal{A}_\Psi \rightarrow \mathbb{C}$ by

$$\begin{aligned} \forall A, B \in \mathcal{A}_\Psi, \forall a, b \in \mathbb{C} \mid \omega(aA + bB) &= a\omega(A) + b\omega(B) \\ \omega(A^\dagger) &= (\omega(A))^*. \end{aligned} \quad (6.8)$$

We then define an inner-product-like map $\langle \cdot, \cdot \rangle : \mathcal{A}_\Psi \times \mathcal{A}_\Psi \rightarrow \mathbb{C}$ using ω ,

$$\forall A, B \in \mathcal{A}_\Psi \mid \omega(A^\dagger B) = \langle A | B \rangle. \quad (6.9)$$

By quotienting out the null operators and using equivalence classes, the remaining operators can be regarded as vectors in a vector space. We can then construct the GNS Hilbert space by representing the algebra and acting with it on the semiclassical state $|\Psi\rangle$. We also know that if the map ω is a pure state, then

$$\mathcal{B}(\mathcal{H}_\Psi^{\text{GNS}}) = (\pi_\Psi(\mathcal{A}_\Psi))''. \quad (6.10)$$

Statement

In the conformal field theory, given the semiclassical state $|\Psi\rangle$ and the single-trace operators forming the algebra $\mathcal{S}$, the Hilbert space constructed through the GNS procedure is denoted by $\mathcal{H}_{\Psi, \mathcal{S}}^{\text{GNS}}$. Furthermore,

$$\mathcal{Y} = (\pi_\Psi(\mathcal{S}))''. \quad (6.11)$$

On the other hand, if we consider a region O of the boundary on which the theory is defined, then

$$\mathcal{Y}_O = (\pi_\Psi(\mathcal{S}_O))''. \quad (6.12)$$

It is important to note that, since $\mathcal{S} \subseteq \mathcal{A}_\Psi$, the Hilbert spaces constructed from these two algebras are completely different.

As stated above, in the boundary theory, given a semiclassical state and repeatedly acting on it with all operators that remain well-defined in the limit $N \rightarrow \infty$, belonging to the algebra $\mathcal{A}_\Psi$, we can construct the Hilbert space $\mathcal{H}_\Psi^{\text{GNS}}$ around $|\Psi\rangle$. In this sense, by constructing a Hilbert space around the state $|\Psi\rangle$, we have effectively introduced a perturbation. To clarify this statement, let us turn to the gravitational theory in the bulk.

For the theory in the bulk, we consider the field corresponding to the classical geometry, or equivalently the classical field, ϕ_c , coupled to gravity. As we know, this coupling is governed by the factor $\frac{1}{16\pi G_N}$ appearing in the gravitational action in the bulk. Accordingly,

the perturbative expansion of the field takes the form

$$\phi_c \rightarrow \phi = \phi_c + k\delta\phi, \quad k = \sqrt{8\pi G_N}. \quad (6.13)$$

Thus, considering the scalar-field action, we obtain

$$S[\phi] = S[\phi_c] + S_2[\delta\phi] + kS_3[\delta\phi] + k^2S_4[\delta\phi] + \dots \quad (6.14)$$

Since $k \propto G_N \rightarrow 0$, the above action becomes

$$\lim_{G_N \rightarrow 0} S[\phi] = S[\phi_c] + S_2[\delta\phi]. \quad (6.15)$$

The presence of the quadratic action indicates the quantization of the fields in the theory. Quantizing these fields therefore gives rise to a Fock space, $\mathcal{H}_\Psi^{\text{Fock}}$. Moreover, acting with $\delta\phi$ on the classical field ϕ_c generates this Fock space. We thus encounter an interesting observation:

Statement

In the boundary theory, by considering the semiclassical state $|\Psi\rangle$ and acting with operators that remain well-defined in the limit $N \rightarrow \infty$, we can construct the Hilbert space $\mathcal{H}_\Psi^{\text{GNS}}$. On the other hand, by quantizing the fields in the bulk gravitational theory in the limit $G_N \rightarrow 0$, we obtain the Fock space $\mathcal{H}_\Psi^{\text{Fock}}$. The duality therefore requires

$$\mathcal{H}_\Psi^{\text{GNS}} \cong \mathcal{H}_\Psi^{\text{Fock}}. \quad (6.16)$$

Furthermore, since the state $|\Psi\rangle$ and the classical field ϕ_c respectively generate the Hilbert space and Fock space by acting with operators belonging to $\mathcal{A}_\Psi$ and with $\delta\phi$, it is natural to identify

$$|\Psi\rangle \cong |0\rangle_{\delta\phi}, \quad (6.17)$$

where $|\Psi\rangle$ is a cyclic and separating vector with respect to the algebra $\mathcal{A}_\Psi$, while $|0\rangle_{\delta\phi}$ is the vacuum state of the bulk theory. Thus,

$$\mathcal{B}(\mathcal{H}_\Psi^{\text{GNS}}) \cong \mathcal{B}(\mathcal{H}_\Psi^{\text{Fock}}). \quad (6.18)$$

We have therefore been able to reformulate the AdS/CFT duality using the algebraic language. This reformulation in the limit $N \rightarrow \infty$ possesses several distinctive features:

1. As we have seen, in the limit $N \rightarrow \infty$, for every semiclassical state $|\Psi\rangle$, the Hilbert space $\mathcal{H}_\Psi^{\text{GNS}}$ can be identified with the Fock space associated with the vacuum state corresponding to $|\Psi\rangle$. Therefore, as illustrated in Fig. 6.1, in the limit $N \rightarrow \infty$ these spaces are distinct from one another and have no common intersection.
2. These semiclassical geometries have energy differences of the form

$$\Delta E \simeq \frac{1}{G_N}. \quad (6.19)$$



Figure 6.1: In the limit $N \rightarrow \infty$, the state spaces, known as Fock spaces, are distinct and separately equivalent to the spaces constructed around the semiclassical state $|\Psi\rangle$.

Therefore, in the semiclassical limit $G_N \rightarrow 0$, any two distinct semiclassical geometries have an infinite energy difference. Consequently, no process can induce a transition between these two geometries.

3. From the algebraic viewpoint, semiclassical states have energy differences of order $O(G_N^0)$. Therefore, even if two semiclassical states have the same energy, because the energy difference between the corresponding semiclassical geometries is infinite, these states cannot have any overlap with one another.

Thus, in the limit $N \rightarrow \infty$, one cannot use a single large Hilbert space to describe the system. Instead, we encounter a collection of mutually unrelated Hilbert spaces, each describing a semiclassical geometry. A special feature therefore emerges in the semiclassical limit. Suppose that we are working at finite N in the AdS/CFT duality. As illustrated in Fig. 6.2, because of quantum effects, there is a fluctuating layer in the bulk as well as a band on the boundary. If we consider the algebra corresponding to the band on the boundary, $\mathcal{M}_T$, it can be identified with a single Cauchy slice $\mathcal{B}_{\Sigma_T}^N$.

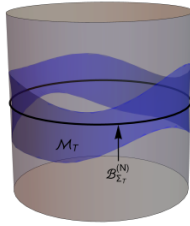


Figure 6.2: At finite N , the algebra associated with the timelike band on the boundary shown in the figure is equivalent to the algebra on a single Cauchy slice.

However, in the limit $N \rightarrow \infty$, according to the properties discussed above, every semiclassical state has its own associated algebra. Thus, as shown in Fig. 6.3, a variety of different algebras emerge. This feature does not occur at finite N ; it is only in the semiclassical regime that different algebras, rather than a single algebra associated with the boundary band, emerge. Therefore, in the limit $N \rightarrow \infty$, we have access to a subalgebra of the full algebra.

Mathematically, we know that the Hamiltonian of the gravitational theory can be written

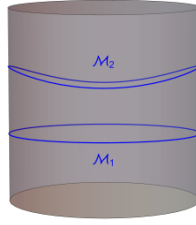


Figure 6.3: The algebras $\mathcal{M}_1$ and $\mathcal{M}_2$ generated by the fundamental fields on two different Cauchy slices are not equivalent.

as

$$H = \int_{\sigma} d^{d-1}x T^{00} \quad (6.20)$$

where σ is the spatial slice and $T^{\mu\nu}$ is the stress-energy tensor, defined by

$$T^{\mu\nu} = N \text{tr}(\dots). \quad (6.21)$$

As is clear, this definition is well-defined at finite N but becomes problematic in the limit $N \rightarrow \infty$. Consequently, the Hamiltonian in Eq. 6.20 also fails to remain well-defined in this limit. We therefore need to redefine the Hamiltonian and stress-energy tensor. Let

$$\hat{T}^{\mu\nu} \equiv \frac{1}{N} T^{\mu\nu}, \quad \hat{H} \equiv \frac{1}{N} H. \quad (6.22)$$

With these definitions, the above quantities remain well-defined in the limit $N \rightarrow \infty$. Now consider an arbitrary boundary operator such as $O(x)$. Its evolution can be written as

$$i[\hat{H}, O(x)] = \frac{1}{N} \partial_t O(x). \quad (6.23)$$

Taking the limit $N \rightarrow \infty$, we obtain

$$\lim_{N \rightarrow \infty} i[\hat{H}, O(x)] = \lim_{N \rightarrow \infty} \frac{1}{N} \partial_t O(x) = 0. \quad (6.24)$$

We may therefore conclude that, in the semiclassical regime, operators on a spatial slice of the boundary do not evolve under the Heisenberg picture. Thus, at different time slices, the corresponding operators are not related to one another by any dynamical process.

Statement

Therefore, at finite N , the algebra associated with the timelike band on the boundary is equivalent to the algebra on a Cauchy slice. However, in the limit $N \rightarrow \infty$, every time slice has its own algebra and these algebras are independent of one another.

An important point is that, as shown in Fig. 6.4, $\mathcal{H}_{\Psi}^{\text{GNS}}$ is a Hilbert space for the entire spacetime of the CFT, whereas $\mathcal{H}_{\Psi}^{\text{Fock}}$ corresponds only to a single Cauchy slice Σ in the bulk theory.

This is in a sense natural for the bulk-boundary correspondence, since, given the equiv-

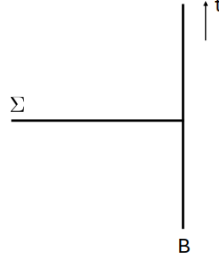


Figure 6.4: For a general asymptotically AdS geometry, the bulk algebra $\mathcal{B}(\mathcal{H}_\Psi^{\text{Fock}})$ can be defined on a single bulk Cauchy slice Σ , whereas the boundary algebra $\mathcal{B}(\mathcal{H}_\Psi^{\text{GNS}})$ is defined over the entire boundary spacetime. In the diagram, the vertical line represents the boundary.

alence between $\mathcal{B}(\mathcal{H}_\Psi^{\text{Fock}})$ and $\mathcal{B}(\mathcal{H}_\Psi^{\text{GNS}})$, they must have the same dimensionality.

Having introduced the AdS/CFT duality and reformulated it in the algebraic language, in the next section we shall consider its simplest case, consisting of a vacuum state in the gravitational theory and a cyclic and separating vector in the boundary theory, and apply the algebraic framework to it.

6.2 Vacuum State

As we know, in the AdS/CFT duality, the CFT_d defined on the geometry $\mathbb{R} \times S_{d-1}$ is dual to a gravitational theory in AdS_{d+1} . This geometry is defined by the metric

$$ds^2 = -(1 + \frac{\rho^2}{R^2})dt^2 + (1 + \frac{\rho^2}{R^2})^{-1}d\rho^2 + \rho^2 d\Omega_{d-1}^2. \quad (6.25)$$

In this case, the vacuum state in the gravitational theory, $|0\rangle_{\text{AdS}}$, is equivalent to the cyclic and separating state $|\Omega\rangle$ with respect to the algebra of the theory,

$$|\Omega\rangle \cong |0\rangle_{\text{AdS}}. \quad (6.26)$$

Thus, we can reconstruct both the Hilbert space and the Fock space explicitly.

Let us now turn to the gravitational theory. By quantizing the fundamental fields in the gravitational theory, we know that the Fock space contains creation and annihilation operators, denoted respectively by $a_k^\dagger$ and a_k . Acting with these operators on the vacuum generates the Fock space. Therefore, an arbitrary fundamental field in the gravitational theory can be written as

$$\phi(X) = \sum_k [u_k(X)a_k + u_k^*(X)a_k^\dagger], \quad (6.27)$$

where $\{u_k(X)\}$ are wavefunctions propagating in the bulk geometry.

As we know, the fundamental bulk field appears as a source in the boundary theory. Therefore, the boundary operators and their correlation functions play a crucial role in the limit $N \rightarrow \infty$. Suppose that the boundary theory contains operators such as $O(x)$. Their

correlation functions are

$$\langle O \rangle = 0 \quad (6.28)$$

$$\langle O_1 O_2 \rangle = O(N^0) \quad (6.29)$$

$$\langle O_1 \dots O_n \rangle = N^{2-n}. \quad (6.30)$$

Therefore, at leading order in the limit $N \rightarrow \infty$,

$$\langle O_1 \dots O_n \rangle = \sum (\text{products of two-point functions}) \sim O(N^0),$$

which defines a Gaussian theory in terms of two-point functions. These functions are explicitly determined, up to a normalization constant, by conformal symmetry. Every single-trace operator can be regarded as a generalized free field, and consequently the Hilbert space $\mathcal{H}_\Omega^{(\text{GNS})}$, obtained by acting with the algebra of single-trace operators on $|\Omega\rangle$, has a Fock-like structure. In this case, one can show that

$$\mathcal{B}(\mathcal{H}_\Omega^{\text{GNS}}) \cong \mathcal{B}(\mathcal{H}_{\text{AdS}}^{\text{Fock}}). \quad (6.31)$$

On the other hand, using the Hilbert space obtained through the GNS construction, the representation of this algebra can be written as

$$\pi_\Omega(O(x)) = \sum_k [v_k(x)b_k + v_k^*(x)b_k^\dagger], \quad (6.32)$$

where, as in the Fock space, $b_k |\Omega\rangle = 0$. Furthermore, $v_k(x)$ are basis functions on the geometry $\mathbb{R} \times S_{d-1}$ on which the CFT is defined. Using Eq. 6.4, we then have

$$\sum_k [v_k(x)b_k + v_k^*(x)b_k^\dagger] = \lim_{r \rightarrow \infty} r^{-\Delta} \sum_k [u_k(X)a_k + u_k^*(X)a_k^\dagger]. \quad (6.33)$$

Since $u_k(X)$ and $v_k(x)$ are corresponding basis sets for the two theories, we may identify

$$a_k = b_k, \quad a_k^\dagger = b_k^\dagger. \quad (6.34)$$

Consequently, the Hilbert space $\mathcal{H}_\Omega^{\text{GNS}}$ is equivalent to the Fock space $\mathcal{H}_{\text{AdS}}^{\text{Fock}}$. Thus,

Statement

Using Eq. 6.34, the AdS/CFT duality can be expressed as

$$\mathcal{H}_\Omega^{\text{GNS}} \cong \mathcal{H}_{\text{AdS}}^{\text{Fock}}, \quad |\Omega\rangle \cong |0\rangle_{\text{AdS}}, \quad \mathcal{B}(\mathcal{H}_\Omega^{\text{GNS}}) \cong \mathcal{B}(\mathcal{H}_{\text{AdS}}^{\text{Fock}}). \quad (6.35)$$

It is also possible to establish the relation between the fundamental fields in the gravitational theory, $\phi(X)$, and the boundary operators, represented here in their algebraic form, $\pi_\Omega(O(x))$, using a bulk-to-boundary propagator. We have

$$\phi(X) = \int d^d x K(X; x) \pi_\Omega(O(x)) \quad (6.36)$$

where $K(X; x)$ is the kernel of the above integral, which behaves as a Green's function

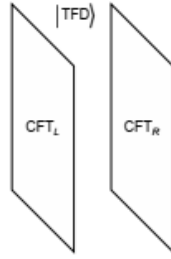


Figure 6.5: The thermofield double state constructed from the two boundaries, left and right, on which the conformal field theories are defined.

(propagator), and is required to satisfy Eq. 6.4.

6.3 Thermofield Double State

We now turn to the thermofield double state. To begin this discussion and understand what this state is, we first need to recall the physical problem that leads to it. As we know, in high-energy physics we have the AdS/CFT duality. If we wish to study an eternal AdS black hole [34], holography allows us to use two boundary theories, denoted by CFT^*R and CFT^*L . Thus, in the bulk there is an eternal black hole, while on the boundary there are two conformal field theories.

However, this setup is not valid under all circumstances and depends crucially on the temperature of the eternal black hole. To examine this statement schematically, let the temperature of the eternal AdS black hole be T . Due to the CFT duality, this quantity has an explicit dependence on the degrees of freedom of the theory. As T increases, the number of excited degrees of freedom also increases, and therefore one cannot study the interior bulk geometry at arbitrary temperatures without taking entanglement into account. Consequently, the eternal black hole has a critical temperature known as the Hawking–Page temperature, denoted by T_{HP} [35]. This temperature and its associated phase diagram arise in discussions of the black-hole paradox, which we shall address in later chapters.

Thus, on the two sides of the boundary, we have the boundary theories $\text{CFT}_{R,L}$, each with its own Hilbert space $\mathcal{H}_{R,L}$. The Hilbert space of the complete boundary theory is therefore

$$\mathcal{H}_R \otimes \mathcal{H}_L.$$

The thermofield double state is a state in this Hilbert space and is defined as

$$|TFD_\beta\rangle = \frac{1}{\sqrt{Z_\beta}} \sum_n \exp(-\frac{\beta}{2} E_n) |n\rangle_R |n\rangle_L, \quad (6.37)$$

where $Z_\beta = \sum_n \exp(-\beta E_n)$ (see Fig. 6.5). By taking a partial trace, one can also obtain the density operator for either of the $\text{CFT}_{R,L}$ theories.

However, an important question remains:

Question

Can the Hilbert space still be written as $\mathcal{H} * R \otimes \mathcal{H} * L$ when we take the limit $N \rightarrow \infty$?

As we know, the Gibbs free energy can be written as

$$F(\beta) = -\frac{1}{\beta} \log Z_\beta. \quad (6.38)$$

We also know that at temperatures below T_{HP} , corresponding to finite N , the Gibbs free energy is of order $O(N^0)$, whereas at temperatures above T_{HP} , corresponding to the limit $N \rightarrow \infty$, the free energy is of order $O(N^2)$. The discontinuity in this quantity signals a significant change in the bulk theory and consequently affects the boundary theory and its description.

We are therefore interested in studying these changes in the physics of the system, caused by the number of degrees of freedom and the effects of entanglement, from the algebraic viewpoint. We first specify the algebraic structure of the problem and then consider the different limits. As before, we turn to the boundary theory. We denote the algebras associated with the boundary theories $\text{CFT} * R$ and $\text{CFT} * L$ by $\mathcal{M} * R$ and $\mathcal{M} * L$, respectively. We also require a cyclic and separating vector with respect to these two algebras. The natural choice is the thermofield double state, with the caveat that its definition contains the parameter N ; nevertheless, in general, we require such a vector.

We now consider the finite- and infinite-degree-of-freedom limits:

1. $T < T_{HP}$ In this temperature regime, corresponding to finite N , the Gibbs free energy is of order $O(N^0)$. The boundary theory consists of two CFTs, on the right and left sides. According to the AdS/CFT duality, each side has its own separate AdS geometry.

Let us focus on CFT_R . As before, it contains single-trace operators that generate the algebra $\mathcal{M}_R$. By considering a cyclic and separating vector such as $|\Omega\rangle$, we can construct the corresponding Hilbert space, denoted by $\mathcal{H}_\Omega^{\text{GNS},R}$. One can also show that

$$\mathcal{M}_R = \mathcal{B}(\mathcal{H}_\Omega^{\text{GNS},R}) = (\pi_\omega(\mathcal{S}_R))''. \quad (6.39)$$

Here, ω is the map defined on the algebra and $\mathcal{S}_R$ is the algebra of single-trace operators. Similarly, one can construct the corresponding Hilbert space, cyclic and separating vector, and algebra for CFT_L . Hence the total Hilbert space of the boundary theory is

$$\mathcal{H}_{TFD_\beta}^{\text{GNS}} = \mathcal{H}_\Omega^{\text{GNS},L} \otimes \mathcal{H}_\Omega^{\text{GNS},R}. \quad (6.40)$$

Moreover,

$$\mathcal{M}_L = \mathcal{B}(\mathcal{H}_\Omega^{\text{GNS},L}) \quad , \quad \mathcal{M}_R = \mathcal{B}(\mathcal{H}_\Omega^{\text{GNS},R}). \quad (6.41)$$

We might therefore conclude that at finite N , the algebras associated with the two theories are equal to the full operator algebras, indicating the absence of entanglement

between CFT_R and CFT_L (see Fig. 6.6).

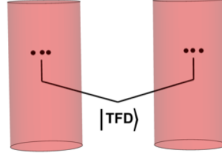


Figure 6.6: For $T < T_{HP}$, the thermofield double state is dual to two disconnected AdS spacetimes.

It is worth noting that, because there exists a cyclic and separating vector with respect to the algebras under consideration, a modular structure is automatically present. Thus, the modular operator can be written as

$$-\log \Delta_\beta = \beta(\hat{h}_R - \hat{h}_L) \quad (6.42)$$

where $\hat{h}_{R,L}$ are the Hamiltonians associated with the bulk theory.

On the other hand, we conclude that the algebras $\mathcal{M}_{R,L}$ are von Neumann algebras of type I. Thus, if we examine the gravitational theory in the bulk, we can again claim that, due to the duality, in addition to the equivalence of the Hilbert spaces with the corresponding Fock spaces, the thermofield double state is equivalent to the vacuum state in a pair of disconnected AdS geometries. We also have

$$\widetilde{\mathcal{M}}_L \cong \mathcal{M}_L, \quad \widetilde{\mathcal{M}}_R \cong \mathcal{M}_R, \quad (6.43)$$

where $\widetilde{\mathcal{M}}_{R,L}$ are the algebras associated with the gravitational theories in the AdS geometries. Thus, in summary,

Statement

At finite N , corresponding to $T < T_{HP}$, knowing that the boundary theory is dual to two disconnected AdS spacetimes, we have the following algebraic reformulation:

$$\mathcal{H}_\Omega^{\text{GNS}, R(L)} \cong \mathcal{H}_{\text{AdS}}^{\text{Fock}, R(L)} \quad (6.44)$$

$$|TFD_\beta\rangle \cong |0\rangle_{\text{AdS} \times \text{AdS}} \quad (6.45)$$

$$\widetilde{\mathcal{M}}_{R,L} \cong \mathcal{M}_{R,L} \cong \mathcal{B}(\mathcal{H}_\Omega^{\text{GNS}, R(L)}). \quad (6.46)$$

As we can see, there are no significant difficulties in this temperature regime, and we have been able to provide a new algebraic formulation.

2. $T > T_{HP}$

In this temperature regime, corresponding to the limit $N \rightarrow \infty$, the Gibbs free energy and the energy are both of order $O(N^2)$. As we know, in this temperature range the bulk geometry is an eternal AdS black hole with metric

$$ds^2 = -f(\rho)dt^2 + \frac{d\rho^2}{f(\rho)} + \rho^2 d\Omega_{d-1}^2, \quad f(\rho) = 1 + \frac{\rho^2}{R^2} - \frac{\mu}{\rho^{d-2}}. \quad (6.47)$$

Thus, in this setup, the boundary theory consisting of $\text{CFT}_{R,L}$ gives rise to a single bulk geometry. This is in contrast with the regime $T < T_{HP}$, where each boundary theory gives rise to its own spacetime geometry. The corresponding Penrose diagram is shown in Fig. 6.7.

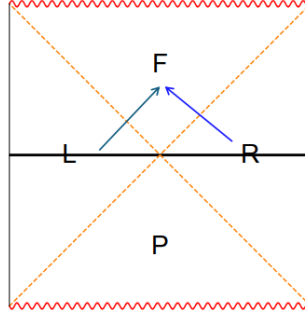


Figure 6.7: For $T > T_{HP}$, the thermofield double state is dual to an eternal AdS black hole. The horizontal black line represents a Wilson line extending from the left boundary to the right boundary. Bulk degrees of freedom originating from the R and L regions of the black hole can pass through the horizon, as indicated by the arrows, and interact in region F .

The existence of a unique bulk geometry associated with the two boundary theories leads to two important problems:

Question

- **The Factorization Problem**

Suppose that, in the boundary theory, we have operators O_R and O_L belonging respectively to CFT_R and CFT_L . By virtue of the duality, we can associate bulk operators $\tilde{O}_R$ and $\tilde{O}_L$ with the corresponding boundary operators. One might then expect that the bulk theory contains the operator $\tilde{O}_L \otimes \tilde{O}_R$. However, the bulk contains a single spacetime geometry, and factorization of the operators is fundamentally in tension with the uniqueness of the bulk geometry [36].

- **The Behind-the-Horizon Problem**

As stated above, the boundary theories CFT_R and CFT_L can be assigned Hilbert spaces $\mathcal{H}_R$ and $\mathcal{H}_L$. Thus the boundary theory has Hilbert space

$$\mathcal{H} = \mathcal{H}_L \otimes \mathcal{H}_R. \quad (6.48)$$

The Hamiltonian of this boundary system can then be written as

$$H = H_L + H_R, \quad (6.49)$$

which indicates that the two boundary theories do not interact with each other. However, in the bulk gravitational theory, if a particle is moved from the right side to the left side [37], the Hamiltonians must interact with one

another, leading to an apparent contradiction.

To resolve these problems, we need to approach them from a different perspective. Because of the role played by entanglement in this system, we can use the algebraic framework to study these problems, since, as we have seen in previous chapters, this formulation provides a suitable tool for understanding and analyzing problems involving entanglement.

We first consider a cyclic and separating vector $|\Omega\rangle$ for the boundary theory, which consists of two CFTs on the right and left sides, and take the thermofield double state $|TFD\rangle_\beta$ for the complete boundary theory. As before, we denote the algebras of the two theories by $\mathcal{M}_R$ and $\mathcal{M}_L$. Let us focus on a single-trace operator O_R belonging to the algebra $\mathcal{M}_R$. In the limit $N \rightarrow \infty$, we have

$$\langle TFD_\beta | O_R | TFD_\beta \rangle = O(N). \quad (6.50)$$

This means that the state $|TFD_\beta\rangle$ is not well-defined for the single-trace operators forming the algebra $\mathcal{S}_R$. We can therefore define modified single-trace operators that remain well-defined in the limit $N \rightarrow \infty$:

$$\forall O_R \in \mathcal{S}_R \mid \hat{O}_R \equiv O_R - \langle TFD_\beta | O_R | TFD_\beta \rangle. \quad (6.51)$$

We then obtain well-defined correlation functions in the semiclassical regime,

$$\langle TFD_\beta | \hat{O}_R | TFD_\beta \rangle = 0 \quad (6.52)$$

$$\langle TFD_\beta | \hat{O}_R^1 \hat{O}_R^2 | TFD_\beta \rangle = O(N^0). \quad (6.53)$$

Thus, in both theories, we can define generalized single-trace operators $\hat{O}_{R,L}$ and corresponding algebras $\mathcal{S}_{R,L}$ (although these algebras are generalized versions of the original algebras, we use the same notation). The complete algebra of single-trace operators in the boundary theory is therefore

$$\mathcal{S} = \mathcal{S}_L \otimes \mathcal{S}_R. \quad (6.54)$$

As we know, $\mathcal{S}_{R,L}$ and $\mathcal{S}$ are C^* -algebras. Hence, given $\mathcal{S}$ and the state $|TFD\rangle_\beta$, we can reconstruct the Hilbert space $\mathcal{H}_{TFD_\beta}^{\text{GNS}}$. We can also show that, using this construction, the algebras associated with the two theories are

$$\begin{aligned} \mathcal{M}_L &= (\pi_\omega(\mathcal{S}_L))'' \quad , \quad \mathcal{M}_R = (\pi_\omega(\mathcal{S}_R))'' = \mathcal{M}'_L \\ \mathcal{B}(\mathcal{H}_{TFD_\beta}^{\text{GNS}}) &= \mathcal{M}_L \vee \mathcal{M}_R. \end{aligned} \quad (6.55)$$

Another important point is that, at finite N , the algebras $\mathcal{M}_{R,L}$ are, as stated in the previous section, von Neumann algebras of type I. However, since the semiclassical limit $N \rightarrow \infty$ is taken in the present system, these algebras become type III₁ algebras [38, 29]. Therefore,

Statement

The Hilbert space associated with the boundary theory, denoted by $\mathcal{H}_{TFD_\beta}^{\text{GNS}}$, is not factorized for temperatures $T > T_{HP}$. Thus, both the factorization problem and the behind-the-horizon problem, which manifests itself as the apparent absence of interaction between the two boundary Hamiltonians, are resolved.

We now turn to the gravitational theory in the bulk. As we know, for $T > T_{HP}$, the bulk spacetime is an eternal AdS black hole. We previously constructed the Hilbert space using the GNS construction, which has a structure analogous to Fock space. Accordingly, by quantizing the fundamental fields in the black-hole geometry, we can construct the Fock space. The vacuum state of the Fock space obtained by quantizing the eternal AdS black-hole geometry is the Hartle–Hawking state, which we shall use. On the other hand, according to Fig. 6.7, for each side of the bulk region, the corresponding algebra can be identified with the algebra of boundary operators on the same side. Thus,

Statement

In the limit $N \rightarrow \infty$, corresponding to $T > T_{HP}$, the algebraic correspondence between the boundary theory and the bulk gravitational theory, which is an eternal AdS black hole, can be written as

$$\mathcal{H}_{TFD_\beta}^{\text{GNS}} \cong \mathcal{H}_{\text{BH}}^{\text{Fock}} \quad (6.56)$$

$$|TFD_\beta\rangle \cong |HH\rangle \quad (6.57)$$

$$\mathcal{M}_{R,L} \cong \widetilde{\mathcal{M}}_{R,L}, \quad (6.58)$$

where $\widetilde{\mathcal{M}}_{R,L}$ denotes the algebra associated with the bulk regions.

Thus, we have seen that the algebraic framework allows us to study the different temperature regimes corresponding to different limits of the number of degrees of freedom. The algebraic approach provides a new formulation through which problems such as the factorization problem and the behind-the-horizon problem can be analyzed and addressed. Several points remain to be emphasized:

- As we know from the AdS/CFT duality, every single-trace operator $O_{R,L}$ in the theories CFT $\ast R, L$ can be expanded in terms of basis functions $v \ast k(x)$,

$$O_{R,L}(x) = \sum_k v^{(R,L)} \ast k(x) a^{(R,L)} \ast k, \quad (6.59)$$

where $a_k^R |TFD_\beta\rangle = \exp(-\frac{\beta\omega}{2}) a_{-k}^L |TFD_\beta\rangle$. The corresponding fundamental bulk fields can be expanded in the basis $u^{(R,L)} \ast k$ as

$$\phi \ast R, L(X) = \sum_k u^{(R,L)} \ast k(X) a^{(R,L)} \ast k. \quad (6.60)$$

““

- Although we already know from the usual holographic description what geometries are dual to the boundary theory at different temperatures, the algebraic viewpoint reveals that for $T < T_{HP}$, the algebra of operators on the two sides of the boundary is a type I von Neumann algebra. In the regime $T > T_{HP}$, on the other hand, it becomes a type III₁ von Neumann algebra.
- As discussed in the previous sections, in the limit $N \rightarrow \infty$, the correspondence between the boundary Hilbert space and the Fock space is specific to a cyclic and separating vector. The same phenomenon occurs here. For $T < T_{HP}$, for every β , one can construct a Hilbert space through the GNS construction that is dual to the bulk Fock space.

However, for $T > T_{HP}$, which corresponds to the limit $N \rightarrow \infty$, for every β the constructed Hilbert space is equivalent to the corresponding Fock space, and none of these Hilbert spaces or Fock spaces has a common intersection with another one (see Sec. 6.1).

- Although we used the GNS construction and modular theory in both temperature regimes, in the regime $T < T_{HP}$ we found that the Hilbert space associated with the boundary theory can be factorized. This allows us to define reduced density operators for the two sides of the boundary theory and, as we have shown, the modular operator takes the form given in Eq. 6.42.

However, for $T > T_{HP}$, although we still use the GNS construction and a modular structure exists, because the algebra under consideration is of type III₁ and a reduced density operator cannot be defined, the modular operator cannot be expressed in the form of Eq. 6.42. This provides a resolution of the behind-the-horizon problem. “

To conclude this section, it is appropriate to emphasize a fundamental point. As we have seen, if we have two boundary theories $\text{CFT}_{R,L}$, then at temperatures $T < T_{HP}$, corresponding to finite N , the bulk geometry consists of two disconnected AdS spacetimes. If we increase the temperature, corresponding to increasing entanglement, an eternal AdS black hole emerges in the bulk. Thus, this provides an example of

Statement

the emergence of geometry in the limit of infinite entanglement.

6.4 Generic Black Hole

We now turn to generic black holes. Unlike the thermofield double setup, these black holes possess additional bulk regions at every temperature. An example of such a black hole is shown in Fig. 6.8. Region I does not belong to either of the boundary theories on the right and left.

In the thermofield double state, by taking the limit $N \rightarrow \infty$, we were able to define generalized single-trace operators and their associated algebra and establish a correspondence between the boundary and bulk algebras:

$$\mathcal{M}_{R,L} = (\pi_\omega(\mathcal{S}_{R,L}))'' = \widetilde{\mathcal{M}}_{R,L}. \quad (6.61)$$

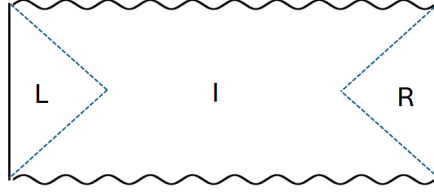


Figure 6.8: A two-sided elongated black hole whose interior region is denoted by I .

Here $\widetilde{\mathcal{M}}_{R,L}$ are the algebras associated with the bulk regions denoted by R and L in the figure. However, unlike the thermofield double case, the regions R and L do not cover an entire Cauchy slice, and the interior region I remains unknown. Therefore, although we expect the operators inside this region, which are also well-defined in the limit $N \rightarrow \infty$, to be encoded in the boundary theory, they do not belong to the algebra generated by the single-trace operators.

Thus, in these systems,

$$\mathcal{S} \subset \mathcal{A}_\Psi, \quad (6.62)$$

where $|\Psi\rangle$ is a cyclic and separating vector with respect to the algebra $\mathcal{A}_\Psi$. We therefore expect the operators located in region I to belong to $\mathcal{A}_\Psi$ but not to $\mathcal{S}$.

For generic black holes, which are not unlike the setups discussed above, we would like to use the algebraic viewpoint. Furthermore, as we saw in the previous section, infinite entanglement leads to the emergence of an eternal AdS black hole. We are therefore interested in exploring the semiclassical regime in what follows.

- **One-sided Black Hole** We first consider a one-sided black hole. The Penrose diagram of this type of black hole is shown below.

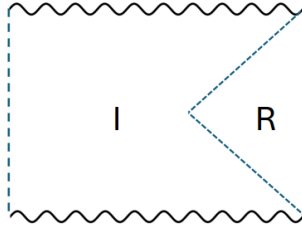


Figure 6.9: A one-sided elongated black hole. The vertical dashed line indicates the location where spacetime either ends smoothly or terminates on an end-of-the-world brane.

As can be seen, the boundary CFT is located on the right, while its dual, namely the bulk gravitational theory, occupies region R . In the boundary theory, many operators remain well-defined in the limit $N \rightarrow \infty$ and generate the algebra $\mathcal{A}_\Psi$. On the other hand, since in the AdS/CFT duality our attention is focused on single-trace operators, these operators generate a subalgebra

$$\mathcal{S} \subset \mathcal{A}_\Psi.$$

Thus, given the state $|\Psi\rangle$ in the boundary theory and constructing the corresponding Hilbert space, we can establish a correspondence between the algebra of boundary single-trace operators,

$$\mathcal{M} = (\pi_\omega(\mathcal{S}))'',$$

and the algebra of the bulk region R , denoted by $\widetilde{\mathcal{M}}$, in the form

$$\mathcal{M} = \widetilde{\mathcal{M}}.$$

However, region I remains unknown! Naturally, the information in this region must be encoded on the boundary, but its precise structure depends on the geometry and physical quantities involved. In other words, it is unlikely that one can make a general statement about region I. The only information we can extract about this region from the algebraic formulation is that the fundamental fields in this region correspond to boundary operators that belong to $\mathcal{A}_\Psi$ but not to the algebra generated by the single-trace operators. Therefore, we may claim that the information contained in the interior region I is not represented by single-trace operators.

At first sight, this conclusion may seem to indicate a failure of the algebraic language to describe the phenomenon. However, precisely this lack of information about region I leads to a new formulation of an important concept in black-hole physics. As we know, in the black-hole information paradox, one approach to this puzzle involves the concept of a firewall [39]. This concept states that if information is encoded in the radiation, then at the black-hole horizon a firewall forms, as illustrated in the Penrose diagram in Fig. 6.10.

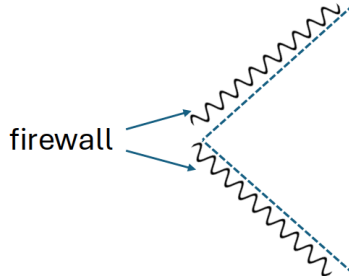


Figure 6.10: Generic states do not possess a smooth geometry beyond the horizon; namely, a firewall is present at the horizon.

The firewall carries extremely high energy, so any object crossing the horizon is immediately destroyed. Therefore, in the presence of this feature, no information can be extracted from inside the black hole, namely from region I. We can regard this phenomenon as analogous to what happens in the algebraic description and write

Statement

Suppose that, in the boundary theory of a one-sided black hole, the algebra of operators that remain well-defined in the limit $N \rightarrow \infty$ is identical to the algebra

of single-trace operators,

$$S = \mathcal{A}_\Psi.$$

Then there are no operators available to describe the interior region I. In this case, the black hole possesses a firewall at its horizon.

Thus, using the algebraic framework, we have been able to formulate the concept of a firewall, which plays an important role in black-hole physics.

6.5 Perturbative $\frac{1}{N}$ Corrections

As we saw in the algebraic formulation of the AdS/CFT duality, in the boundary theory, given a state such as $|\Omega\rangle$, we constructed a Hilbert space through the GNS construction. This Hilbert space has a structure analogous to the Fock space obtained by quantizing the fundamental fields in the bulk gravitational theory. In the limit $N \rightarrow \infty$, only the zeroth- and second-order terms in the action survive. If we wish to incorporate $\frac{1}{N}$ corrections into our calculations [40, 41], the scalar-field action in the bulk can be written as

$$\begin{aligned} \phi_c \rightarrow \phi = \phi_c + k\delta\phi, \quad k = \sqrt{8\pi G_N} \\ S[\phi] = S[\phi_c] + S_2[\delta\phi] + kS_3[\delta\phi] + k^2S_4[\delta\phi] + \dots \end{aligned} \quad (6.63)$$

where

$$S_2[\delta\phi] = -\frac{1}{2} \int d^{d+1}X \sqrt{-g} [(\partial\phi)^2 + m^2\phi^2], \quad S_3[\delta\phi] = -\frac{1}{3} \int d^{d+1}X \sqrt{-g} \phi^3. \quad (6.64)$$

Thus,

$$\begin{aligned} S[\phi] &= S[\phi_c] + S_2[\delta\phi] + kS_3[\delta\phi] + k^2S_4[\delta\phi] + \dots \\ &= S[\phi_c] - \int d^{d+1}X \sqrt{-g} \frac{1}{2} [(\partial\phi)^2 + m^2\phi^2] + \frac{k}{3}\phi^3 + \dots + \frac{k^{N-2}}{N}\phi^N + \dots \end{aligned} \quad (6.65)$$

From the above action, including the $G_N \rightarrow 0$ corrections¹, we can obtain the equation of motion for the scalar field:

$$\frac{\partial S[\phi]}{\partial \phi} - \partial_\mu \left(\frac{\partial S[\phi]}{\partial (\partial_\mu \phi)} \right) = 0 \rightarrow \partial^2 \phi = m^2 \phi + k\phi^2 + \dots \quad (6.66)$$

Expanding the scalar field perturbatively in the parameter k ,

$$\phi = \phi^{(0)} + k\phi^{(1)} + \dots,$$

the above equation of motion becomes

$$\begin{aligned} \phi^{(0)} : (\partial^2 - m^2)\phi^{(0)} &= 0 \\ \phi^{(1)} : (\partial^2 - m^2)\phi^{(1)} &= (\phi^{(0)})^2 \\ &\vdots \end{aligned}$$

¹Since we are working in the gravitational theory, we use G_N in place of N .

The coupled equations above can be solved using a Green's function. Thus,

$$\phi^{(1)}(X) = \int d^{d+1} X' \sqrt{-g} G(X, X') (\phi^{(0)}(X'))^2. \quad (6.67)$$

Here, $G(X, X')$ is the Green's function, i.e. the bulk propagator for the scalar field. As we know from Eq. 6.36, each perturbative order of the field can be expressed as

$$\phi^{(0)}(X) \equiv \phi(X) = \int d^d x K(X; x) \pi_\Omega(O(x)). \quad (6.68)$$

Furthermore,

$$\begin{aligned} \phi^{(1)}(X) &= \int d^{d+1} X' \sqrt{-g} G(X, X') \left(\phi^{(0)}(X') \right)^2 \\ &= \int d^{d+1} X' \sqrt{-g} G(X, X') \int d^d x_1 K(X'; x_1) \pi_\Omega(O(x_1)) \int d^d x_2 K(X'; x_2) \pi_\Omega(O(x_2)) \\ &= \int d^d x_1 d^d x_2 K(X'; x_1, x_2) \pi_\Omega(O(x_1)) \pi_\Omega(O(x_2)), \end{aligned} \quad (6.69)$$

where

$$K(X'; x_1, x_2) \equiv \int d^{d+1} X' \sqrt{-g} G(X, X') K(X'; x_1) K(X'; x_2).$$

Thus, including the G_N corrections, the scalar field in the bulk can be written as

$$\begin{aligned} \phi(X) &= \phi^{(0)}(X) + k\phi^{(1)}(X) + \dots \quad (6.70) \\ &= \int d^d x K(X; x) \pi_\Omega(O(x)) + k \int d^d x_1 d^d x_2 K(X'; x_1, x_2) \pi_\Omega(O(x_1)) \pi_\Omega(O(x_2)) + \dots \\ &= \int d^d x K(X; x) \pi_\Omega(O(x)) + \sum_{n=1}^{\infty} \frac{1}{N^n} \int \prod_{j=1}^{n+1} d^d x_j K_{i_1 \dots i_j}(X; x_1, \dots, x_j) \pi_\Omega(O(x_j)) \end{aligned} \quad (6.71)$$

It is important to note that the operators $\pi_\Omega(O(x))$ appearing in the above expressions are zeroth-order operators. We can therefore apply the corresponding corrections to the boundary operators as well.

12 Show that the perturbative correction to the boundary operator $\pi_\Omega(O(x))$ can be written as

$$\pi_\Omega(O(x)) = \pi_\Omega^0(O(x)) + \sum_{n=1}^{\infty} \frac{1}{N^n} \int \prod_{j=1}^{n+1} d^d x_j \tilde{K}_{i_1 \dots i_j}(X; x_1, \dots, x_j) \pi_\Omega(O(x_j)) \quad (6.72)$$

As can be seen from Eqs. 6.71 and 6.72, these expressions behave exactly as expected in the limit $N \rightarrow \infty$. These results describe the operators and scalar fields in a regime where quantum effects are still present in the bulk and the spacetime geometry is not purely classical.

We can also consider these corrections when a charge and current are present in the boundary theory, corresponding to a gauge field in the gravitational theory. We begin with

the boundary theory. As stated above, the current $J^0(x)$ gives rise to a conserved charge, denoted by Q ,

$$Q = \int_{\sigma} d^{d-1} J^0(x). \quad (6.73)$$

In this case, the single-trace operators on the boundary remain invariant under the presence of such a charge,

$$O(x) = i[Q, O(x)]. \quad (6.74)$$

However, as before, the single-trace operators and the charge are not well-defined in the limit $N \rightarrow \infty$. We therefore need to construct well-defined quantities by dividing them by N . Defining

$$\hat{Q} \equiv \frac{Q}{N}, \quad \hat{O} \equiv \frac{O}{N}, \quad \hat{J}^\mu(x) \equiv \frac{J^\mu}{N},$$

we obtain

$$\hat{O}(x) = \frac{O(x)}{N} = i[\hat{Q}, O(x)]. \quad (6.75)$$

Thus, as is clear, quantum corrections introduce a degree of nonlocality in the boundary theory, since the two operators do not commute.

On the other hand, the presence of a charge in the boundary theory gives rise to a gauge field such as

$$A_M = (A_z, A_\mu)$$

in the bulk gravitational theory. Using the AdS/CFT dictionary, the current generated on the boundary corresponds to the field strength F_{MN} , which is

$$F_{MN} = \partial_M A_N - \partial_N A_M.$$

Thus,

$$\hat{J}^\mu(x) = \lim_{z \rightarrow 0} J^\mu(z, x) = \lim_{z \rightarrow 0} \sqrt{-g} F^{z\mu}(z, x). \quad (6.76)$$

Now suppose that the scalar field $\phi(z, x)$ corresponds to a boundary operator $O(x)$. Then

$$i[\hat{Q}, \phi(z, x)] = \frac{1}{N} \phi(z, x) \equiv k \phi(z, x). \quad (6.77)$$

Therefore, the action in the bulk gravitational theory, including quantum corrections, takes the form

$$S[\phi, A_\mu] = S_2[\delta\phi, \delta A_\mu] + k S_3[\delta\phi, \delta A_\mu] + \dots, \quad k = \sqrt{8\pi G_N}, \quad (6.78)$$

where

$$\begin{aligned} S_2[\delta\phi, \delta A_\mu] &= - \int d^{d+1}x \sqrt{-g} [(\partial\phi)^2 + m^2\phi^2 + \frac{1}{4}F_{MN}F^{MN}] \\ S_3[\delta\phi, \delta A_\mu] &= A_M J^M + \dots \end{aligned} \tag{6.79}$$

$$\begin{aligned} &\vdots \\ J^M &= i(\phi^* \partial^M \phi - \phi \partial^M \phi^*). \end{aligned} \tag{6.80}$$

The equation of motion is then

$$\nabla_M F^{MN} = k J^N + O(k^2) + \dots \tag{6.81}$$

One can then show that

$$A_M(X) = A_M^{(0)}(X) + k \int d^{d+1}X' \sqrt{-g} G_{MN}(X, X') J^N(X') + \dots, \tag{6.82}$$

which behaves in the same way as Eq. 6.71 [42].

Thus, in this chapter we studied the duality between gravitational theories and gauge field theories, AdS/CFT, from the algebraic viewpoint. As we have seen, this new formulation allows us to offer a different perspective on problems such as factorization and, with its help, provide an appropriate response to them. It is important to emphasize that, at this stage, this approach provides only a new formulation of holography. Nevertheless, we expect it to be useful for studying correspondences between subregions of boundary and gravitational theories.

We should also emphasize that the fundamental motivation for using the algebraic language was that in the regime of strong entanglement, $N \rightarrow \infty$, it is fundamentally problematic to assign a single Hilbert space to the system. By contrast, working with operators and their associated algebras turns this limitation into a useful feature. For example, in the thermofield double system at temperatures above T_{HP} , the conventional Hilbert-space-based approach was insufficient to resolve the problem. The algebraic formulation, however, provides the necessary framework for analyzing the system.

Chapter 7

Subregion-Subalgebra Duality

It may have been necessary in this chapter to examine de Sitter space using the algebraic language, but we found it appropriate to introduce an important discussion after the AdS/CFT duality, known as subregion-subalgebra duality. This discussion originates from the previous chapter but is treated here in greater detail. As we understood from the algebraic language of AdS/CFT duality, the boundary CFT theory in the limit $N \rightarrow \infty$ gives rise to the bulk AdS theory. It is therefore natural to ask whether, instead of considering the entire boundary theory, one can take only a portion of this boundary theory and recover the corresponding subregion of the bulk.

This possibility is already partially present in the discussion of entanglement entropy, but in this chapter we review and examine the question more carefully.

7.1 Initial Setup

We begin this section by reviewing what we obtained from the algebraic language of AdS/CFT duality. As we stated, the boundary theory contains numerous well-defined operators in the limit $N \rightarrow \infty$; these operators form an algebra $\mathcal{A}_\Psi$, where $|\Psi\rangle$ is a well-defined state in the semi-classical regime. By restricting attention to single-trace operators only, one can consider a subalgebra of the above algebra that is a C^* -algebra; we denote it by $\mathcal{S}$.

Given the algebras $\mathcal{A}_\Psi$ and $\mathcal{S}$ together with the state $|\Psi\rangle$, one can construct, via the GNS construction, a Hilbert space (built around $|\Psi\rangle$) that we denote by $\mathcal{H}_\Psi^{\text{GNS}}$. Moreover, the state $|\Psi\rangle$ is both cyclic and separating for the algebra $\mathcal{A}_\Psi(\mathcal{S})$. Consequently one has

$$\mathcal{B}(\mathcal{H}_\Psi^{\text{GNS}}) = (\pi_\omega(\mathcal{A}_\Psi))''. \quad (7.1)$$

In the bulk gravitational theory, by quantizing the fundamental fields and acting with the resulting creation and annihilation operators on the vacuum $|0\rangle$, one reconstructs a Fock space that we denote by $\mathcal{H}_\Psi^{\text{Fock}}$.¹ The duality between the boundary theory and the gravitational theory implies

$$\mathcal{H}_\Psi^{\text{GNS}} \cong \mathcal{H}_\Psi^{\text{Fock}}. \quad (7.2)$$

¹Usually the index $|\Psi\rangle$ is used instead of the vacuum state.

Hence the vacuum state in AdS space may be identified with the cyclic and separating vector $|\Psi\rangle$. We also have

$$\mathcal{B}(\mathcal{H}_\Psi^{\text{GNS}}) = \mathcal{B}(\mathcal{H}_\Psi^{\text{Fock}}). \quad (7.3)$$

Furthermore, in the limit $N \rightarrow \infty$ every Hilbert space constructed via the GNS construction corresponds to the same Fock space and they share no common points (see the review in Sec. 6.1). They can therefore be written as a tensor product of factors:

$$\begin{aligned} \mathcal{H}_{\Psi,i}^{\text{GNS}} &\cong \mathcal{H}_{\Psi,i}^{\text{Fock}}, & \mathcal{H}_\Psi^{\text{GNS}} &= \bigotimes_i \mathcal{H}_{\Psi,i}^{\text{GNS}}, \\ \mathcal{B}(\mathcal{H}_{\Psi,i}^{\text{GNS}}) &\cong \mathcal{B}(\mathcal{H}_{\Psi,i}^{\text{Fock}}), & \mathcal{B}(\mathcal{H}_\Psi^{\text{GNS}}) &= \bigotimes_i \mathcal{B}(\mathcal{H}_{\Psi,i}^{\text{GNS}}). \end{aligned} \quad (7.4)$$

The essential content of Eq. (7.3) is that the subalgebras acting on $\mathcal{H}_\Psi^{\text{GNS}}$ and on $\mathcal{H}_\Psi^{\text{Fock}}$ must be in one-to-one correspondence. In particular, let $\mathfrak{b}$ be an open subregion on a Cauchy slice in the bulk, and let $\widetilde{\mathcal{M}}_{\mathfrak{b}} \subset \mathcal{B}(\mathcal{H}_\Psi^{\text{Fock}})$ be the von Neumann algebra associated with this region. Then there must exist a subalgebra of the boundary theory, $\mathcal{M}_{\mathfrak{b}} \subset \mathcal{B}(\mathcal{H}_\Psi^{\text{GNS}})$, such that

$$\widetilde{\mathcal{M}}_{\mathfrak{b}} = \mathcal{M}_{\mathfrak{b}}. \quad (7.5)$$

Thus the boundary subalgebra $\mathcal{M}_{\mathfrak{b}}$ contains all the quantum information and the local and causal properties that the bulk algebra $\widetilde{\mathcal{M}}_{\mathfrak{b}}$ possesses. Consequently,

Statement

If a bulk gravitational subregion $\mathfrak{b}$ with algebra $\widetilde{\mathcal{M}}_{\mathfrak{b}}$ is equivalent to a boundary subalgebra $\mathcal{M}_{\mathfrak{b}}$, then relation (7.5) holds. The meaning of this equivalence is that the bulk region $\mathfrak{b}$ can be reconstructed from the boundary subalgebra $\mathcal{M}_{\mathfrak{b}}$. This correspondence is called subregion-subalgebra duality [30].

As we remarked in the previous chapter, if the subregion $\mathfrak{b}$ is taken to be the entire region, the correspondence reduces to ordinary AdS/CFT duality, to the two-sided thermal state, and to the study of generic black holes. The equality (7.5) shows that the boundary subalgebra $\mathcal{M}_{\mathfrak{b}}$ reconstructs every bulk operation that can be performed inside $\mathfrak{b}$. Moreover, as argued in the previous chapter, $\mathcal{M}_{\mathfrak{b}}$ also encodes the causal structure, geometric information, and the entanglement structure of the region $\mathfrak{b}$.

For example, the entanglement of a bulk subregion $\mathfrak{b}$ with its causal complement is described by the spectrum of the modular operators (or modular Hamiltonians), which can be studied through modular flows. Subregion-subalgebra duality (7.5) identifies the bulk modular operator belonging to $\widetilde{\mathcal{M}}_{\mathfrak{b}}$ with the boundary modular operator belonging to $\mathcal{M}_{\mathfrak{b}}$. Consequently the entanglement structure of the bulk can be extracted from the modular flows of $\mathcal{M}_{\mathfrak{b}}$.

Because of the limit $G_N \rightarrow 0$ in the bulk theory, the bulk may be regarded as a quantum field theory (without the relativistic qualifier, as previously noted). Accordingly the bulk subregion $\mathfrak{b}$ with algebra $\mathcal{M}_{\mathfrak{b}}$ inherits the properties discussed for quantum field theory (see the review in Chap. 5):

1. In quantum field theory the Reeh–Schlieder theorem asserts that if $\mathfrak{b}$ is an open region, then the vacuum $|0\rangle \in \mathcal{H}_\Psi^{\text{Fock}}$ is cyclic and separating for the algebra $\widetilde{\mathcal{M}}_{\mathfrak{b}}$. By the correspondence (7.5) the state $|\Psi\rangle \in \mathcal{H}_\Psi^{\text{GNS}}$ is therefore cyclic and separating for $\mathcal{M}_{\mathfrak{b}}$.
2. As we saw in Chap. 5, causality in quantum field theory implies that for any region $\mathfrak{b}$

$$\widetilde{\mathcal{M}}_{\mathfrak{b}} = \widetilde{\mathcal{M}}_{\hat{\mathfrak{b}}}, \quad (7.6)$$

where $\hat{\mathfrak{b}}$ denotes the causal completion of $\mathfrak{b}$. Subregion-subalgebra duality therefore associates this causal region with a boundary subalgebra:

$$\widetilde{\mathcal{M}}_{\mathfrak{b}} = \widetilde{\mathcal{M}}_{\hat{\mathfrak{b}}} = \mathcal{M}_{\mathfrak{b}}. \quad (7.7)$$

Hence

Statement

From the subalgebra $\mathcal{M}_{\mathfrak{b}}$ one can reconstruct the subalgebras corresponding to the causal regions $\hat{\mathfrak{b}}$.

3. Since $\widetilde{\mathcal{M}}_{\mathfrak{b}}$ is a subalgebra of a bulk quantum-field-theory region, and every such algebra is a type-III₁ von Neumann algebra, relation (7.5) implies that the boundary subalgebra $\mathcal{M}_{\mathfrak{b}}$ is likewise of type III₁.
4. By the Haag duality for regions of trivial topology one has

$$\widetilde{\mathcal{M}}_{\mathfrak{b}'} = (\widetilde{\mathcal{M}}_{\mathfrak{b}})'. \quad (7.8)$$

5. Additivity in quantum field theory (again for regions of trivial topology) yields

$$\widetilde{\mathcal{M}}_{\mathfrak{b}_1 \cup \mathfrak{b}_2} = \widetilde{\mathcal{M}}_{\mathfrak{b}_1} \vee \widetilde{\mathcal{M}}_{\mathfrak{b}_2}. \quad (7.9)$$

Thus the duality itself generates bulk subregions from information encoded in the boundary theory. One deep consequence is that the causal structure of the bulk is encoded in the commutation relations of the boundary algebras. For instance, if two regions O_1 and O_2 are spacelike separated, the corresponding boundary algebras $\mathcal{A}_{O_1}$ and $\mathcal{A}_{O_2}$ satisfy

$$[\mathcal{A}_{O_1}, \mathcal{A}_{O_2}] = 0. \quad (7.10)$$

By subregion-subalgebra duality this is precisely the algebraic statement of locality in quantum field theory. Therefore

Statement

Causality in the bulk emerges from the commutation structure on the boundary.

In general, however, one cannot choose an arbitrary region and reconstruct it in the bulk. The sufficiency and completeness of the boundary operators in the limit $N \rightarrow \infty$ are essential. In the next section we examine the precise conditions under which reconstruction is possible.

7.2 Entanglement-Wedge Reconstruction: Algebraic Formulation

For subregion-subalgebra duality there exist two classes of reconstruction: entanglement-wedge reconstruction [43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54] and causal-entanglement-wedge reconstruction [55, 56, 57, 58, 59, 40, 42, 60, 61, 62, 63]. These two notions answer the question of which region of the bulk geometry can be reconstructed solely from the algebra of operators associated with a given boundary region. We now define each of them.

- **Entanglement-wedge reconstruction.** We examine the reconstruction purely in algebraic language. For this purpose we assume familiarity with the notions that appear in the physics of entanglement entropy. Briefly, in the boundary CFT one considers an arbitrary region A . The entanglement entropy of the boundary region A , denoted S_A , is

$$S_A = \frac{\text{Area}(\gamma_A)}{4G_N}, \quad (7.11)$$

where γ_A is a surface that

1. lies in the bulk,
2. lies on a Cauchy slice of the bulk,
3. has minimal area among all surfaces that end on ∂A .

This surface is called the Ryu–Takayanagi surface. As illustrated in Fig. 7.1, the surface γ_A bounds a wedge-shaped region $\mathfrak{b}_A$ in the bulk. The entanglement entropy of the boundary region A therefore carries information about the bulk region $\mathfrak{b}_A$.

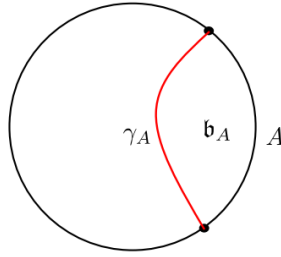


Figure 7.1: A constant-time Cauchy slice Σ of an asymptotically static AdS spacetime is shown, with the boundary indicated by the black circle. γ_A is the Ryu–Takayanagi surface for the boundary region A , and the region between γ_A and A is the subregion $\mathfrak{b}_A$ associated with A . The entanglement wedge $\hat{\mathfrak{b}}_A$ (not shown in the figure) is the domain of dependence of $\mathfrak{b}_A$.

If one respects the initial setup of Sec. 7.1 and takes the limit $N \rightarrow \infty$, one can associate with the boundary region A a subalgebra $\mathcal{X}_A \subset \mathcal{B}(\mathcal{H}_\Psi^{\text{GNS}})$. One then has

$$\mathcal{X}_A = \widetilde{\mathcal{M}}_{\mathfrak{b}_A}. \quad (7.12)$$

Because of causality one may replace the algebra of $\mathfrak{b}_A$ by that of its causal completion $\hat{\mathfrak{b}}_A$, writing $\mathcal{X}_A = \widetilde{\mathcal{M}}_{\mathfrak{b}_A} = \widetilde{\mathcal{M}}_{\hat{\mathfrak{b}}_A}$.

We now examine the algebra of operators supported on the boundary region A . The algebra $\mathcal{X}_A$ is defined in the limit $N \rightarrow \infty$. At finite N the algebra of local operators on A is denoted $\mathcal{B}_A^{(N)}$ and acts on the CFT Hilbert space $\mathcal{H}_{\text{CFT}}$. Causality implies

$$\mathcal{B}_A^{(N)} = \mathcal{B}_{\hat{A}}^{(N)}. \quad (7.13)$$

In the limit $N \rightarrow \infty$ the operators that constitute $\mathcal{B}_A^{(N)}$ are no longer well-defined. Consequently the naive identification $\mathcal{X}_A = \lim_{N \rightarrow \infty, \Psi} \mathcal{B}_A^{(N)}$ is in general incorrect. Following [30] we therefore define

$$\mathcal{X}_A = \left(\pi_\Psi \left(\lim_{N \rightarrow \infty, \Psi} \mathcal{B}_A^{(N)} \right) \right)'' = \mathcal{X}_{\hat{A}}. \quad (7.14)$$

As recalled from earlier chapters, all operators that remain well-defined as $N \rightarrow \infty$ generate the algebra $\mathcal{A}_\Psi$. Hence $\lim_{N \rightarrow \infty, \Psi} \mathcal{B}_A^{(N)} \subset \mathcal{A}_\Psi$.

Statement

From the entanglement entropy of a boundary region A one obtains, via the Ryu–Takayanagi surface γ_A , a bulk region $\mathfrak{b}_A$. This entropy encodes information shared by the two regions. The algebra $\mathcal{X}_A$ associated with A is then equal to the algebra of the bulk region $\mathfrak{b}_A$:

$$\mathcal{X}_A = \widetilde{\mathcal{M}}_{\mathfrak{b}_A}. \quad (7.15)$$

This reconstruction is called entanglement-wedge reconstruction.

Relation (7.15) is precisely the statement of subregion-subalgebra duality realized by entanglement-wedge reconstruction.

- **Causal-entanglement-wedge reconstruction.** The second type of reconstruction relies on causality. In AdS/CFT we showed that if $\mathcal{S}$ denotes the algebra of all single-trace operators that remain well-defined as $N \rightarrow \infty$, then $\mathcal{Y} = (\pi_\Psi(\mathcal{S}))''$. For a boundary subregion A with single-trace algebra $\mathcal{S}_A$ one likewise has

$$\mathcal{Y}_A = (\pi_\Psi(\mathcal{S}_A))'' = \mathcal{Y}_{\hat{A}}. \quad (7.16)$$

Thus one may introduce a second algebra $\mathcal{Y}_{\hat{A}}$ associated with A that is well-defined as $N \rightarrow \infty$. Since this algebra is generated by single-trace operators only, one necessarily has

$$\mathcal{Y}_{\hat{A}} \subseteq \mathcal{X}_A. \quad (7.17)$$

Consequently

Statement

Causal-entanglement-wedge reconstruction asserts that local operators inside the causal wedge of the region $\hat{A}$ can be expressed in terms of the single-trace opera-

tors on $\hat{A}$ that generate the algebra $\mathcal{Y}_{\hat{A}}$. Therefore

$$\mathcal{Y}_{\hat{A}} = \widetilde{\mathcal{M}}_{\mathcal{C}_{\hat{A}}}. \quad (7.18)$$

More explicitly, if $\phi(X)$ is a bulk field inside the causal wedge of A , one has

$$\phi(X) = \int_{\hat{A}} d^d x K_A^{(C)}(X; x) \pi_{\Psi}(O(x)), \quad X \in \mathcal{C}_{\hat{A}}, \quad x \in \hat{A}, \quad (7.19)$$

where $O(x)$ is the single-trace operator dual to $\phi(X)$ and

$$\mathcal{C}_{\hat{A}} = (\tilde{\mathcal{J}}^+(\hat{A}) \cap \tilde{\mathcal{J}}^-(\hat{A}))''$$

is the causal wedge of $\hat{A}$. The kernel $K_A^{(C)}$ is the same as previously. Because of the one-to-one correspondence between the single-trace operators inside $\mathcal{Y}_{\hat{A}}$ and the fundamental fields reconstructed by the above integral, one obtains

$$\mathcal{Y}_{\hat{A}} = \widetilde{\mathcal{M}}_{\mathcal{C}_{\hat{A}}}. \quad (7.20)$$

Thus one again has a subregion-subalgebra duality, now realized through causal regions.

Although the presence of the causal region makes the analysis more delicate than in the previous case, special instances of this reconstruction were already encountered in Chap. 6. For the two-sided thermal state the right and left boundaries R and L possess causal regions $\hat{R}$ and $\hat{L}$ that correspond to bulk regions, yielding

$$\mathcal{Y}_{R,L} = \widetilde{\mathcal{M}}_{\mathcal{C}_{R,L}} \equiv \widetilde{\mathcal{M}}_{R,L}. \quad (7.21)$$

Combining (7.17) with the entanglement-wedge reconstruction (7.15) one finds $\widetilde{\mathcal{M}}_{\mathcal{C}_{\hat{A}}} \subset \widetilde{\mathcal{M}}_{\hat{\mathbf{b}}_A}$, and therefore

$$\mathcal{C}_{\hat{A}} \subseteq \hat{\mathbf{b}}_A. \quad (7.22)$$

Thus, to reconstruct a bulk region from boundary data one may employ either entanglement-wedge or causal-entanglement-wedge reconstruction. The algebras associated with a given boundary region A satisfy the inclusion (7.17), which implies (7.22). The inclusion (7.17) recalls a lemma stated in Chap. 4.4: if two von Neumann algebras satisfy $\mathcal{Y} \subset \mathcal{X}$ and both admit a common cyclic and separating vector $|\Psi\rangle$, then the modular flow of $\mathcal{X}$ acting on $\mathcal{Y}$ generates the whole of $\mathcal{X}$.

This lemma relates the two reconstructions in a striking way. The algebra $\mathcal{Y}_{\hat{A}}$ is generated by single-trace operators that remain well-defined as $N \rightarrow \infty$, while all well-defined operators generate $\mathcal{X}_A$. With the common cyclic and separating vector $|\Psi\rangle$ the modular flow of $\mathcal{X}_A$ can therefore produce the full set of well-defined operators from the single-trace ones. Concretely, consider a boundary region A . As shown in Fig. 7.2, $\hat{A}$ is the causal completion of A . One may introduce an infinitesimal time band of width ε about A and denote by $\mathcal{Y}_{\hat{A}}^{\varepsilon}$ the algebra generated by single-trace operators inside that band. Both $\mathcal{Y}_{\hat{A}}^{\varepsilon} \subset \mathcal{X}_A$ and $\mathcal{Y}_{\hat{A}}$ admit the same cyclic and separating vector $|\Psi\rangle$. The lemma then implies that $\mathcal{X}_A$ is

generated by the modular flow of $\mathcal{X}_A$ acting on $\mathcal{Y}_{\hat{A}}^\varepsilon$:

$$\mathcal{X}_A = \{O(s, x) \equiv \Delta_{\mathcal{X}_A}^{-is} O_\varepsilon(x) \Delta_{\mathcal{X}_A}^{+is}, \quad x \in A, s \in \mathbb{R}\}. \quad (7.23)$$

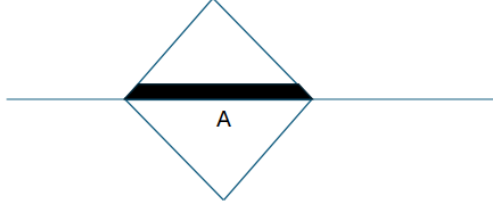


Figure 7.2: Consider an infinitesimal time band of width ε around A inside $\hat{A}$, indicated by the shaded region, and denote the algebra generated by single-trace operators in this strip by $\mathcal{Y}_{A_\varepsilon}$. The semi-classical state under consideration is cyclic and separating for both $\mathcal{Y}_{A_\varepsilon} \subset \mathcal{X}_A$ and $\mathcal{Y}_A$. The lemma then implies that $\mathcal{X}_A$ can be generated by acting with the modular flow operators associated with $\mathcal{X}_A$ on $\mathcal{Y}_{A_\varepsilon}$, yielding Eq. (7.23), in which $O(\vec{x})$ is to be understood as smeared inside the time band (local operators must be smeared in order to be well-defined).

Applying the representation π_Ψ to the right-hand side and invoking (7.5) reconstructs the fundamental bulk fields inside $\mathfrak{b}_A$ [50]:

$$\begin{aligned} \phi(X) \in \widetilde{\mathcal{M}}_{\mathfrak{b}_A} &= \pi_\Psi(O(s, x)) = \Delta_{\mathcal{X}_A}^{-is} O_\varepsilon \Delta_{\mathcal{X}_A}^{+is} \\ &= \int_{-\infty}^{+\infty} ds \int dx K_A^{(E)}(X; s, x) O(s, x), \quad X \in \hat{\mathfrak{b}}_A. \end{aligned} \quad (7.24)$$

The same relation can be examined at finite N . The local-operator algebra on A is still $\mathcal{B}_A^{(N)}$, for which $|\Psi\rangle$ remains cyclic and separating, so a modular operator $\Delta_{A, \Psi}$ exists. Combining (7.14) and (7.23) yields [30]

$$\pi_\Psi\left(\lim_{N \rightarrow \infty, \Psi} \Delta_{A, \Psi}^{-is} O(x) \Delta_{A, \Psi}^{+is}\right) = \Delta_{\mathcal{X}_A}^{-is} \pi_\Psi(O(x)) \Delta_{\mathcal{X}_A}^{+is}. \quad (7.25)$$

The left-hand side produces operators that belong to $\mathcal{X}_A$ but not to $\mathcal{Y}_{\hat{A}}$. When the modular operator $\Delta_{A, \Psi}$ acts geometrically one obtains the equality

$$\mathcal{Y}_{\hat{A}} = \mathcal{X}_A, \quad \mathcal{C}_{\hat{A}} = \mathfrak{b}_A. \quad (7.26)$$

This occurs, for example, when A is a half-space or a spherical region, and also for the full right or left boundary in the two-sided thermal state. In general, however, $\Delta_{A, \Psi}$ need not be geometric and one has only the inclusion $\mathcal{Y}_{\hat{A}} \subseteq \mathcal{X}_A$.

At finite N the local-operator algebra on a boundary region A is $\mathcal{B}_A^{(N)}$. Because the Hilbert space factorizes,

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_{\bar{A}}, \quad (7.27)$$

this algebra is of type I. Taking the limit $N \rightarrow \infty$ produces the type-III₁ algebra $\mathcal{X}_A$. By

the entanglement-wedge reconstruction (7.12) the latter is identified with the bulk algebra of $\mathfrak{b}_A$, which is also of type III₁. The same statement holds for $\mathcal{Y}_{\hat{A}}$.

This raises the question

Question

We have seen that the modular operator $\Delta_{\mathcal{X}_A}$ acting on $\mathcal{Y}_{\hat{A}}$ generates the whole of $\mathcal{X}_A$. Can one therefore discuss entanglement-wedge reconstructions without any geometric input?

The question is equivalent to asking whether spacetime itself is constructed solely from quantum information—an important and deep issue. The answer is affirmative: these reconstructions are independent of any geometric description of the bulk gravitational theory. One can go further and assert that the boundary region algebras need not correspond to any geometric bulk region at all. The algebraic viewpoint therefore opens the door to situations in which geometric limitations prevent a conventional analysis.

As an illustration consider the system $B \cup R$ consisting of a gravitational subsystem B and a non-gravitational subsystem R . The non-gravitational subsystem R may be treated as fully quantum for any value of G_N . The gravitational subsystem B , however, is well-defined only in the semi-classical limit $G_N \rightarrow 0$. Their mutual entanglement therefore appears paradoxical. From the algebraic perspective one defines, for any subsystem $Q \subset B \cup R$,

$$\mathcal{X}_Q \equiv \lim_{G_N \rightarrow 0, \Psi} \mathcal{B}_Q. \quad (7.28)$$

The definition is identical in form to (7.14) and carries no geometric data. The same construction will be used later for the algebraic treatment of black-hole evaporation, in which the black hole plays the role of B and the Hawking radiation the role of R .

7.3 Quantum-Information Aspects of Entanglement-Wedge Reconstruction

We now examine the quantum-information aspects of entanglement-wedge reconstruction from the algebraic viewpoint. One of the most important geometric ingredients is the Ryu–Takayanagi surface γ_A [6, 46]. For nested boundary regions $A_1 \subseteq A_2$ the corresponding bulk regions satisfy $\hat{\mathfrak{b}}_{A_1} \subseteq \hat{\mathfrak{b}}_{A_2}$. Entanglement-wedge reconstruction (7.12) then implies

$$\hat{\mathfrak{b}}_{A_1} \subseteq \hat{\mathfrak{b}}_{A_2} \implies \mathcal{X}_{A_1} \subseteq \mathcal{X}_{A_2} \implies \widetilde{\mathcal{M}}_{\hat{\mathfrak{b}}_{A_1}} \subseteq \widetilde{\mathcal{M}}_{\hat{\mathfrak{b}}_{A_2}}.$$

All of the subregion-subalgebra dualities under discussion hold in the limit $N \rightarrow \infty$. In that limit the bulk is semi-classical and may be treated as a quantum field theory. In Chap. 5 we discussed the additivity property of quantum field theory. With the aid of entanglement-wedge reconstruction we can now examine the same property in the holographic setting. Consider two boundary regions A_1 and A_2 . As illustrated in Fig. 7.3, entanglement-wedge reconstruction yields the inclusions

$$\mathfrak{b}_{A_1} \cup \mathfrak{b}_{A_2} \subseteq \mathfrak{b}_{A_1 \cup A_2}, \quad \mathfrak{b}_{A_1 \cap A_2} \subseteq \mathfrak{b}_{A_1} \cap \mathfrak{b}_{A_2}. \quad (7.29)$$

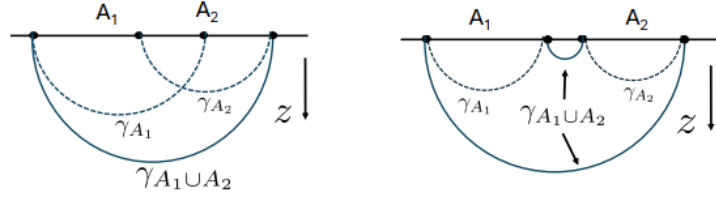


Figure 7.3: Examples of superadditivity of entanglement wedges (Eq. (7.29)) in AdS/CFT, shown on a constant-time Cauchy slice in the Poincaré patch. On the right: two overlapping intervals A_1 and A_2 . On the left: two disjoint intervals A_1 and A_2 that are sufficiently close to each other. The Ryu–Takayanagi surfaces for $A_1 \cup A_2$ are drawn as solid lines.

Passing to the associated algebras one obtains

$$\widetilde{\mathcal{M}}(\mathfrak{b}_{A_1} \cup \mathfrak{b}_{A_2}) \subseteq \widetilde{\mathcal{M}}(\mathfrak{b}_{A_1 \cup A_2}) \implies \widetilde{\mathcal{M}}(\mathfrak{b}_{A_1}) \vee \widetilde{\mathcal{M}}(\mathfrak{b}_{A_2}) \subseteq \widetilde{\mathcal{M}}(\mathfrak{b}_{A_1 \cup A_2}), \quad (7.30)$$

$$\widetilde{\mathcal{M}}(\mathfrak{b}_{A_1 \cap A_2}) \subseteq \widetilde{\mathcal{M}}(\mathfrak{b}_{A_1} \cap \mathfrak{b}_{A_2}) \implies \widetilde{\mathcal{M}}(\mathfrak{b}_{A_1 \cap A_2}) \subseteq \widetilde{\mathcal{M}}(\mathfrak{b}_{A_1}) \wedge \widetilde{\mathcal{M}}(\mathfrak{b}_{A_2}). \quad (7.31)$$

By (7.12) the corresponding boundary algebras in the limit $N \rightarrow \infty$ satisfy

$$\mathcal{X}_{A_1} \vee \mathcal{X}_{A_2} \subseteq \mathcal{X}_{A_1 \cup A_2}, \quad \mathcal{X}_{A_1 \cap A_2} \subseteq \mathcal{X}_{A_1} \wedge \mathcal{X}_{A_2}. \quad (7.32)$$

At finite N the algebras $\mathcal{X}_{A_i}$ become $\mathcal{B}_{A_i}^{(N)}$ and ordinary additivity holds:

$$\mathcal{B}_{A_1 \cup A_2}^{(N)} = \mathcal{B}_{A_1}^{(N)} \vee \mathcal{B}_{A_2}^{(N)}. \quad (7.33)$$

Comparing (7.32) with (7.33) one sees that additivity is violated in the limit $N \rightarrow \infty$ [30, 64]. In other words, knowledge of the operators on two regions separately does not give complete access to the operators on their union. Therefore

Statement

In the limit $N \rightarrow \infty$ the additivity property (7.33) of the boundary theory (or of quantum field theory) is violated. Relation (7.32) shows that separate access to two regions does not exhaust the algebra of their union; one obtains the stronger property of superadditivity.

Additivity is a strong form of locality. Its violation in the large- N limit therefore implies that the notion of locality becomes ambiguous:

Statement

In ordinary quantum field theory all operators and quantum information are local by virtue of additivity. In the limit $N \rightarrow \infty$, where additivity fails, the information becomes non-local.

Haag duality continues to hold. At finite N one has $\mathcal{B}_{\bar{A}}^{(N)} = (\mathcal{B}_A^{(N)})'$. Taking the limit $N \rightarrow \infty$ and using (7.14) yields $\mathcal{X}_{\bar{A}} = (\mathcal{X}_A)'$. Entanglement-wedge reconstruction then implies

$$\mathcal{X}_{\bar{A}} = \widetilde{\mathcal{M}}_{\mathfrak{b}_{\bar{A}}} = (\widetilde{\mathcal{M}}_{\mathfrak{b}_A})' = \widetilde{\mathcal{M}}_{\bar{\mathfrak{b}}_A} = (\mathcal{X}_A)', \quad (7.34)$$

where the pure-state identification $\bar{\mathfrak{b}}_A = \mathfrak{b}_{\bar{A}}$ has been used. For the single-trace algebras $\mathcal{Y}_{\bar{A}}$ additivity still holds, but Haag duality fails.

The failure of additivity and the consequent non-locality have an important consequence:

Statement

If, in subregion-subalgebra duality, information in the bulk is stored non-locally, then the loss of a portion of the boundary does not erase that information.

This is illustrated in Fig. 7.3. Operators supported inside the region bounded by $\gamma_{A_1 \cup A_2}$ but outside the regions bounded by γ_{A_1} and γ_{A_2} are precisely those that appear because of the violation of additivity; they are distributed non-locally on the boundary. Superadditivity therefore implies

Statement

Non-locality of information increases the amount of accessible information.

Fig. 7.4 shows two complementary situations. On the left an operator cannot be reconstructed from either $\mathcal{X}_{A_1}$ or $\mathcal{X}_{A_2}$ separately, yet it can be reconstructed from their intersection. On the right an operator that lives in the intersection can be reconstructed from either algebra alone. The existence of such a common region that houses a set of operators is a manifestation of redundancy, which again follows from non-locality:

Statement

Non-locality of information produces redundancy in the common region.

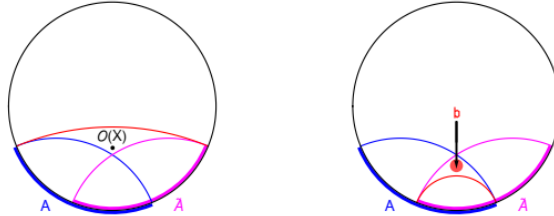


Figure 7.4: Left: the indicated operator cannot be reconstructed separately from the algebras associated with the subregions A_1 and A_2 . Right: the operator can also be reconstructed separately from either of the boundary algebras. This occurs because of superadditivity.

All of these features are reminiscent of quantum error correction [47]. In that context information is stored so that it remains recoverable even after part of the system is erased. In holographic entanglement-wedge reconstruction the same phenomenon occurs because of the large- N limit and the resulting non-locality; the situation can therefore be analysed from the viewpoint of quantum error correction. Concretely, suppose a fundamental field $\phi(X)$ is supported in a common bulk region $\mathfrak{b}$. Using the GNS representation associated with a common semi-classical state $|\Psi\rangle$ one has, for operators $O_{A_1} \in \mathcal{X}_{A_1}$ and $O_{A_2} \in \mathcal{X}_{A_2}$,

$$\phi(X) = \pi_{\Psi}(O_{A_1}(x)) = \pi_{\Psi}(O_{A_2}(x)), \quad (7.35)$$

where

$$O_{A_i}(x) = \int_{\hat{A}_i} d^d x K_{A_i}^{(C)}(X; x) O(x). \quad (7.36)$$

Thus the same bulk information can be reconstructed from infinitely many different boundary regions. The same algebraic language can be used to formulate the island rule, but we defer concrete physical applications to a separate section.

7.4 Reconstruction of Causal Bulk Regions from the Boundary

We have seen that AdS/CFT duality, in the algebraic language, identifies a bulk subregion with a boundary subalgebra via either entanglement-wedge or causal-entanglement-wedge reconstruction. In the standard causal reconstruction the causal wedge of a boundary region A is

$$C_{\hat{A}} = (\mathcal{J}^+(\hat{A}) \cap \mathcal{J}^-(\hat{A}))''. \quad (7.37)$$

The algebraic formulation is deeper: the modular flow of $\mathcal{Y}_{\hat{A}}$ generates a von Neumann algebra $\mathcal{X}_A$ without any geometric input on the boundary. We now examine the same idea for causal diamonds.

A causal diamond in the bulk is the set

$$D(p, q) = \mathcal{J}^+(p) \cap \mathcal{J}^-(q), \quad (7.38)$$

where p and q are timelike-related points with $q > p$. Geometrically it consists of all points that can both be influenced by p and influence q .

Take the vacuum (identified with the cyclic and separating vector $|\Psi\rangle$) of global AdS. Let I_ω be a time band of width $\omega < \pi R$ on the boundary, corresponding to the interval $t \in (-\omega/2, +\omega/2)$. Its causal wedge is

$$\mathcal{W}_{\rho_\omega} = \tilde{\mathcal{J}}^+(I_\omega) \cap \tilde{\mathcal{J}}^-(I_\omega),$$

where

$$\rho_\omega = R \tan\left(\frac{\pi}{2} - \frac{\omega}{2R}\right). \quad (7.39)$$

Causal reconstruction asserts that $\mathcal{W}_{\rho_\omega} = \mathcal{W}_{\rho_\omega}''$ and that the algebra of the time band satisfies [30, 65]

$$\mathcal{Y}_{I_\omega} = \widetilde{\mathcal{M}}_{\mathcal{W}_{\rho_\omega}}. \quad (7.40)$$

If $\omega \geq \pi R$ the causal wedge covers an entire Cauchy slice and $\mathcal{W}_{\rho_\omega}''$ is the whole of global AdS; consequently $\mathcal{Y}_{I_\omega} = \mathcal{Y}_B$.

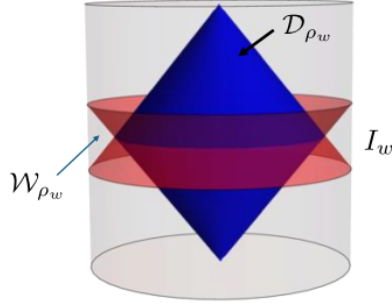


Figure 7.5: $\mathcal{W}_{\rho_w}$ and $\mathcal{D}_{\rho_w}$ are respectively the bulk duals of the time-band algebra $\mathcal{Y}_{I_w}$ and of its commutant $\mathcal{Y}'_{I_w}$ in the vacuum sector.

Taking the commutant of both sides of (7.40) and using Haag duality (7.8) yields

$$\mathcal{Y}'_{I_w} = \widetilde{\mathcal{M}}'_{\mathcal{W}_{\rho_w}} = \widetilde{\mathcal{M}}_{\mathcal{D}_{\rho_w}}, \quad (7.41)$$

where $\mathcal{W}'_{\rho_w} = \mathcal{D}_{\rho_w}$ (see Fig. 7.5). Thus the commutant of the time-band algebra is identified with the algebra of a bulk region $\mathcal{D}_{\rho_w}$ that does not reach the boundary. Therefore

Statement

In global AdS some causal diamonds with $\omega < \pi R$ never reach the boundary (Fig. 7.5). There is no geometric boundary subregion whose dual is precisely this bulk region. The bulk subregion therefore corresponds to a von Neumann algebra that has no obvious geometric interpretation on the boundary.

Previously every bulk subregion O was paired with a geometric boundary algebra $\mathcal{A}_O$. The observation above shows that $\mathcal{A}_O$ need not admit a local geometric interpretation on the boundary. In the ordinary causal reconstruction one expects a bulk field $\phi(X)$ inside the causal wedge of A to belong to $\mathcal{A}_A$. In the generalized reconstruction any suitable von Neumann subalgebra of the full boundary algebra may correspond to a causal bulk region:

Statement

The notion of reconstruction is generalized to

$$O \longleftrightarrow \mathcal{A}_O \subseteq \mathcal{A}_B, \quad (7.42)$$

where $\mathcal{A}_B$ is the algebra of the entire boundary. The causal structure of the bulk is thereby extracted directly from the algebraic structure of the boundary.

As the width ω approaches πR , more operators enter $\mathcal{Y}_{I_w}$, the commutant $\mathcal{Y}'_{I_w}$ shrinks, and the dual region $\mathcal{D}_{\rho_w}$ becomes smaller.

7.5 Generalized Entropy at Finite N

We finally discuss the definition of entropy from the viewpoint of subregion-subalgebra duality at finite N [66, 67]. In earlier chapters we introduced relative entropy as the algebraic

quantity that replaces ordinary entropy for type-II and type-III von Neumann algebras. In AdS/CFT the large- N limit corresponds to the classical limit $G_N \rightarrow 0$ of the bulk. The bulk is then a quantum field theory whose algebras are of type III. For the two-sided thermal state one finds

$$\begin{aligned}\mathcal{Y}_{R,L} &= \mathcal{B}(\mathcal{H}_{R,L}^{\text{GNS}}) \quad (N \not\rightarrow \infty), \\ \mathcal{Y}_{R,L} &= \widetilde{\mathcal{M}}_{R,L} \neq \mathcal{B}(\mathcal{H}_{R,L}^{\text{GNS}}) \quad (N \rightarrow \infty).\end{aligned}\tag{7.43}$$

Thus at temperatures below the Hawking–Page temperature the boundary algebras are of type I and ordinary statistical-mechanical entropy is well-defined, while above that temperature the algebras are of type III and statistical entropy ceases to exist.

The same phenomenon appears for a bulk region such as R whose algebra $\widetilde{\mathcal{M}}_R$ is of type III at infinite N . One can regularize the bulk by introducing a splitting of width ε , converting the type-III algebra into a type-I algebra. The resulting bulk entropy diverges as

$$S_{\text{bulk}}(\varepsilon) = a \frac{A_{\text{hor}}}{\varepsilon^{d-1}} + \dots, \quad \varepsilon \rightarrow 0.\tag{7.44}$$

The generalized entropy

$$S_{\text{gen}} \equiv \frac{A_{\text{hor}}}{4G_N(\varepsilon)} + S_{\text{bulk}}\tag{7.45}$$

cancels the divergences and remains finite [68]. The same construction shows how a type-III algebra $\mathcal{Y}_R$ at infinite N becomes a type-I algebra at finite N .

In the semi-classical regime one introduces a splitting and writes $\widetilde{\mathcal{M}}_R^\varepsilon$. Subregion-subalgebra duality then identifies

$$\mathcal{Y}_R^\varepsilon = \widetilde{\mathcal{M}}_R^\varepsilon,\tag{7.46}$$

where the right-hand side is of type I. Although an explicit regularization for a general boundary theory is not known, the duality guarantees its existence. At a small but non-zero value of G_N (of order the Planck length) the separation between the area term and the bulk entropy loses meaning; the whole expression is simply the entropy of the type-I algebra $\mathcal{Y}_R^\varepsilon = \mathcal{B}(\mathcal{H}_R)$. Thus

Statement

The introduction of a splitting ε in the limit $G_N \rightarrow 0$ is equivalent to working at finite G_N . The type-III von Neumann algebra becomes type I and the entropy is given by the generalized entropy

$$S_{\text{gen}} = \frac{A_{\text{hor}}}{4G_N(\varepsilon)} + S_{\text{bulk}}(\varepsilon).\tag{7.47}$$

Analogous statements hold for the entanglement wedge. The Ryu–Takayanagi entropy S_A is precisely the generalized entropy of the bulk region $\mathfrak{b}_A$:

$$S_{\text{gen}}(\mathfrak{b}_A) = S_A.\tag{7.48}$$

At $\varepsilon \sim \ell_P$ the algebra $\mathcal{X}_A^\varepsilon$ reduces to the finite- N algebra $\mathcal{B}_A^{(N)}$. The same construction applies to causal wedges. For the causal wedge $\mathcal{W}_{\rho_\omega}$ dual to the time band I_ω one has

$$S_{\text{gen}}(\mathcal{W}_{\rho_\omega}) = \lim_{\varepsilon \rightarrow 0} \left[\frac{A(\mathcal{W}_{\rho_\omega})}{4G_N(\varepsilon)} + \tilde{S}_{\mathcal{W}_{\rho_\omega}}(\varepsilon) \right]. \quad (7.49)$$

Each term diverges in the ultraviolet, but their sum is finite. Two important features follow:

- The quantity $S_{\text{gen}}(\mathcal{W}_{\rho_\omega})$ admits a well-defined finite- G_N extension.
- The algebra $\mathcal{Y}_{I_\omega}$ expands at finite N into a type-I subalgebra $\mathcal{B}_{I_\omega}$ whose entropy is given by $S_{\text{gen}}[\mathcal{W}_{\rho_\omega}]$. This subalgebra cannot coincide with the full boundary algebra $\mathcal{B}(\mathcal{H}_{\text{CFT}})$, which has vanishing vacuum entropy. Likewise the commutant $\mathcal{Y}'_{I_\omega}$ expands into a type-I algebra $\mathcal{B}_{\mathcal{D}_{\rho_\omega}}$.

At finite N the bulk regions $\mathcal{W}_{\rho_\omega}$ and $\mathcal{D}_{\rho_\omega}$ themselves fluctuate and cannot be defined sharply; the algebras of well-defined operators nevertheless provide a natural definition of the corresponding subsystems in the quantum-gravity regime.

Chapter 8

Volume as an Index of a Subalgebra

In the previous chapter we introduced subregion-subalgebra duality and saw that, by means of entanglement-wedge and causal-entanglement-wedge reconstructions, information about a particular bulk subregion can be encoded in the boundary theory. In the present chapter we introduce an index associated with a subalgebra and use it, via inclusion indices, to compute the volume of the bulk subregion dual to that subalgebra. This feature leads us naturally to the notion of complexity and allows us, in various settings, to process bulk AdS subregions using only the boundary theory.

8.1 Structure of Half-Sided Modular Inclusion

As we explained in Chap. 4, given von Neumann algebras $\mathcal{M}$ and $\mathcal{N}$ with $\mathcal{N} \subseteq \mathcal{M}$, one can define a notion of time. Taking a state $|\Psi\rangle$ that is cyclic and separating for both algebras, one can introduce the operator

$$D(t) = \Delta_{\mathcal{M}}^{-it} \Delta_{\mathcal{N}}^{+it}$$

which satisfies

$$\mathcal{N}_t = D(t)\mathcal{N}D^\dagger(t) \subset \mathcal{M}, \quad \forall t \leq 0. \quad (8.1)$$

This statement asserts that the subalgebras generated from $\mathcal{N}$ by the modular flow remain invariant in the half-plane $t \leq 0$. The structure is known as a half-sided modular inclusion. It implies three remarkable properties [14, 15, 16, 17]:

Statement

1. The operators $K_{\mathcal{M}}$ and $K_{\mathcal{N}}$ obey the algebra

$$\begin{aligned} [K_{\mathcal{M}}, K_{\mathcal{N}}] &= -2\pi i(K_{\mathcal{M}} - K_{\mathcal{N}}) = -4\pi^2 i G, \\ J_{\mathcal{M}} \exp(iGs) J_{\mathcal{M}} &= \exp(-iGs), \quad J_{\mathcal{N}} \exp(iGs) J_{\mathcal{N}} = \exp(-iGs). \end{aligned} \quad (8.2)$$

2. The same statement implies that the generator of the algebra $\mathcal{M}$ produces the time half-plane s inside $\mathcal{M}$ itself:

$$\exp(iGs)\mathcal{M}\exp(-iGs) \subset \mathcal{M}, \quad \forall s \leq 0. \quad (8.3)$$

Consequently

$$\mathcal{N} = \exp(-iG)\mathcal{M}\exp(iG). \quad (8.4)$$

3. $\mathcal{M}$ must be a type-III₁ von Neumann algebra.

The third property is especially interesting: the half-sided modular inclusion and its consequences hold only for type-III₁ algebras. This immediately motivates the study of quantum field theories, which possess algebras of this type. Throughout the chapter we continue the convention that bulk algebras carry a tilde ($\widetilde{\mathcal{M}}$) while boundary algebras are written without a tilde.

Consider therefore a 1 + 1-dimensional quantum field theory in the vacuum state $|\Psi\rangle$. As shown in Fig. 8.1, the right Rindler wedge (the pink triangle) carries the algebra $\mathcal{M}$, for which $|\Psi\rangle$ is cyclic and separating.

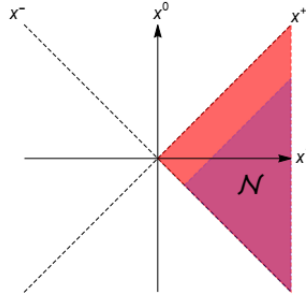


Figure 8.1: $\mathcal{N}$ is the algebra of the indicated subregion that admits a half-sided modular inclusion. The inclusion generates a translation along the x^- direction.

Let $\mathcal{N}$ be the subalgebra associated with the region

$$\mathcal{N} = \{x_+ > 0, x_- < -1\} \quad (8.5)$$

(in light-cone coordinates $x_{\pm} = x_0 \pm x_1$). Because $|\Psi\rangle$ is cyclic and separating for $\mathcal{M}$, the latter possesses a modular structure. Acting with the modular Hamiltonian $K_{\mathcal{M}}$ and using causality produces a family of subalgebras

$$\mathcal{N}_t = \Delta_{\mathcal{M}}^{-it} \mathcal{N} \Delta_{\mathcal{M}}^{+it} = \{x_+ > 0, x_- < -e^{-2\pi t}\} \subset \mathcal{M}. \quad (8.6)$$

In quantum field theory $P^0 = H$ and $P^1 = P$, so the light-cone momenta are $P_{\pm} = \frac{1}{2}(P_0 \pm P_1)$. The modular operator of $\mathcal{N}$ must leave the point $a^{\mu} = (a^+, a^-) = (0, -1)$

invariant. Hence

$$\exp(iK_{\mathcal{N}}t) = \exp(-i\alpha^\mu P_\mu) \exp(iK_{\mathcal{N}}t) \exp(+i\alpha^\mu P_\mu) = \exp(-iP_+) \exp(iK_{\mathcal{N}}t) \exp(+iP_+). \quad (8.7)$$

The Baker–Campbell–Hausdorff formula then yields

$$K_{\mathcal{N}} = K_{\mathcal{M}} + i[K_{\mathcal{M}}, P_+]. \quad (8.8)$$

Using $K_{\mathcal{M}} = 2\pi K$ and the quantum-field-theory commutator $[K, P_+] = iP_+$ one obtains

$$K_{\mathcal{N}} = K_{\mathcal{M}} - 2\pi P_+, \quad (8.9)$$

and therefore the generator

$$G = \frac{1}{2\pi}(K_{\mathcal{M}} - K_{\mathcal{N}}) = P_+. \quad (8.10)$$

An analogous computation for the opposite half-plane gives $G = P_-$.

It is convenient to work in Kruskal coordinates. The unitary $\exp(isP_+)$ maps a bulk point (U, V) to

$$U \rightarrow U_s = U + s, \quad V \rightarrow V_s = \frac{V}{1 - sV}. \quad (8.11)$$

A bulk field of vanishing spatial momentum transforms as $\phi_{\vec{k}=0}(U, V) \rightarrow \lambda_s \phi_{\vec{k}=0}(U_s, V_s)$. The corresponding boundary operator transforms as $O(t) \rightarrow \lambda_s O(f(t; s))$ with

$$f(t; s) = t + \log(1 + se^{-t}). \quad (8.12)$$

For $s \leq 0$ one replaces $s \rightarrow -s$ and uses $f(t; s) = t + \log(1 - se^{-t})$.

Acting with the unitary $\exp[is(P_+ + P_-)]$ produces the bulk transformation [29, 69]

$$U \rightarrow U_s = \frac{U \cos s + \sin s}{\cos s - \sin s U}, \quad V \rightarrow V_s = \frac{V \cos s + \sin s}{\cos s - \sin s V}, \quad (8.13)$$

and the boundary operator transforms with

$$f(t; s) = t - \log \left[\frac{\cos s - \sin s e^t}{\cos s + \sin s e^t} \right]. \quad (8.14)$$

This structure furnishes a convenient and rigorous algebraic framework in the limit $N \rightarrow \infty$. As we shall see, it is particularly useful for the study of operators in the eternal AdS black hole.

8.2 Theory of Index of Inclusions

We now review the theory of indices of inclusions of von Neumann algebras. The study of inclusions $\mathcal{N} \subseteq \mathcal{M}$ is a fundamental topic in operator algebras. A natural question is how to quantify the relative size of $\mathcal{M}$ with respect to $\mathcal{N}$. The index theory initiated by Vaughan

Jones supplies a precise answer. For a suitable inclusion of factors one defines the Jones index

$$[\mathcal{M} : \mathcal{N}]. \quad (8.15)$$

This quantity plays a role analogous to the degree of a field extension and encodes structural information about the relation between the two algebras.

A striking result is that the possible values of the index are highly constrained. Jones showed that if the index is less than 4 it can take only the discrete values

$$[\mathcal{M} : \mathcal{N}] \in \{4 \cos^2(\pi/n)\}_{n \geq 3}, \quad (8.16)$$

while every real number in the interval $[4, \infty)$ is attainable. This discovery opened deep connections between von Neumann algebras, knot theory, mathematical physics and representation theory.

The Jones index is defined for type-II₁ factors. Kosaki's index and Longo's index extend the notion to all factors and, respectively, to type-III₁ factors. We begin with Kosaki's index.

- **Kosaki index.**

The index for inclusions of type-II₁ factors was first introduced by Jones. We restrict attention to the standard representation, which is the most relevant for physics; a more general definition can be found in [70].

Let $\mathcal{N} \subset \mathcal{M}$ be an inclusion of von Neumann factors acting on a common Hilbert space $\mathcal{H}_{\text{tot}}$ that realises a standard representation of $\mathcal{M}$. Suppose there exists a conditional expectation

$$E : \mathcal{M} \rightarrow \mathcal{N} \quad (8.17)$$

satisfying

- $E(n) = n$ for all $n \in \mathcal{N}$,
- $E(n_1 m n_2) = n_1 E(m) n_2$ for $n_1, n_2 \in \mathcal{N}$ and $m \in \mathcal{M}$.

Let $|\phi\rangle, |\psi\rangle \in \mathcal{H}_{\text{tot}}$ with $|\phi\rangle$ cyclic and separating for $\mathcal{M}$. The associated weights are denoted $\phi, \psi : \mathcal{M} \rightarrow \mathbb{C}$. The inverse weight $\phi^{-1} : \mathcal{B}(\mathcal{H}_{\text{tot}}) \rightarrow \mathcal{M}'$ is defined by

$$\langle \psi | \phi^{-1}(a) | \psi \rangle = \text{tr}(\Delta_{\psi\phi} a), \quad a \in \mathcal{B}(\mathcal{H}_{\text{tot}}), \quad (8.18)$$

where $\Delta_{\psi\phi}$ is the relative modular operator of ϕ with respect to ψ and the trace is taken on $\mathcal{H}_{\text{tot}}$.

For a finite type-I_m factor $\mathcal{M} = \mathcal{B}(\mathcal{H}_m) \otimes \mathbf{1}_{\bar{\mathcal{H}}_m}$ the cyclic and separating vector $|\phi\rangle$ may be represented by a full-rank reduced density operator $\rho_\phi \in \mathcal{M}$. The relative modular operator is then

$$\Delta_{\psi\phi} = \rho_\psi^{-1} \otimes \rho'_\psi, \quad (8.19)$$

and Eq. (8.18) yields

$$\phi^{-1}(a) = \text{tr}_{\mathcal{H}_m}(\rho_\phi^{-1}a). \quad (8.20)$$

Thus the inverse weight depends only on data associated with ϕ .

An analogous construction defines the inverse of a general operator-valued weight. For the conditional expectation E one obtains an inverse $E^{-1} : \mathcal{N}' \rightarrow \mathcal{M}'$ characterised by

$$(\psi^{-1} \circ E^{-1})^{-1} = E \circ \psi'^{-1} \quad (8.21)$$

for every weight $\psi' : \mathcal{M}' \rightarrow \mathbb{C}$. When $\mathcal{M}$ and $\mathcal{N}$ are factors, $E^{-1}(\mathbf{1}_{\mathcal{N}'})$ is proportional to $\mathbf{1}_{\mathcal{M}'}$. Kosaki's index is then defined by

$$E^{-1}(\mathbf{1}_{\mathcal{N}'}) = \text{Ind } E \cdot \mathbf{1}_{\mathcal{M}'}. \quad (8.22)$$

The definition makes sense for any conditional expectation between factors, irrespective of type.

As an illustration consider a finite-dimensional type-I inclusion

$$\mathcal{N} = \mathbf{1}_k \otimes \mathcal{B}(\mathcal{H}_n) \subset \mathcal{M} = \mathcal{B}(\mathcal{H}_m) = \mathcal{B}(\mathcal{H}_n) \otimes \mathcal{B}(\mathcal{H}_k).$$

A natural conditional expectation is the normalised partial trace

$$E = \frac{1}{k} \text{tr}_{\mathcal{H}_k} : \mathcal{B}(\mathcal{H}_n) \otimes \mathcal{B}(\mathcal{H}_k) \rightarrow \mathcal{B}(\mathcal{H}_n). \quad (8.23)$$

On the doubled Hilbert space $\mathcal{H}_{\text{ext}} = \mathcal{H}_m \otimes \mathcal{H}_m$ one finds

$$E^{-1}(n') = k \text{tr}_{\mathcal{H}_k}(n'), \quad (8.24)$$

and therefore

$$\text{Ind } E = k^2, \quad (8.25)$$

which counts the number of copies of $\mathcal{N}$ inside $\mathcal{M}$.

- **Jones index.**

The Jones index [71] characterises an inclusion $\mathcal{N} \subset \mathcal{M}$ of type-II₁ factors. One works in the GNS Hilbert space $\mathcal{H}_{\mathcal{M}} = L^2(\mathcal{M})$ associated with the unique normalised trace $\text{tr}_{\mathcal{M}}$. The corresponding cyclic and separating vector $|\Omega\rangle$ satisfies

$$\langle \Omega | \cdot | \Omega \rangle = \text{tr}_{\mathcal{M}}(\cdot). \quad (8.26)$$

The action of $\mathcal{N}$ on $|\Omega\rangle$ generates a subspace $V_{\mathcal{N}} = \overline{\pi_{\text{tr}}(\mathcal{N})|\Omega\rangle} \subset \mathcal{H}_{\mathcal{M}}$. The orthogonal projection onto this subspace is the Jones projection $e_{\mathcal{N}} \in \mathcal{N}'$. The Jones index is

defined by

$$[\mathcal{M} : \mathcal{N}] = \frac{1}{\text{tr}_{\mathcal{N}'}(e_{\mathcal{N}})}. \quad (8.27)$$

When the conditional expectation is the one induced by the trace, Kosaki's index coincides with the Jones index. The proof proceeds by verifying that the relative modular operators are trivial and that

$$\text{tr}_{\mathcal{N}'}(e_{\mathcal{N}}) = \frac{1}{\text{Ind } E}, \quad (8.28)$$

which immediately yields equality of the two indices.

- **Longo index.**

Longo's index [72] applies to inclusions of infinite factors and, in particular, to type-III₁ factors. Consider an inclusion $\mathcal{N} \subset \mathcal{M}$ of type-III₁ factors acting on a common Hilbert space that realises a standard representation for both algebras, with a common cyclic and separating vector $|\Omega\rangle$. Let $E : \mathcal{M} \rightarrow \mathcal{N}$ be a normal conditional expectation. The modular automorphism groups of a faithful state ϕ on $\mathcal{N}$ and of $\phi \circ E$ on $\mathcal{M}$ coincide when restricted to $\mathcal{N}$. Consequently the modular crossed products preserve the inclusion:

$$\widehat{\mathcal{N}} = \mathcal{N} \rtimes_{\sigma^\phi} \mathbb{R} \subset \widetilde{\mathcal{M}} = \mathcal{M} \rtimes_{\sigma^{\phi \circ E}} \mathbb{R}. \quad (8.29)$$

The canonical endomorphism of $\widehat{\mathcal{M}}$ into $\widehat{\mathcal{N}}$ is

$$\widehat{\gamma}(\widehat{m}) = J_{\widehat{\phi}} J_{\widehat{\phi \circ E}} \widehat{m} J_{\widehat{\phi \circ E}} J_{\widehat{\phi}}. \quad (8.30)$$

The composition $\text{tr}_{\widehat{\mathcal{M}}} \circ \widehat{\gamma}$ is necessarily a scalar multiple of $\text{tr}_{\widehat{\mathcal{M}}}$. Longo's index is defined to be this scalar:

$$\text{tr}_{\widehat{\mathcal{M}}} \circ \widehat{\gamma} = \lambda \text{tr}_{\widehat{\mathcal{M}}}, \quad \text{Ind } E \equiv \lambda. \quad (8.31)$$

All three indices coincide whenever they are simultaneously defined.

8.3 Complexity: Algebraic Description

The index of an inclusion measures the relative size of two von Neumann algebras. In the preceding chapters we established an algebraic formulation of AdS/CFT in which boundary single-trace operators are dual to bulk operators, and we reconstructed bulk regions via entanglement wedges and causal wedges. It is therefore natural to propose that the index of a boundary inclusion is related to the volume of the corresponding bulk subregion, and that the growth of this volume with time is a measure of complexity.

We illustrate the idea with the eternal AdS black hole, dual to a pair of boundary CFTs at a temperature above the Hawking–Page temperature (i.e., in the large- N limit). The two-sided thermofield-double state $|\text{TFD}_\beta\rangle$ is cyclic and separating. In every Cauchy slice the GNS Hilbert space built on $|\text{TFD}_\beta(t)\rangle$ coincides with the Fock space of the bulk fields. While

the individual Hamiltonians H_R and H_L are not well-defined at infinite N , their difference $K = H_R - H_L$ remains a well-defined generator of time evolution. The system admits a half-sided modular inclusion, and the operators $P_{\pm}$ play the role of modular Hamiltonians [29].

Let $\mathcal{N}_{0,\omega}$ be the subalgebra of single-trace operators supported in a time band of width ω centred at $t = 0$. The unitaries $U_{\pm}(s) = \exp(iP_{\pm}s)$ act on right- and left-moving operators according to

$$\begin{aligned} U_{\pm}(s)O_R(t)U_{\pm}^{\dagger}(s) &\propto O_R(f^{\pm}(t;s)), & f^{\pm}(t;s) &= t \mp \log(1 \mp se^{\pm t}), \\ U_{\pm}(s)O_L(t)U_{\pm}^{\dagger}(s) &\propto O_L(f^{\pm}(t;s)), & f^{\pm}(t;s) &= t \pm \log(1 \pm se^{\mp t}). \end{aligned} \quad (8.32)$$

At the special value $s_* = -\sinh t / \cosh \omega$ the unitary $U_+(s_*)$ realises ordinary time evolution generated by K . The widths of the transformed bands become

$$\begin{aligned} \omega_R(s_*) &= \omega - \frac{1}{2} \log \left(\frac{\cosh(t+\omega)}{\cosh(t-\omega)} \right) \simeq (1 - \tanh t)\omega + O(\omega^2), \\ \omega_L(s_*) &= \omega + \frac{1}{2} \log \left(\frac{\cosh(t+\omega)}{\cosh(t-\omega)} \right) \simeq (1 + \tanh t)\omega + O(\omega^2). \end{aligned} \quad (8.33)$$

Thus the right band shrinks while the left band expands, and a simple identification $\mathcal{N}_{t,\omega}$ is no longer possible.

The problem is resolved by the unitary generated by $P_+ + P_-$:

$$U(s) = \exp(is(P_+ + P_-)). \quad (8.34)$$

It maps operators according to

$$U(s)O_{R,L}(t)U^{\dagger}(s) \propto O_{R,L}(f(t;s)), \quad f(t;s) = t - \log \left[\frac{\cos s - \sin s e^{-t}}{\cos s + \sin s e^{-t}} \right]. \quad (8.35)$$

Choosing

$$s_* = -\frac{1}{2} \tan^{-1} \left(\frac{\sinh t}{\cosh \omega} \right) \quad (8.36)$$

establishes a unitary equivalence between $\mathcal{N}_{t,\omega}$ and a subalgebra of $\mathcal{N}_{0,\omega}$:

$$\mathcal{N}_{0,\omega_*(t)} = U(s_*)\mathcal{N}_{t,\omega}U^{\dagger}(s_*) \subset \mathcal{N}_{0,\omega}, \quad (8.37)$$

where

$$\omega_*(t) = \log \left(\frac{\sqrt{\cosh^2 \omega + \sinh^2 t} + \sinh \omega}{\cosh t} \right) = \frac{\omega}{\cosh t} + O(\omega^2). \quad (8.38)$$

Both bands now shrink symmetrically with time.

Because the index is unitarily invariant,

$$[\mathcal{B}(\mathcal{H}^{\text{GNS}}) : \mathcal{N}_{t,\omega}] = [\mathcal{B}(\mathcal{H}^{\text{GNS}}) : \mathcal{N}_{0,\omega_*(t)}] \geq [\mathcal{B}(\mathcal{H}^{\text{GNS}}) : \mathcal{N}_{0,\omega}]. \quad (8.39)$$

Hence

Statement

As time t increases, the relative size of the algebra associated with a fixed time band ω grows.

At infinite N the GNS Hilbert spaces built on $|\text{TFD}_\beta(0)\rangle$ and $|\text{TFD}_\beta(t)\rangle$ are disjoint. Let $\mathcal{S} = \mathcal{S}_L \vee \mathcal{S}_R$ be the algebra of all single-trace operators and $\mathcal{S}_{t_1}$ the same algebra centred at time t_1 . The GNS construction yields

$$\mathcal{H}_{|\text{TFD}_\beta(t)\rangle}^{\text{GNS}} = \{\mathcal{N}_{t_1, \omega}^t |\text{TFD}_\beta(0)\rangle \mid \mathcal{N}_{t_1, \omega}^t = \pi_t(\mathcal{S}_{t_1})\}. \quad (8.40)$$

The two Hilbert spaces are unitarily inequivalent, yet the two-point functions that remain finite at large N induce a bijection $b_{0,t}$ between the representations:

$$b_{0,t}(\pi(O(t_1))) = \pi_t(O(t_1 - t)). \quad (8.41)$$

In particular

$$b_{t,0}(\mathcal{N}_{0,\omega}^t) = \mathcal{N}_{t,\omega}^0 \simeq \mathcal{N}_{0,\omega_*(t)}^0 \subset \mathcal{N}_{0,\omega}^0. \quad (8.42)$$

Thus the representation centred at time t becomes smaller, which means that the evolution from $|\text{TFD}_\beta(0)\rangle$ to $|\text{TFD}_\beta(t)\rangle$ increases complexity.

Statement

As time t increases, the state $|\text{TFD}_\beta(t)\rangle$ becomes more complex.

The simultaneous growth of the relative size of the algebras and of the complexity of the state suggests that the inclusion index may be identified with a bulk volume:

Statement

If a boundary algebra $\mathcal{N} \subset \mathcal{M}$ with relative commutant $\mathcal{N}' \cap \mathcal{M}$ is dual to a bulk region $\mathbf{B}$, and if $\mathbf{b}$ denotes a maximal-volume Cauchy slice of $\mathbf{B}$, then the immersion of $\mathcal{N}$ into $\mathcal{M}$ is quantified by the inclusion index

$$(\mathcal{M} : \mathcal{N}) = \exp(C \text{Vol}(\mathbf{b})), \quad (8.43)$$

where $C \propto 1/G_N$ is a dimensionful constant [67].

The index $(\cdot : \cdot)$ defined above does not satisfy the exact multiplicativity of the classical Jones/Kosaki/Longo indices. For a chain $\mathcal{L} \subset \mathcal{N} \subset \mathcal{M}$ one has only the inequality

$$(\mathcal{M} : \mathcal{L}) \geq (\mathcal{M} : \mathcal{N})(\mathcal{N} : \mathcal{L}). \quad (8.44)$$

We therefore employ parentheses rather than square brackets to avoid confusion. The same statements remain valid in AdS_2 , where the bulk dual of $\mathcal{N}'_{t,\omega} \cap \mathcal{M}$ coincides with the dual of $\mathcal{N}'_{0,\omega_*(t)} \cap \mathcal{M}$.

8.4 Applications

We now illustrate the utility of the boundary inclusion index as a function of bulk volume and show that, in special cases, the relative size of two boundary algebras is independent of all boundary data.

Consider first the geometry of Fig. 7.5. Let $\mathcal{N}$ be the algebra of a time band I_ω inside the full boundary algebra $\mathcal{M}$. Its bulk dual is the causal wedge $\mathcal{W}_{\rho_\omega}$, while the relative commutant $\mathcal{N}' \cap \mathcal{M}$ is dual to the diamond $\mathcal{D}_{\rho_\omega}$. The maximal-volume slice $\mathfrak{d}_\omega$ of the latter has volume

$$\text{Vol}(\mathfrak{d}_\omega) = \frac{l_{\text{AdS}}^d \text{Vol}(S^{d-1})}{d} (\sin^2 \rho_\omega)^{d/2} F\left(\frac{d}{2}, \frac{d+1}{2}; \frac{d}{2} + 1; \sin^2 \rho_\omega\right). \quad (8.45)$$

As the band width ω increases the inclusion index $(\mathcal{M} : \mathcal{N})$ decreases.

The manner in which a boundary algebra is immersed into the full algebra affects the dual bulk region (Fig. 8.2).

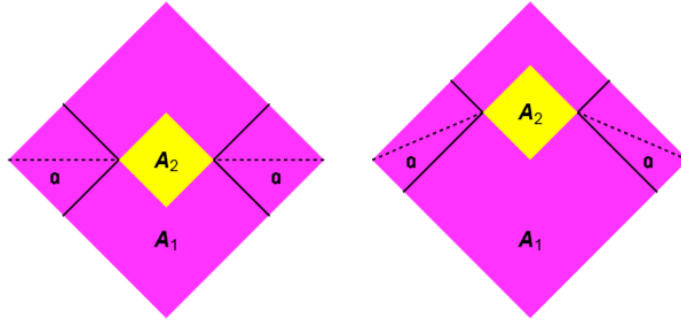


Figure 8.2: The way a boundary algebra is immersed into the full algebra determines the dual bulk region.

We next compare entanglement-wedge and causal-wedge reconstructions. Let R be a bulk subregion (Fig. 8.3). Entanglement-wedge reconstruction supplies an algebra $\mathcal{X}_R$, while causal-wedge reconstruction supplies a smaller algebra $\mathcal{Y}_R \subseteq \mathcal{X}_R$. Their dual regions are $\hat{\mathfrak{b}}_R$ and $\hat{\mathfrak{c}}_R$ respectively. The inclusion index $(\mathcal{X}_R : \mathcal{Y}_R)$ is determined by the volume of $(\hat{\mathfrak{c}}_R)' \cap \hat{\mathfrak{b}}_R$. For a time-reversal-symmetric region this reduces to

$$\text{Vol}(\bar{\mathfrak{c}}_R \cap \mathfrak{b}_R) = \text{Vol}(\mathfrak{b}_R) - \text{Vol}(\mathfrak{c}_R). \quad (8.46)$$

Two concrete cases are of interest.

- **Two disjoint intervals in the vacuum.**

Let $R = R_1 \cup R_2$ with

$$R_1 = \{(t, \theta) \mid t = 0, -\omega < \theta + \pi/2 < \omega\}, \quad R_2 = \{(t, \theta) \mid t = 0, -\omega < \theta - \pi/2 < \omega\}.$$

The volumes are $\text{Vol}(\bar{\mathfrak{c}}_R) = 4\omega l_{\text{AdS}}^2/\epsilon$ and $\text{Vol}(\mathfrak{b}_R) = 4\omega l_{\text{AdS}}^2/\epsilon - 2\pi l_{\text{AdS}}^2$. Consequently

$$\text{Vol}(\bar{\mathfrak{c}}_R \cap \mathfrak{b}_R) = 2\pi l_{\text{AdS}}^2, \quad (8.47)$$

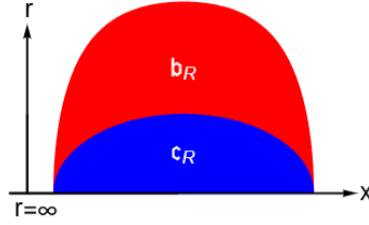


Figure 8.3: A time-reversal-symmetric Cauchy slice in the bulk. The maximal-volume slices (union of the red and blue regions) of $\mathfrak{b}_R$ and $\mathfrak{c}_R$ correspond to the entanglement wedge and the causal wedge of a boundary subregion R .

which is independent of the boundary width ω .

- **A single interval at finite temperature.**

Let $R = \{(t, \theta) \mid t = 0, -\omega < \theta < \omega\}$ in a thermal state above the Hawking–Page temperature (dual to a BTZ black hole). The volumes are $\text{Vol}(\bar{\mathfrak{c}}_R) = 2\omega l_{\text{AdS}}^2/\epsilon + \pi l_{\text{AdS}}^2$ and $\text{Vol}(\mathfrak{b}_R) = 2\omega l_{\text{AdS}}^2/\epsilon - \pi l_{\text{AdS}}^2$. Again

$$\text{Vol}(\bar{\mathfrak{c}}_R \cap \mathfrak{b}_R) = 2\pi l_{\text{AdS}}^2, \quad (8.48)$$

independent of both ω and temperature.

A further application concerns superadditivity. In the large- N limit the algebras of two boundary regions satisfy

$$\mathcal{X}_{R_1} \vee \mathcal{X}_{R_2} \subsetneq \mathcal{X}_{R_1 \cup R_2}. \quad (8.49)$$

The difference is measured by the volume of $\overline{\mathfrak{b}_{R_1} \cup \mathfrak{b}_{R_2}} \cap \mathfrak{b}_{R_1 \cup R_2}$ (Fig. 8.4).

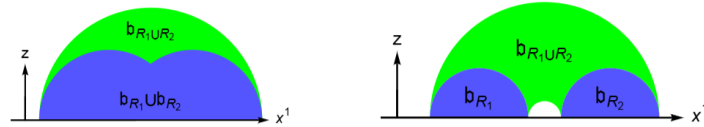


Figure 8.4: Superadditivity realised by bulk volumes: $\mathcal{X}_{R_1} \vee \mathcal{X}_{R_2} \subsetneq \mathcal{X}_{R_1 \cup R_2}$. A constant-time slice of AdS_3 with maximal-volume slices is shown. The light-blue region is the maximal volume of $\mathfrak{b}_{R_1} \cup \mathfrak{b}_{R_2}$ (dual to $\mathcal{X}_{R_1} \vee \mathcal{X}_{R_2}$); the union of the blue and green regions is the maximal volume of $\mathfrak{b}_{R_1 \cup R_2}$ (dual to $\mathcal{X}_{R_1 \cup R_2}$). Left: overlapping intervals. Right: disjoint intervals.

- **Overlapping intervals.**

Take $R = \{(t, x) \mid t = 0, x \in (-a-b, a+b)\}$, $R_1 = \{(t, x) \mid t = 0, x \in (-a-b, a-b)\}$ and $R_2 = \{(t, x) \mid t = 0, x \in (-a+b, a+b)\}$. Then

$$\begin{aligned} \text{Vol}(\mathfrak{b}_{R_1, R_2}) &= \frac{2a l_{\text{AdS}}^2}{\epsilon} - \pi l_{\text{AdS}}^2, \\ \text{Vol}(\mathfrak{b}_R) &= \frac{2(a+b) l_{\text{AdS}}^2}{\epsilon} - \pi l_{\text{AdS}}^2, \end{aligned} \quad (8.50)$$

and

$$\text{Vol}(\overline{\mathfrak{b}_{R_1} \cup \mathfrak{b}_{R_2}} \cap \mathfrak{b}_{R_1 \cup R_2}) = 2l_{\text{AdS}}^2 \arctan\left(\frac{b}{\sqrt{a^2 - b^2}}\right). \quad (8.51)$$

- **Disjoint intervals in the vacuum.**

For $b > a$ the intervals become disjoint. In the window $a < b < \sqrt{2}a$ one recovers the same constant volume found earlier,

$$\text{Vol}(\overline{\mathfrak{b}_{R_1} \cup \mathfrak{b}_{R_2}} \cap \mathfrak{b}_{R_1 \cup R_2}) = 2\pi l_{\text{AdS}}^2. \quad (8.52)$$

When $b > \sqrt{2}a$ superadditivity ceases to hold.

Chapter 9

Algebraic Approach in the Quantum Gravity Regime

Up to this chapter, we have examined only selected topics in high-energy physics, most of which involved AdS/CFT duality. The strength of the algebraic language in this context has been the emergence of the so-called subregion-subalgebra duality. We then addressed, in a certain sense, our initial question about what is called spacetime: whether spacetime is a physical object or merely a particular consequence of entanglement.

In the present chapter we employ the algebraic approach to study several problems in quantum gravity and attempt to apply everything we have introduced so far to this physical regime.

9.1 The Page Curve and the Existence of Islands

As we examined in Chapter 7, we were able to use quantum-information aspects, and in particular quantum error correction, within the algebraic formulation of entanglement-wedge reconstruction. One of the important subjects connected with this discussion is the existence of islands [73, 74, 75]. We first review the historical development of the subject and then look at it from the algebraic viewpoint.

The story begins with Hawking’s observation that black holes are not completely black (i.e., perfect black bodies) but emit radiation that we call Hawking radiation. Consequently, when a black hole evaporates completely its information is lost. This statement attracted great attention at the time, yet an important point remained unstated: if a black hole is a quantum system, the loss of information would contradict the unitarity of time-evolution operators in quantum mechanics. The resulting contradiction is known as the black-hole information paradox.

Hawking’s claim, being at odds with quantum mechanics, implies (as illustrated in Fig. 9.1) that the entropy always increases and information is gradually lost. On the other hand we know that the black-hole entropy is proportional to the area of its horizon, $S = A/(4\hbar G_N)$. During the evaporation process one therefore expects the black-hole entropy to decrease, as shown in Fig. 9.2.

During black-hole evaporation both the emission of Hawking radiation (which increases

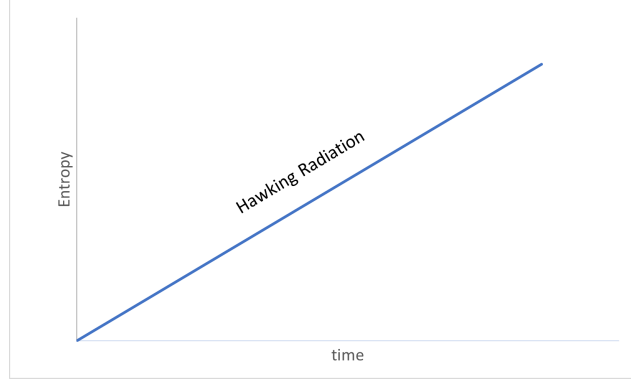


Figure 9.1: The entropy of the radiation emitted by a black hole (Hawking radiation) increases with time.

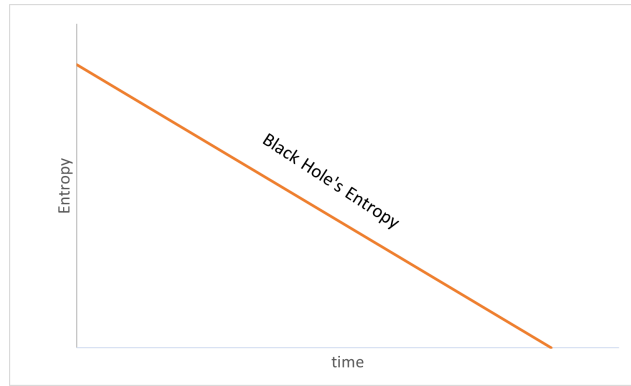


Figure 9.2: The entropy of the black hole, being proportional to its horizon area, must decrease as radiation is emitted.

entropy) and the decrease of the horizon area must be taken into account simultaneously; the combination is depicted in Fig. 9.3.

The physical mechanism of the problem requires that the combination of the two curves be valid so that the entropy remains finite and the principles of quantum mechanics are preserved. Don Page subsequently argued that if the evaporation process is unitary then the entropy must behave as follows:

1. At early times the black hole is large and its entropy increases; consequently the entropy of the Hawking radiation also increases.
2. After a characteristic time called the Page time, roughly half of the information has been radiated. Thereafter the outgoing Hawking radiation is entangled with the earlier radiation, so the entropy of the radiation begins to decrease.

The resulting curve, known as the Page curve, is shown in black in Fig. 9.4 as the combination of the two previous diagrams. If Page's claim is correct, the information paradox is resolved. In 2019 it was shown, within AdS/CFT, that when the entropy of the Hawking radiation is computed with the quantum extremal surface, after the Page time t_p a region inside the black hole is added to the radiation region.

One therefore concludes:

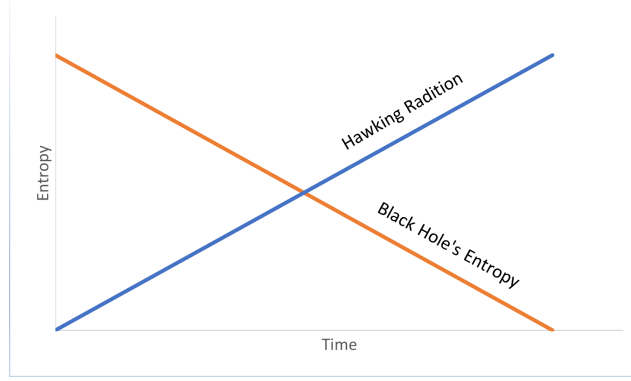


Figure 9.3: The two processes—emission of radiation and decrease of the black-hole area—must be considered simultaneously. The two curves must therefore occur together.

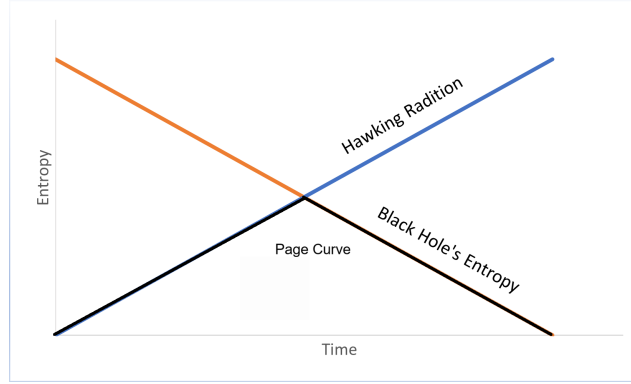


Figure 9.4: The Page curve. At early times it follows the entropy of the Hawking radiation; after the Page time it follows the decreasing black-hole area. In other words, after the Page time the emitted radiation must contain information that resides inside the black hole.

Statement

A black hole shares part of its own information through Hawking radiation. The region inside the black hole that is correlated with the radiation is called an island.

We now formulate the same statement algebraically. In Section 7.2 we introduced the necessary machinery. Let B denote the black-hole system and R the Hawking-radiation system. For $t < t_p$ the radiation has not yet carried away appreciable information, so all information still resides inside the black hole. From the region dual to B one can reconstruct both an interior region I and an exterior region O :

$$\mathcal{X}_B = \widetilde{\mathcal{M}}(O) \vee \widetilde{\mathcal{M}}(I), \quad t < t_p. \quad (9.1)$$

For $t > t_p$, according to Page's argument, the radiation is correlated with the black-hole interior (the island). The boundary region dual to B therefore contains only the exterior O :

$$\mathcal{X}_B = \widetilde{\mathcal{M}}(O), \quad t > t_p. \quad (9.2)$$

Hence

$$\mathcal{X}_B = \begin{cases} \widetilde{\mathcal{M}}(O) \vee \widetilde{\mathcal{M}}(I) & t < t_p, \\ \widetilde{\mathcal{M}}(O) & t > t_p. \end{cases} \quad (9.3)$$

The total system is $B \cup R$. For $t > t_p$ the island I no longer belongs to B and must be included in R . The algebra of R therefore becomes

$$\mathcal{X}_R = \begin{cases} \mathcal{Y}_R & t < t_p, \\ \mathcal{Y}_R \vee \widetilde{\mathcal{M}}(I) & t > t_p. \end{cases} \quad (9.4)$$

This is precisely the statement that a region I inside the black hole is encoded in the Hawking radiation. The algebra associated with the island may be defined by

$$\mathcal{I}_R \equiv \mathcal{Y}'_R \cap \mathcal{X}_R. \quad (9.5)$$

For $t > t_p$ one obtains

$$\mathcal{I}_R \equiv \mathcal{Y}'_R \cap (\mathcal{Y}_R \vee \widetilde{\mathcal{M}}(I)) = \widetilde{\mathcal{M}}(I). \quad (9.6)$$

If $\mathcal{I}_R$ is non-trivial, an island exists and contains independent operators that describe low-energy physics from the radiation. Using the lemma of modular flow one can reconstruct $\mathcal{X}_R$ from $\mathcal{Y}_R$ alone and thereby obtain $\mathcal{I}_R$.

The same definition applies to any holographic subsystem Q :

$$\mathcal{I}_Q = \mathcal{Y}'_Q \cap \mathcal{X}_Q, \quad (9.7)$$

where $\mathcal{Y}_Q$ is the algebra of the causal wedge of Q .

In AdS/CFT we take Q to be a boundary subregion A . Then $\mathcal{Y}_Q = \mathcal{Y}_A$. The operators belonging to $\mathcal{I}_A$ are those that lie in the entanglement wedge of A but not in its causal wedge (Fig. 9.5). The region between the boundary interval A and the Ryu–Takayanagi surface γ_A is denoted $\mathfrak{b}_A$.

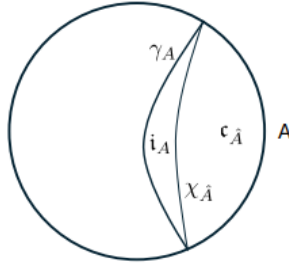


Figure 9.5: A constant-time bulk Cauchy slice. $\chi_{\hat{A}}$ is the edge of the causal wedge and $\gamma_{\hat{A}}$ the RT surface (edge of the entanglement wedge). $c_{\hat{A}}$ is the region between $\chi_{\hat{A}}$ and A (a Cauchy slice of the causal wedge); $i_{\hat{A}}$ is the region between $\chi_{\hat{A}}$ and $\gamma_{\hat{A}}$. The regions $i_{\hat{A}}$ and $c_{\hat{A}}$ are strongly entangled by short-range entanglement near $\chi_{\hat{A}}$.

The bulk region dual to the boundary algebra $\mathcal{I}_A$ is i_A , while the remainder is the causal

wedge c_A . Thus

$$\mathfrak{b}_A = c_A \cup i_A \quad (9.8)$$

and

$$\widetilde{\mathcal{M}}(\mathfrak{b}_A) = \widetilde{\mathcal{M}}(c_A) \vee \widetilde{\mathcal{M}}(i_A). \quad (9.9)$$

By entanglement-wedge reconstruction

$$\mathcal{X}_A = \mathcal{Y}_A \vee \widetilde{\mathcal{M}}(i_A), \quad (9.10)$$

so

$$\mathcal{I}_A = \mathcal{Y}'_A \cap \mathcal{X}_A = \widetilde{\mathcal{M}}(i_A). \quad (9.11)$$

Hence

Statement

The bulk region i_A is an island for the boundary region A .

The island phenomenon shares similarities and differences with Hawking radiation:

- **Similarity.** Both rely on subregion-subalgebra duality and produce entanglement between regions that are not necessarily causally connected.
- **Difference.** For Hawking radiation the entanglement between $\mathcal{Y}_R$ and $\widetilde{\mathcal{M}}(I)$ is non-local, whereas the entanglement between $\mathcal{Y}_A$ and $\widetilde{\mathcal{M}}(i_A)$ is short-range.

We can now ask a sharper question.

Question

Why, according to Eqs. (9.3) and (9.4), does the algebra $\widetilde{\mathcal{M}}(I)$ jump after the Page time $t \geq t_p$?

The answer lies in a boundary phenomenon. Consider two boundary intervals A_1 and A_2 as in Fig. 9.6.

When $|A| = \pi + \epsilon$ the region O is an island for A ; when $|A| = \pi - \epsilon$ it is an island for $\bar{A}$. Therefore

Statement

Changing the size of A from $\pi + \epsilon$ to $\pi - \epsilon$ transfers the island algebra $\widetilde{\mathcal{M}}(O)$ between $\mathcal{X}_A$ and $\mathcal{X}_{\bar{A}}$ [76].

This explains the discontinuous jump of the island algebra near the Page time and relates it, algebraically, to a change in the modular flow.

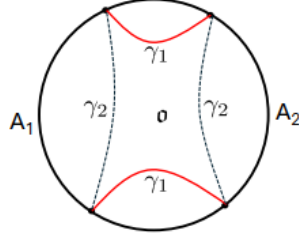


Figure 9.6: Transfer of an island between $A = A_1 \cup A_2$ and $\bar{A}$ in the vacuum of a boundary CFT_2 when the size of A is varied. When $|A| = \pi + \epsilon$ the RT surface is γ_1 , yielding a connected entanglement wedge for A ; the middle region labelled O is an island for A . When $|A| = \pi - \epsilon$ the RT surface is γ_2 and O becomes an island for $\bar{A}$. Shrinking $|A|$ from $\pi + \epsilon$ to $\pi - \epsilon$ transfers the island algebra $\mathcal{M}_O$ from $\mathcal{X}_A$ to $\mathcal{X}_{\bar{A}}$.

9.2 Gravitational Dressing

In ordinary quantum field theory physical observables are defined by local operators. On a manifold M one can introduce local operators $\phi(x)$. When gravity is turned on, diffeomorphism invariance $x^\mu \rightarrow x^\mu + \epsilon^\mu$ implies

$$\delta_\epsilon \phi(x) = -\epsilon^\mu \partial_\mu \phi(x). \quad (9.12)$$

A local operator is therefore no longer a physical observable. Defining local operators (and hence physical observables) in the presence of gravity is a central problem of quantum gravity.

One standard solution is to attach the operator to a gravitational Wilson line or to a geodesic, thereby removing the coordinate dependence. From the algebraic viewpoint the same construction is realised by crossed-product algebras.

Let $\mathcal{H}_Q$ be the Hilbert space of a quantum field theory on a fixed background and $\mathcal{B}(\mathcal{H}_Q)$ its algebra of operators. On a Cauchy slice one has complementary type-III algebras $\mathcal{M}$ and $\mathcal{M}'$ with $\mathcal{M} \vee \mathcal{M}' = \mathcal{B}(\mathcal{H}_Q)$. Let $|\Psi\rangle$ be cyclic and separating for $\mathcal{M}$. Modular flow acts as

$$A(t) = e^{iKt} A e^{-iKt}, \quad A'(t) = e^{-iKt} A' e^{iKt}. \quad (9.13)$$

When gravity is present one introduces two observers R and L with Hamiltonians $\hat{q}_R$ and $\hat{q}_L$ acting on Hilbert spaces $\mathcal{H}_R$ and $\mathcal{H}_L$. Their conjugate momenta $\hat{p}_R$ and $\hat{p}_L$ measure a parameter τ . The total Hilbert space is

$$\mathcal{H} = \mathcal{H}_L \otimes \mathcal{H}_Q \otimes \mathcal{H}_R \quad (9.14)$$

with Hamiltonian

$$H = K - \hat{q}_L + \hat{q}_R. \quad (9.15)$$

Physical states must be annihilated by H . The corresponding projection is

$$\Pi = \int_{-\infty}^{+\infty} ds e^{-isH}. \quad (9.16)$$

Physical states are equivalence classes $\Pi|\Psi\rangle$ and the physical inner product is

$$(\Psi, \Phi) = \langle \Psi | \Pi | \Phi \rangle. \quad (9.17)$$

Gauge-invariant operators are obtained by dressing:

$$\begin{aligned} \hat{A}_R(\tau) &= \int ds e^{isH} (|\tau\rangle\langle\tau|_R \otimes A) e^{-isH} = e^{iK\hat{p}_R} A(-\tau) e^{-iK\hat{p}_R}, \\ \hat{A}'_L(\tau) &= \int ds e^{isH} (|\tau\rangle\langle\tau|_L \otimes A') e^{-isH} = e^{-iK\hat{p}_L} A'(-\tau) e^{iK\hat{p}_L}. \end{aligned} \quad (9.18)$$

These operators commute with the total Hamiltonian and are therefore physical observables. The algebras they generate are crossed products:

$$\widehat{\mathcal{M}} = \{\hat{A}_R(\tau), e^{is\hat{q}_R}\}'' , \quad \widehat{\mathcal{M}}' = \{\hat{A}'_L(\tau), e^{is\hat{q}_L}\}'' . \quad (9.19)$$

They are type-II $_{\infty}$ factors, so density operators and relative entropy are well-defined.

Statement

Operators dressed to observers, which define local operators and physical observables in the presence of gravity, generate crossed-product algebras.

9.3 Entropy of dS Space

For this discussion, we shall begin from its point of origin, which is related to black holes. As we know, closed universes with $\Lambda > 0$, which are referred to as de Sitter (dS) spacetime, in contrast to open universes with $\Lambda < 0$, require the existence of an observer. In other words, after physicists realized that the entropy of a black hole is proportional to the area of the black-hole horizon [77, 78], Gibbons and Hawking [79] proposed a similar idea for the cosmological horizon, which is also proportional to the entropy of closed universes. In particular, for dS spacetime with radius r_{dS} , they assume that an observer moves along a timelike worldline. In this case, there is a region of dS spacetime enclosed by the past and future of the observer and upon which the observer can have an influence. This region is referred to as the static patch. Gibbons and Hawking considered the entropy of dS spacetime, analogous to what was obtained from black-hole physics, to be $\frac{A}{4G_N}$, where A is the area of the region accessible to the observer, and the temperature of dS spacetime is defined as $T = \frac{1}{\beta_{\text{dS}}}$. It is worth noting that the main difference between a black-hole horizon and a cosmological horizon is the dependence of the latter on the existence of an observer. In other words, the black-hole horizon at large distances in an asymptotically flat spacetime gives rise to a spacetime region that can be defined for any observer located at infinity and is, in a sense, independent of the observer. In contrast, in closed universes, every observer has access to different regions of space and will have different experiences and different abilities to influence the spacetime. Therefore, the cosmological horizon is observer-

dependent. The D -dimensional dS spacetime, which is the D -dimensional sphere, due to its large number of symmetries, allows every observer to have the same cosmological horizon. For this reason, it provides a simple and calculable example of closed universes. Therefore, as stated above, to calculate the entropy in closed universes, or equivalently in dS spacetime, we need an observer. It is also not unreasonable, when addressing the problem of eliminating the imaginary phase in the gravitational path integral in this spacetime, to investigate the presence of an observer in this spacetime. Based on this, we have established the necessity of an observer, and now we should address what an observer is within the physical framework. As discussed in the previous section, and as we briefly review here, in quantum field theory one can define local operators in a well-defined manner. Mathematically, let us consider a manifold M as the spacetime on which the quantum field theory is defined, and denote the coordinates on this manifold by x^i , $i = 0, \dots, D$. Now suppose that a source $J(x)$ is placed on M , which is defined in a region U . Also, consider an arbitrary operator such as O . Then, to define locality, we have

$$\langle \psi_J | O'(x') | \psi_J \rangle = \langle 0 | O(x) | 0 \rangle \quad (9.20)$$

where

$$|\psi_J\rangle = \exp(-i \int dx J(x) O(x)) |0\rangle \quad (9.21)$$

Therefore, in an ordinary quantum field theory, one can define local operators. Moreover, as we have understood in previous chapters, the algebra of local operators in any arbitrary region of the manifold M in an ordinary quantum field theory is a type III von Neumann algebra, which we denote by $\mathcal{A}$. We also know that, in this type of algebra, all local operators are infinite-dimensional, and therefore it is not possible to define the map tr . Consequently, one cannot provide a proper interpretation in quantum field theory of what is known as entropy in statistical mechanics. Now, if we introduce gravity into this theory, which in other words gives rise to a theory of quantum gravity, how can we define the entropy in the form

$$S_{\text{gen}} = \frac{A}{4G_N} + S_{\text{out}} \quad (9.22)$$

where S_{out} is the entropy of the matter fields and everything else that is present, and S_{gen} is the generalized entropy? Essentially, it is intriguing to ask whether adding gravity to an ordinary quantum field theory, in which a type III von Neumann algebra governs the theory and entropy cannot be defined, can lead to a well-defined notion of entropy. Accordingly, we begin the setup as follows. Suppose, as discussed above, that a quantum field theory is defined on a manifold M whose coordinates are labeled by x^i , $i = 0, \dots, D$. If we now introduce gravity into this theory, then due to the presence of diffeomorphism symmetry, we can no longer have local operators in the theory. In other words, as we know from quantum gravity [80], a local operator is no longer a pre-existing quantity. Mathematically, due to the diffeomorphism symmetry $x^\mu \rightarrow x'^\mu = x^\mu + \epsilon^\mu$, we have

$$\delta_\epsilon O(x) = -\epsilon^\mu \partial_\mu O(x) \quad (9.23)$$

As we can see, the arbitrary operator O at the two points x^μ and x'^μ is related to each other and, in other words, forms a nonlocal operator. Therefore, when gravity is taken into account in a quantum field theory, the operators are neither local nor gauge invariant. Now, as stated above, in dS spacetime, where the existence of an observer is essential and the observer has access only to a limited region of space known as the static patch, the existence of nonlocal operators becomes problematic. The importance of de Sitter spacetime lies in the fact that every observer has access to only a limited region, and therefore concepts such as locality need to be redefined according to what the observer has access to and possesses. In that case, we also do not have a proper definition of physical observables. Therefore, according to Eq. 9.23, the operator O at the spatial point x becomes nonlocal. If we wish to redefine the concept of locality in a way that is specific to the observer, we can instead express each operator not in terms of a spatial coordinate, but in terms of a new degree of freedom associated with the observer, known as the observer's clock. This approach allows us, as long as an observer exists, to express the operator in terms of this new variable and avoid its dependence on the spatial coordinate, which is responsible for its nonlocality, as shown in Eq. 9.23. In this case, one can define a simple model for describing an observer with mass m [81, 82],

$$H_{\text{obs}} = m + q \quad (9.24)$$

where q is the new degree of freedom, which can be interpreted as the Hamiltonian of the observer's clock. As shown in Fig. 9.7, the Penrose diagram for dS spacetime is displayed. An observer located at a point such as p at past infinity moves along the path γ toward future infinity, q . The region P shown in the figure represents the region on which the observer can have an influence and to which the observer has access, and this same region contains the local operators accessible to the observer. Furthermore, according to the Hamiltonian introduced to define the observer, the observer's action is given by

$$I[q] = \oint_{\gamma} d\tau \left(p \frac{dq}{d\tau} - q \right) \quad (9.25)$$

where γ is the worldline along which the observer moves. In this case, this observer model leads to the definition of the local operators that we require in de Sitter spacetime. It should also be noted that the redefined local operators are specific to the observer, or in other words, they belong to the observer's static patch, and due to the properties of dS spacetime, they cannot be regarded as absolutely local operators. Accordingly, our initial question concerning generalized entropy remains: how does a type III von Neumann algebra, which has no notion of entropy, together with an observer whose physics follows the simple model in Eq. 9.24, lead to the emergence of the notion of entropy? Moreover, given that entropy can only be defined in type I and type II von Neumann algebras¹, and because we are working in a quantum field theory background and the Hilbert space is not factorized, a type II algebra is essentially relevant. In this case, the question can be formulated differently: how does the existence of an observer change a type III von Neumann algebra into a type II von Neumann algebra? The answer to this question can be found in the crossed-product algebra [27, 81]. As discussed in the previous chapter, the crossed-

¹In a type II von Neumann algebra, relative entropy is defined.

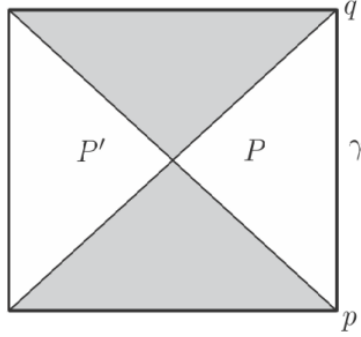


Figure 9.7: The Penrose diagram of de Sitter spacetime is shown. An observer enters from the point p at past infinity and exits at the point q at future infinity. The coordinate system can be chosen such that these points lie at the corners of the diagram, as shown in the figure. The observer's worldline is a timelike curve such as γ from p to q , analogous to the geodesic connecting these two points, which is represented here as a vertical line on the right boundary of the diagram. The region P is the observer's static patch, while P' is its complementary static patch, which is spatially separated from P . The shaded regions behind the past and future horizons lie outside the regions accessible to an observer located in either P or P' .

product algebra structure makes it possible to change the algebra into a type II algebra if we have a locally compact group such as G that contains continuous unitary actions α of this group. Since the model we have considered for the observer has a Hamiltonian and is a one-dimensional quantum system, the continuous unitary actions for the group on which it acts can be determined, and therefore a crossed-product algebra structure can be considered for it. Suppose that in an ordinary quantum field theory, we denote the set of local operators $\phi(\tau)$ by $\mathcal{A}$, which is a type III von Neumann algebra, and denote the cyclic and separating vector with respect to this algebra by $|\Psi\rangle$. Then, according to Tomita–Takesaki theory, one can consider the modular structure for this system. If we take the Hilbert space of this system to be $\mathcal{H}_{\text{grav}}$, then H_{bulk} is known as the generator or Hamiltonian of this space (because the existence of an observer in dS spacetime is important, this Hamiltonian can be regarded as the Hamiltonian of dS spacetime and interpreted as the generator of time evolution in this spacetime). According to the previous section, a one-dimensional quantum system with the Hamiltonian in Eq. 9.24 was considered as the proposed model for the observer. In this model, q , which is referred to as the Hamiltonian of the observer's clock, is the position operator, while p , as the momentum operator, is represented by $p = -i \frac{d}{dq}$. We also expect the Hamiltonian of the observer's clock always to move forward, in other words, to satisfy the condition $q \geq 0$, and we denote its Hilbert space by $L^2(\mathbb{R})$. By adding this one-dimensional quantum system to the original system, we can reconstruct the algebra of the entire system as $\mathcal{M} = \mathcal{A} \otimes \mathcal{B}(L^2(\mathbb{R}))$, which from a physical point of view contains nonlocal operators. Therefore, the total Hamiltonian for this system is

$$\hat{H} = H_{\text{bulk}} + H_{\text{obs}} = H_{\text{bulk}} + m + q \quad (9.26)$$

Then, in order to generate local operators, the only requirement is that the total Hamiltonian of the system satisfy a commutation constraint with the operators,

$$\forall \phi(\tau) \in \mathcal{M}, \quad [\hat{H}, \phi(\tau)] = 0 \quad (9.27)$$

In this case, the above relation implies

$$[H_{\text{bulk}}, \phi(\tau)] = -i\dot{\phi}(\tau), \quad [q, \phi(\tau)] = +i\dot{\phi}(\tau) \quad (9.28)$$

Then, using $q = +i\frac{d}{dp}$, one can state that the above relation and satisfaction of the stated constraint require

$$p = \tau \quad (9.29)$$

Therefore, modeling the observer as a one-dimensional quantum system requires the operator q to be interpreted as the Hamiltonian of the observer's clock and p as the time variable of the observer's clock. Furthermore, to describe local operators, we need to take into account the fact that the condition $q \geq 0$ requires us to use the Heaviside² step function $\Theta(q)$, so that

$$\forall \phi(\tau) \in \mathcal{M}, \quad \hat{\phi}(\tau) = \Theta(q)\phi(\tau)\Theta(q) \quad (9.30)$$

This process of redefining local operators in the presence of gravity in a quantum field theory is referred to as gravitational dressing, which was discussed in the previous section. Therefore, by adding a one-dimensional quantum system to the original system, we created a new algebra called $\mathcal{M}$. We then imposed a constraint on the Hamiltonian and, using gravitational dressing of the operators, were able to reproduce local operators. However, an important point is that imposing a constraint on the Hamiltonian, similar to the BRST procedure, amounts to selecting a subset of the entire set. In other words, the operators in Eq. 9.30 belong to a subset of the algebra $\mathcal{M}$ whose elements all commute with the sum of the generators of the two systems, $\hat{H} = H_{\text{bulk}} + H_{\text{obs}}$, namely

$$\widehat{\mathcal{M}} = \hat{\phi}(\tau) = \Theta(q)\phi(\tau)\Theta(q) \mid \forall \phi(\tau) \in \mathcal{M}, \quad [\hat{H}, \phi(\tau)] = 0 \quad (9.31)$$

As can be seen from the above expression, we conclude that the algebra of all operators accessible to the observer is a crossed-product algebra and, consequently, a type II_1 von Neumann algebra. This result also shows that the existence of such an algebra allows us to define a density operator and entropy, which is of great importance in quantum gravity. Now that we understand how the existence of an observer changes a type III von Neumann algebra into a type $\text{II} * 1$ von Neumann algebra, we should calculate the entropy and obtain what we expect to exist in this theory, namely Eq. 9.22. Suppose that in the original system corresponding to the algebra $\mathcal{A}$, the state $|\Psi\rangle$ is a cyclic and separating vector with respect to this algebra, and denote the Hilbert space of this system by $\mathcal{H}_0$. Then, after adding the new system described above, the Hilbert space becomes $\mathcal{H}_0 \otimes L^2(\mathbb{R})$, and an arbitrary state

²Heaviside

in this space can be written as

$$|\hat{\Psi}\rangle = |\Psi\rangle \otimes |q\rangle, \quad |\Psi\rangle \in \mathcal{H}_0, \quad |q\rangle \in L^2(\mathbb{R}) \quad (9.32)$$

Now, since Eq. 9.27 is the necessary constraint for redefining the local operators, as in the BRST procedure, it is necessary to remove the unphysical states from the physical states. Therefore,

$$|\hat{\Psi}\rangle_P = \Pi |\hat{\Psi}\rangle, \quad \Pi = \int_{-\infty}^{+\infty} ds, e^{-is\hat{H}}, \quad \Pi^2 = \Pi \quad (9.33)$$

where the inner product in the Hilbert space can then be expressed as

$$\forall |\hat{\Psi}\rangle_P, |\hat{\Phi}\rangle_P \in \mathcal{H}_0 \otimes L^2(\mathbb{R}), \quad (\hat{\Psi}_P, \hat{\Phi}_P) = \langle \hat{\Psi} | \Pi | \hat{\Phi} \rangle \quad (9.34)$$

To obtain the relative entropy that exists in a type II von Neumann algebra, we first define the generalized density operator of the algebra $\widehat{\mathcal{M}}$ in the state $|\hat{\Psi}\rangle_P$, which is denoted by $\rho_{\hat{\Psi}}$. Since $\widehat{\mathcal{M}}$ is a type II_1 von Neumann algebra, one can define the map $\text{tr} : \widehat{\mathcal{M}} \rightarrow \mathbb{C}$, and therefore

$$\forall A \in \widehat{\mathcal{M}}, \quad \text{tr}(\rho_{\hat{\Psi}} A) = (\hat{\Psi}_P | A | \hat{\Psi}_P) = \langle \hat{\Psi} | \Pi A | \hat{\Psi} \rangle \quad (9.35)$$

Also, suppose that $|\hat{\Phi}\rangle$ is a cyclic and separating vector with respect to the algebra $\widehat{\mathcal{M}}$. Then, according to the previous chapter, we can state

$$\forall A \in \widehat{\mathcal{M}}, \quad \text{tr}(A) = (\hat{\Phi}_P | A \exp(-q) | \hat{\Phi}_P) \quad (9.36)$$

Therefore, by considering Eqs. 9.35 and 9.36, we obtain

$$\text{tr}(\rho_{\hat{\Psi}} A) = (\hat{\Psi}_P | A | \hat{\Psi}_P) = (\hat{\Phi}_P | \rho_{\hat{\Psi}} A \exp(-q) | \hat{\Phi}_P) \quad (9.37)$$

Now, according to Eq. 5.37, one can consider $A \rightarrow \exp(+iKp)A \exp(-iKp) \exp(-iKs)$, and we obtain

$$\begin{aligned} (\hat{\Phi}_P | \rho_{\hat{\Psi}} e^{+iKp} A e^{-iKp} e^{-iKs} e^{-q} | \hat{\Phi}_P) &= (\hat{\Psi}_P | e^{+iKp} A e^{-iKp} e^{-iKs} | \hat{\Psi}_P) \\ &= (\hat{\Psi}_P | e^{+iKp} A e^{-iKs} e^{-iKp} | \hat{\Psi}_P) \\ &= (\hat{\Phi}_P | \Delta_{\Psi\Phi} e^{+iKp} A e^{-iKs} e^{-iKp} | \hat{\Phi}_P) \\ &= \int dq (\Phi_P, q | \Delta_{\Psi\Phi} e^{+iKp} A e^{-iKs} | q) (q | e^{-iKp} | \Phi_P, q) \\ &= \int dq (\Phi_P, q | \Delta_{\Psi\Phi} e^{+iKp} | q) A e^{-iKs} (q | e^{-iKp} | \Phi_P, q) \\ &= \int dq (\Phi_P | \Delta_{\Psi\Phi} e^{+iKp} A e^{-iKs} e^{-iKp} | \Phi_P) \end{aligned} \quad (9.38)$$

Also, from the left-hand side of the above expression, we have

$$\begin{aligned} (\hat{\Phi}_P | \rho_{\hat{\Psi}} e^{+iKp} A e^{-iKp} e^{-iKs} e^{-q} | \hat{\Phi}_P) &= \frac{1}{2\pi} \int dq (\Phi_P | \rho_{\hat{\Psi}} e^{+iKp} A e^{-iKp} e^{-iKs} e^{-q} | \Phi_P) \\ &= \frac{1}{2\pi} \int dq (\Phi_P | \rho_{\hat{\Psi}} e^{+iKp} A e^{-iKs} e^{-iKp} e^{-q} | \Phi_P) \end{aligned} \quad (9.39)$$

Therefore, by comparing the right-hand and left-hand sides of the above expressions, we can obtain the generalized density operator of the algebra $\widehat{\mathcal{M}}$ in the physical state $|\hat{\Psi}\rangle_P$,

$$\rho_{\hat{\Psi}} = 2\pi \Delta_{\Psi\Phi} \exp(+q) \quad (9.40)$$

We can now use this result to obtain the entropy. In this case, according to the definition of the map tr given above, we have

$$\begin{aligned} S_{\widehat{\mathcal{M}}} &= -\text{tr}(\rho_{\hat{\Psi}} \log \rho_{\hat{\Psi}}) = -(\hat{\Psi}_P | \log \rho_{\hat{\Psi}} | \hat{\Psi}_P) = -(\hat{\Psi}_P | \log[2\pi \Delta_{\Psi\Phi} \exp(+q)] | \hat{\Psi}_P) \\ &= -(\hat{\Psi}_P | (\log 2\pi + \log \Delta_{\Psi\Phi} + q) | \hat{\Psi}_P) \simeq (\hat{\Psi}_P | (+K_{\Psi\Phi} - q) | \hat{\Psi}_P) \end{aligned} \quad (9.41)$$

It can then be shown that $K_{\hat{\Psi}\hat{\Phi}} = K + K_{\hat{\Psi}} - K_{\hat{\Phi}\hat{\Psi}}$. Thus, we have

$$\begin{aligned} S_{\widehat{\mathcal{M}}} &\simeq (\hat{\Psi}_P | (+K_{\Psi\Phi} + q) | \hat{\Psi}_P) = (\hat{\Psi}_P | (+K + K_{\hat{\Psi}} - K_{\hat{\Phi}\hat{\Psi}} + q) | \hat{\Psi}_P) \\ &\simeq (\hat{\Psi}_P | (+K_{\hat{\Psi}} - K_{\hat{\Phi}\hat{\Psi}}) | \hat{\Psi}_P) + (\hat{\Psi}_P | q | \hat{\Psi}_P) = -S_{\mathcal{M}}(\Psi || \Phi) + \bar{q} + S_0 \end{aligned} \quad (9.42)$$

where $S_{\mathcal{M}}(\Psi || \Phi)$ is the relative entropy, defined as

$$S_{\mathcal{M}}(\Psi || \Phi) = -(\Psi_P | \log \Delta_{\Psi\Phi} | \Psi_P) \quad (9.43)$$

Therefore, considering the entropy obtained in the crossed-product algebra $\widehat{\mathcal{M}}$, we can conclude that it is precisely the entanglement entropy appearing in Eq. 9.22, where the term $S_{\mathcal{M}}(\Psi || \Phi)$ is identified with the $\frac{A}{4G_N}$ contribution, while the remaining terms in the above expression correspond to the entropy of the fields and other contributions. Thus, as expected, the existence of an observer, which gives rise to a type II_1 von Neumann algebra, can recover the generalized entropy proposed in quantum gravity in an algebraic language.

9.4 Emergence of the Cosmological Horizon

Up to this point, the picture of an observer that we have considered has been that of an object equipped with a dynamical clock, whose Hilbert space contains a space such as $L^2(\mathbb{R})$. However, the point that we are interested in addressing from an algebraic perspective is that, in quantum gravity, the observer itself should be considered as part of the dynamics of the system. In fact, the distinction between these two viewpoints can be summarized as follows:

- **Observer with a clock**

1. The observer only measures time and is not itself part of the gravitational dynamics.

2. The trajectory of the observer is specified in advance. For example, one may consider a static observer in dS.
3. The observer's clock only records the local time parameter τ .
4. Moreover, the presence of the observer has no effect on spacetime or on the fields present in it. This is what is referred to as backreaction.

““

- **Dynamical observer**

1. The observer itself is part of the gravitational system.
2. It has finite mass. Therefore, it can accelerate under the influence of forces and can even cross the horizon.
3. It interacts with the surrounding fields, and consequently its clock and spatial position can no longer be regarded as fixed.
4. In this case, the observer's clock is no longer an external time parameter and instead behaves intrinsically as a quantum degree of freedom.

If we wish to study a dynamical observer [83], it is necessary to introduce the dynamical variable $Q(\tau)$ and its generalized momentum $P(\tau)$. The action describing the dynamical observer is then given by [82],

$$I = \int d\tau [P \partial_\tau Q - Q \sqrt{-g_{\tau\tau}}]. \quad (9.44)$$

The Hilbert space $\mathcal{H}_{\text{obs}}$ for an observer in dS_2 can be obtained from the Hilbert space of a free quantum particle, with the additional inclusion of a quantum number corresponding to the mass. We now consider a free scalar field $\phi(x)$ in dS_2 , with Hilbert space $\mathcal{H}_Q$. As in the first section, we also introduce two observers, R and L . As a simple example, consider

$$|\Psi_{\text{HH}}\rangle = |0\rangle_{\text{BD}} \otimes |f_{\text{HH}}\rangle, \quad (9.45)$$

where $|0\rangle_{\text{BD}}$ denotes the Bunch–Davies vacuum and $|f_{\text{HH}}\rangle$ is referred to as the Hartle–Hawking state in the Hilbert space of the two observers, $\mathcal{H}_L \otimes \mathcal{H}_R$, which is prepared using the Euclidean path integral.

For example, suppose that the operator ϕ is gravitationally dressed with respect to observer R , which we denote by ϕ_R . It can then be expressed as

$$\langle n'_R, s'_R | \phi_R | n_R, s_R \rangle = \int d^2x \sqrt{-g} (\Psi_{n'_R}^{s'_R}(x))' \phi(x) \Psi_{n_R}^{s_R}(x). \quad (9.46)$$

This dressed operator evolves under the Hamiltonian H_R according to

$$\phi_R(\tau) = \exp(+i\tau H_R) \phi_R \exp(-i\tau H_R). \quad (9.47)$$

Therefore, as we know, the algebra in region R , corresponding to the region accessible to observer R , can be generated by all gravitationally dressed operators.

The important point arising from considering a dynamical observer is that, due to the properties described above, the observer can be located at any point on a Cauchy slice.

Consequently, the dressed operators no longer belong to a specific region of spacetime. Instead, the dressed operators can be regarded as being smeared over the entire Cauchy slice³. Thus, one can state that

Statement

Since gravitationally dressed operators relative to a dynamical observer are spread over the entire Cauchy slice, if we denote the algebra generated by these operators by $\mathcal{A}_R$, then one can show that $\mathcal{A}_R$ is a type I_∞ von Neumann algebra.

The reason for this can be understood from the fact that, in quantum field theory defined on dS spacetime, there is no cosmological horizon. In this case, one can argue that, due to the continuity of the operators over a Cauchy slice, $\mathcal{A}_R$ is an algebra of type I. Consequently, the generalized entropy is not well-defined for this algebra. Therefore, our aim is to obtain this quantity and, more importantly, to understand the emergence of the cosmological horizon in quantum gravity in the presence of an observer.

Accordingly, let us consider the parameter Q to be given by

$$Q = m + q, \quad (9.48)$$

where q plays the role of the Hamiltonian of the observer's clock and m represents the mass of the observer. There are then two possible procedures that determine the algebra associated with gravitationally dressed operators relative to the observer. Thus,

- If we first take the commutant twice and then consider the limit $m \rightarrow \infty$, the algebra in region R , denoted by $\mathcal{A}_R$, is a type I von Neumann algebra.
- If we first take the limit $m \rightarrow \infty$ and then complete the algebra by taking its double commutant, the resulting algebra $\mathcal{A}_R$ is a type II von Neumann algebra.

We can therefore conclude that

Statement

If we consider the limit $m \rightarrow \infty$, the observer approaches the static limit in which it possesses a cosmological horizon. In this case, the algebra corresponding to the gravitationally dressed operators associated with the observer is a type II von Neumann algebra, and consequently the trace tr and relative entropy (generalized entropy) can be defined.

On the other hand, if the observer's mass does not approach infinity, or equivalently, if the observer is not completely static, then, in principle, it can have access to the entire spacetime. One may think of this as follows: if the object is sufficiently light, it can move through spacetime, whereas if it is extremely massive, it remains localized at a fixed position together with the region gravitationally bound to it.

Now consider an observer with a large but finite mass, i.e., finite m . At the moment $\tau = 0$, there is an uncertainty of order $\frac{1}{m}$ in the observer's position. Furthermore, due to the exponential expansion of dS spacetime, this uncertainty grows with time as

$$\frac{1}{m} \exp\left(\frac{2\pi}{\beta_{\text{dS}}} \tau\right). \quad (9.49)$$

³In [8], the verb *smeare* is used.

In this case,

Statement

When this uncertainty becomes of order $O(1)$, the corresponding time is called the scrambling time,

$$\tau_s \equiv \frac{\beta_{\text{dS}}}{2\pi} \log m. \quad (9.50)$$

At this point, after the scrambling time, the observer can no longer remain within the same patch. In other words, spacetime becomes scrambled from the perspective of the observer.

Therefore, when the observer has a very large but finite mass, one can state that before the scrambling time τ_s , the observer remains within its patch, and a type II von Neumann algebra can be associated with it.

Now, if the observer experiences a scrambled spacetime after the scrambling time, how should this situation be studied? To answer this question, we make use of a type of correlation function known as the out-of-time-ordered correlator. Although ϕ is a free field and does not interact directly with the observer, gravitational constraints, which arise from gauging the isometry group, nevertheless couple them in a nontrivial way. Once the observer is dynamical, rich physics can emerge from these couplings.

For example, inserting a matter operator $\phi_R(\tau)$ at time τ along the worldline of an observer induces a recoil in its trajectory, leading to nontrivial out-of-time-ordered correlations. In particular, consider two scalar fields labeled by A and B . In this case, we have

$$G_4(\tau) = \langle \Psi_{\text{HH}} \left| \phi_R^B \left(\frac{\tau}{2} \right) \phi_R^A \left(-\frac{\tau}{2} \right) \phi_R^B \left(\frac{\tau}{2} + u_2 \right) \phi_R^A \left(-\frac{\tau}{2} + u_1 \right) \right| \Psi_{\text{HH}} \rangle \quad (9.51)$$

Suppose first that the mass of the observer remains fixed. In this case, in the limit $\tau \rightarrow \infty$, the above expression approaches zero. Therefore, one can conclude that the information between the two fields is lost as the out-of-time-ordered correlation function approaches zero.

On the other hand, suppose that the time τ is fixed while the observer's mass is taken to the limit $m \rightarrow \infty$. In this case, expression 9.51 can be evaluated in the vacuum state, and for an observer with a very large but finite mass, we obtain

$$G_4 - G^0 * 4 \propto \frac{\exp \left(\frac{4\pi}{\beta_{\text{dS}}} \tau \right)}{m^2} \propto \exp \left[\frac{4\pi}{\beta_{\text{dS}}} (\tau - \tau_s) \right]. \quad (9.52)$$

This behavior is characteristic of quantum chaos, with a Lyapunov exponent given by

$$\frac{4\pi}{\beta_{\text{dS}}}. \quad (9.53)$$

Therefore, the important point that can be inferred from the quantum-chaotic behavior in this case is that

Statement

The cosmological horizon emerges for an observer with a very large but finite mass as

a consequence of thermal and chaotic effects.

9.5 Quantum volatile black hole spacetime

Another application of the crossed-product algebra is in eternal black holes. As we discussed in Chapter 6.1, in the two-sided thermal state, by considering temperatures above T_{HP} , this black hole can be realized in the bulk. In this case, if we denote the algebras corresponding to the operators in the regions R and L by $\mathcal{M}_R$ and $\mathcal{M}_L$, respectively, then we have

$$\mathcal{M} = \widetilde{\mathcal{M}}_R = \widetilde{\mathcal{M}}'_L \quad (9.54)$$

We also denote the modular Hamiltonian by K . Now, given the setup of the problem, we can state that the same observers R and L are located on the right and left boundaries of the bulk, respectively. As before, we denote the Hamiltonians corresponding to these observers by $\hat{q}_{R,L}$ and their generalized momenta by $\hat{p}_{R,L}$, whose eigenvalues $p_{R,L}$ represent time on the two boundaries. Schematically, as shown in Fig. 9.8, for temperatures T_{HP} , we have an eternal AdS black hole in the bulk region.

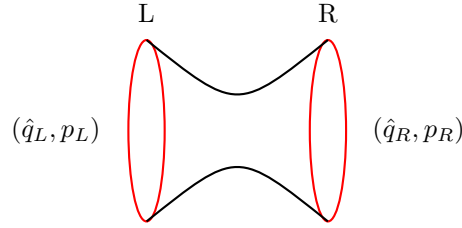


Figure 9.8: Two boundary theories are considered, analogous to the predefined observers located on the two sides. The region between these two boundary theories, which represents the eternal AdS black hole, is also shown.

Now, given the existence of the $N \rightarrow \infty$ limit, the two-sided thermal state can be rewritten as

$$|TFD\rangle = \sum_n f(E_n - E_0) |E_n\rangle_L \otimes |E_n\rangle_R \quad (9.55)$$

where $E_0 \sim O(N^2)$, while $f(E_n - E_0)$ is of order $O(N^0)$. In the bulk, the Hamiltonians corresponding to the observers represent fluctuations of the energy around its background value. In this case, on the boundary, we have

$$\hat{q}_L = H_L - \langle H_L \rangle \quad , \quad \hat{q}_R = H_R - \langle H_R \rangle \quad (9.56)$$

which can be identified with the Hamiltonian of the boundary theory, namely $\text{CFT}_{R,L}$. On the other hand, in some papers (85 and 90) that have studied eternal AdS black holes, the Hamiltonians $\hat{q}_{R,L}$ are interpreted as second-order charges arising from metric fluctuations. The Hamiltonian constraint corresponding to the Killing Vector is also associated with Schwarzschild Time in the bulk. Note that in all these interpretations, the operators $\hat{q}_{R,L}$ are regarded as degrees of freedom of the quantum field theory. On the other hand, as in the previous section, one can fix the gauge and identify the time of the observers with

Schwarzschild time. In this case, we can state that the algebras corresponding to the right and left regions in the bulk are precisely crossed-product algebras.

Statement

The dual of the two-sided thermal state at $T > T_{HP}$ is an eternal AdS black hole whose corresponding bulk algebra is a type III von Neumann algebra. Now, by considering the arrangement of observers on the two sides of the boundary and fixing the gauge, the algebra corresponding to the operators in the bulk can be regarded as a type II von Neumann algebra.

This leads to a change in the entanglement structure of the system. This change can also be described in terms of the geometric structure. In other words, the gravitationally dressed operators corresponding to the observers R and L in these algebras, just like the quantum fields in the bulk, live on a spacetime with quantum time. In this general setting, the time difference between the right and left boundaries, $\langle \hat{p}_L + \hat{p}_R \rangle$, is a gauge-invariant quantity that exhibits fluctuations of order $O(1)$ in the limit $G_N \rightarrow 0$. Consequently,

Statement

This time difference between the boundaries can be represented by a geodesic connecting the right and left boundaries (see the Penrose diagram in Fig. 9.9). Therefore, we can state that the right and left boundaries are connected by a quantum wormhole.

Therefore, as in the previous section, by considering the state $|\hat{\Psi}\rangle = |\Psi\rangle \otimes |g\rangle \in \hat{H}$, where

$$\int dq |g(q)|^2 = 1 \quad (9.57)$$

is the normalization factor.

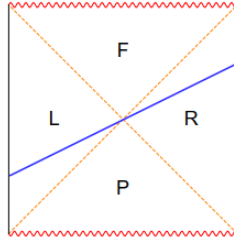


Figure 9.9: Geometrically, the expectation value of $\hat{p}_R + \hat{p}_L$ is equivalent to the time separation between the endpoints of a geodesic extending from the R boundary to the L boundary.

Now, given that the algebras in the bulk on the right and left are crossed-product algebras, which are type II algebras, then, for example, for the right-side bulk algebra, which we denote by $\widehat{\mathcal{M}}_R$, according to the entropy defined in the crossed-product algebra, we have

$$S_{\widehat{\mathcal{M}}_R} = -S_{\mathcal{M}}(\Psi||\Phi) + \text{const.} \quad (9.58)$$

Here, $|\Psi\rangle$ is the two-sided thermal state, whose dual is the Hartle–Hawking state, $|\text{HH}\rangle$, and the constant appearing in the expression is independent of this state. It can also be shown

that

$$S_{\mathcal{M}}(\Psi||\Phi) = -S_{\text{gen}} + \text{const} \quad (9.59)$$

In this case, we obtain

$$S_{\widehat{\mathcal{M}}_R} = S_{\text{gen}} + \text{const} \quad (9.60)$$

Therefore, we can state that in equilibrium, $t \rightarrow \infty$, the entropy corresponding to region R is equal to the generalized entropy of the black hole.

9.6 Generalized Gravitational Entropy for a Local Region

Up to this point, we have discussed various settings in which the modular Hamiltonian acts as the generator of time evolution. However, in general, the modular Hamiltonian does not necessarily coincide with the time-translation generator. For example, in Schwarzschild-dS spacetime, there are two horizons with different temperatures due to the presence of both the black hole and cosmological horizons. Consequently, there is no state for which the modular Hamiltonian can be identified with the generator of time evolution. In other words, the modular Hamiltonian does not correspond to a geometric quantity in this case.

Let us first consider the Penrose diagram shown in Fig. 9.10. The region R lies between the two cosmological and black hole horizons, located at r_c and r_b , respectively, while the region L is the corresponding region on the other side of the diagram. Let us denote the algebra associated with region R by $\mathcal{M} = \mathcal{M}_R$ and the algebra associated with region L by $\mathcal{M}' = \mathcal{M}_L$. Thus, the region R is bounded by two Killing horizons. In this case, one can choose a Cauchy slice for region R such that

$$\Sigma = H_{b,R}^- \cup H_{c,R}^-. \quad (9.61)$$

Since the modular Hamiltonian cannot be interpreted as a geometric quantity, the algebra associated with region R can instead be factorized as

$$\mathcal{M}_R = \mathcal{M}_b \otimes \mathcal{M}_c, \quad (9.62)$$

where $\mathcal{M}_b$ and $\mathcal{M}_c$ denote the algebras associated with the regions bounded by the black hole and cosmological horizons, respectively [84].

With this decomposition in mind, one can determine the modular operator associated with the algebra $\mathcal{M}_R$ such that, for each horizon, it behaves as a geometric quantity. Following the observer construction introduced above, we can choose two observers associated with each horizon. More precisely, for the cosmological horizon, we consider $\hat{q}_{R,c}$ and $\hat{q}_{L,c}$, while for the black hole horizon we consider $\hat{q}_{R,b}$ and $\hat{q}_{L,b}$. We must also take into account the Hamiltonian constraint,

$$\hat{q}_{L,c} = K_c + \hat{q}_{R,c} \quad , \quad \hat{q}_{L,b} = K_b + \hat{q}_{R,b}, \quad (9.63)$$

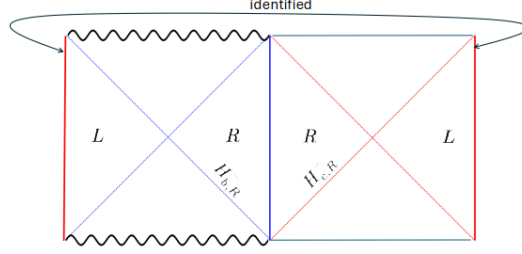


Figure 9.10: Penrose diagram of Schwarzschild-dS spacetime. The vertical red lines on the left and right sides of the diagram are identified. Each point in the diagram represents a $(d-2)$ -dimensional sphere, and therefore a spatial slice has topology $S^1 \times S^{d-2}$. The blue dashed lines represent the black hole horizons at radius r_b , while the red dashed lines represent the cosmological horizons at radius $r_c > r_b$. The trajectories of observers R and L are shown as solid blue and red vertical lines, respectively, and their accessible regions are labeled by R and L . A Cauchy slice Σ of region R can be chosen as the union of the past black hole and cosmological horizons: $\Sigma = H_{b,R}^- \cup H_{c,R}^-$.

where K_b and K_c are the modular Hamiltonians associated with the two algebras $\mathcal{M}_b$ and $\mathcal{M}_c$, respectively.

Consequently, by imposing the modular fixing condition for each observer, one can obtain the modular operators in the geometric description. The resulting algebra $\mathcal{M}$ associated with region R is then a type-II von Neumann algebra. The generalized entropy of this system can therefore be written as

$$S_{\text{gen}}^{(R)} = \frac{A_{\text{cos}}}{4G_N} + \frac{A_{\text{BH}}}{4G_N} + \tilde{S}^{(R)}, \quad (9.64)$$

where $\tilde{S}^{(R)}$ denotes the entropy of the other matter fields contained within region R .

9.7 JT Gravity

As is well known, the action describing the gravitational theory is the Einstein–Hilbert action. In two dimensions, this action is inherently topological and, in fact, is proportional to the Euler characteristic χ . More precisely, we have

$$\chi = \frac{1}{4\pi} \left(\int d^2x \sqrt{g} R + 2 \oint dx \sqrt{h} K \right) = 2 - 2g - n \quad (9.65)$$

where n denotes the number of boundaries and g denotes the number of handles. This fact does not imply that the theory is trivial. In the path-integral formulation, one must carefully take into account the measure and gauge fixing. This leads to a theory with intrinsically strong coupling. To overcome this issue, we need to introduce an additional field coupled to gravity.

This model plays an important role in two-dimensional gravity and in string theory formulated on the worldsheet. Accordingly, the dilaton action can be formulated by introducing

a scalar field, referred to as the dilaton field Φ , as follows [85, 86],

$$S[g, \Phi] = -\frac{1}{16\pi G_N} \int d^2x \sqrt{g} [\Phi R + U(\Phi)] \quad (9.66)$$

where $U(\Phi)$ plays the role of a potential for the theory. We can then write the following generalized action,

$$\begin{aligned} S &= -\frac{S_0}{4\pi} \int d^2x \sqrt{g} R - \frac{1}{16\pi G_N} \int d^2x \sqrt{g} [\Phi R + U(\Phi)] - \frac{1}{16\pi G_N} \oint dx \sqrt{h} K \\ &\equiv -\frac{S_0}{4\pi} \int_M \sqrt{g} R - \frac{1}{16\pi G_N} \int d^2x \sqrt{g} [\Phi R + U(\Phi)] - \frac{1}{16\pi G_N} \oint dx \sqrt{h} (K - 1) \end{aligned} \quad (9.67)$$

where S_0 is a constant, and for convenience, we have included the $(K - 1)$ term in the boundary action, which will simplify the calculations below. As is evident, the first term in the above expression is the topological term appearing in Eq. 9.65; the second term is responsible for the dynamics through the presence of the coupled dilaton field; and the final term is the boundary contribution.

The gravitational action of JT gravity is obtained by choosing $U(\Phi) = -\Lambda\Phi$, giving [87, 88]

$$S_{\text{JT}}[g, \Phi] = -\frac{1}{16\pi G_N} \int d^2x \sqrt{g} \Phi (R - \Lambda). \quad (9.68)$$

Furthermore, taking $\Lambda = -\frac{2}{L^2}$, where L is the AdS length scale, and setting $L \equiv 1$ for simplicity, the action becomes

$$S_{\text{JT}}[g, \Phi] = -\frac{1}{16\pi G_N} \int_{\mathcal{M}} d^2x \sqrt{g} \Phi (R + 2) - \frac{1}{8\pi G_N} \oint_{\partial\mathcal{M}} \sqrt{h} \Phi (K - 1). \quad (9.69)$$

We can now derive the equations of motion from the above action. Varying with respect to the dilaton field gives

$$R = -2, \quad (9.70)$$

while variation with respect to the metric gives an equation that depends entirely on the dilaton field,

$$(\nabla_\mu \nabla_\nu - g_{\mu\nu} \nabla^2) \Phi + g_{\mu\nu} \Phi = 0. \quad (9.71)$$

If we initially take the dilaton to be constant, the equations of motion imply that $\Phi = 0$. This means that, locally, the spacetime is AdS_2 . Globally, however, different physical choices are possible, such as the Poincaré Patch or even the Global Patch.

Thus far, we have assumed that the dilaton is constant and equal to $\Phi = 0$. However, this assumption leads to several difficulties. Which patch of the spacetime should we choose? Moreover, what determines the relation between bulk time and boundary time in the quantum-mechanical description?

The solution [89, 90, 91, 92] is to impose boundary conditions that explicitly break the conformal symmetry of AdS_2 by turning on a source for the dilaton. This is a special feature of two-dimensional theories. In higher-dimensional versions of AdS/CFT, one can work with

theories that are exactly conformally invariant without encountering such an issue.

To implement this construction, we introduce a UV cutoff in the geometry and impose Dirichlet boundary conditions,

$$ds^2|_{\partial M} = \frac{dt^2}{\epsilon^2} \ , \ \Phi|_{\partial M} = \frac{\Phi_r}{\epsilon} \ , \ \epsilon \rightarrow 0 \quad (9.72)$$

This causes the geometry to approach NAdS₂, as shown in Fig. 9.11.

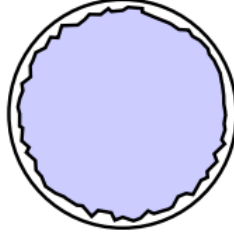


Figure 9.11: The NAdS₂ geometry. The cutoff imposed by the boundary condition breaks the conformal boundary, which is represented by the black line.

As is evident, the presence of this cutoff breaks the conformal boundary. At finite temperature, the Rindler region is selected, whereas at zero temperature, the Poincaré patch is selected. The Global Patch does not admit a solution for the dilaton with the appropriate boundary conditions unless the theory is specially deformed [93]. Nevertheless, the geometry can still be regarded as exactly AdS₂ locally everywhere. One can also work in a gauge in which all information about metric fluctuations is encoded in the shape of the boundary curve consistent with Eq. 9.72.

The presence of this boundary condition gives rise to several important features, which we now discuss:

- **Symmetry** The asymptotic symmetries of AdS₂ are given by the one-dimensional conformal group, corresponding to time reparametrizations. These symmetries are spontaneously broken to the global conformal group $SL(2, \mathbb{R})$ due to the boundary condition 9.72, since Φ_r carries a physical dimension. Another way to see this is to note that, at the classical level, the dilaton behaves as $\Phi \propto \cosh \rho$, and it is manifestly not invariant under conformal transformations.

- **Schwarzian**

The on-shell action for a given boundary curve satisfying the boundary conditions in Eq. 9.72 is equal to an action known as the Schwarzian Action. This analysis can also be carried out in Lorentzian signature.

To elaborate, suppose that we work on a hyperbolic disk with coordinates $(\hat{t}, \hat{\rho})$, both of which depend on the boundary time t ,

$$(\hat{t}, \hat{\rho}) = (\hat{t} = f(t), \hat{\rho} = \rho(t)). \quad (9.73)$$

Since time is periodic in Euclidean signature, the function $f(t)$ must satisfy

$$f(t) \in \text{Diff}(S^1) \ , \ f(t + \beta) = f(t) + \beta. \quad (9.74)$$

The function $\rho(t)$, on the other hand, is determined by imposing the Dirichlet boundary condition on the metric. More precisely,

$$ds^2|_{\partial M} = \left[\rho'(t)^2 + \frac{4\pi^2}{\beta^2} \sinh^2 \rho(t) f'(t)^2 \right] dt^2 \sim \frac{\pi^2}{\beta^2} e^{2\rho} f'(t)^2 dt^2 = \frac{1}{\epsilon^2} dt^2. \quad (9.75)$$

Thus, near the AdS_2 boundary, we obtain the relation between the two functions $f(t)$ and $\rho(t)$,

$$\rho(t) \simeq -\log \epsilon - \log f'(t) \simeq -\log f'(t) \quad (9.76)$$

Consequently, the boundary degrees of freedom are parametrized by a single function $f(t) \in \text{Diff}(S^1)$. However, this parameter space is too large. In particular, two boundary curves related by Eq. 9.76 correspond to equivalent boundary configurations. Therefore, to remove this redundancy, we can simply go to the Poincaré patch and use the fact that the geometry is conformal near the boundary, in which case $Z \simeq 0$. Taking into account the Poincaré patch and its symmetry group $\text{PSL}(2, \mathbb{R})$, we have

$$F = \frac{\beta}{\pi} \tan\left(\frac{\pi f}{\beta}\right). \quad (9.77)$$

Therefore, the physical parameter space of boundary curves is

$$\text{Diff}(S^1)/\text{PSL}(2, \mathbb{R}). \quad (9.78)$$

Consequently, the action in Eq. 9.67 can be written as

$$S = -S_0 \chi - \Phi_r \int dt \tan \frac{\pi f(t)}{\beta}, t, \quad (9.79)$$

where $f(t), t$ denotes the Schwarzian derivative of the function $f(t)$, defined by

$$f(t), t = \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2. \quad (9.80)$$

Thus, as is evident, in JT gravity the only nonvanishing term that gives rise to dynamics is the boundary term. This is natural: since the spacetime has maximal symmetry and the boundary condition breaks this symmetry, it is expected that the dynamics is encoded in the boundary contribution.

Having established the basic properties of JT gravity, we now turn to the algebraic perspective and study the operator algebras residing on its boundary. We begin with the metric

$$ds^2 = -dX^2 - dY^2 + dZ^2, \quad (9.81)$$

which covers the entire AdS spacetime and will be denoted by AdS_2^0 . As is well known,

AdS_2^0 admits a Lie algebra generated by

$$\begin{aligned} j_1 &= X\partial_Y - Y\partial_X \\ j_2 &= Y\partial_Z + Z\partial_Y \\ j_3 &= -X\partial_Z - Z\partial_X, \end{aligned} \tag{9.82}$$

which satisfy

$$[j_a, j_b] = i\epsilon_{ab}^c j_c. \tag{9.83}$$

Introducing the coordinates T and σ ,

$$\begin{aligned} X &= \cos T \cosh \sigma \\ Y &= \sin T \cosh \sigma \\ Z &= \sinh \sigma, \end{aligned} \tag{9.84}$$

the metric can be rewritten as

$$ds^2 = d\sigma^2 - \cosh^2 \sigma, dT^2, \quad -\infty < \sigma, T < +\infty. \tag{9.85}$$

This spacetime has a right conformal boundary at $\sigma \rightarrow \infty$ and a left conformal boundary at $\sigma \rightarrow -\infty$. In fact, JT gravity describes the entire AdS_2 spacetime. Accordingly, for the right boundary we use the coordinates (σ_R, T_R) , while for the left boundary we use (σ_L, T_L) . Naturally, as discussed above, the coordinates T and σ are not independent and, by taking the boundary time to be t and imposing the boundary conditions, both can be regarded as functions of this variable. Thus, one can show that

$$\exp(\sigma_R) = \frac{2}{\epsilon} \frac{1}{\dot{T}_R} \equiv \frac{2\Phi_r}{\epsilon} \exp(\tilde{\sigma}_R) \quad , \quad \exp(-\sigma_L) = \frac{2}{\epsilon} \frac{1}{\dot{T}_L} \equiv \frac{2\Phi_r}{\epsilon} \exp(-\tilde{\sigma}_L). \tag{9.86}$$

Here, $\tilde{\sigma}_{R,L}$ denote the renormalized radial coordinates, which are related to the geodesic length ℓ connecting the right and left boundaries,

$$\ell = \tilde{\sigma}_R - \tilde{\sigma}_L + \text{const} \tag{9.87}$$

where the constant depends only on $T_{R,L}$. The gravitational action can then be rewritten in terms of these coordinates as

$$S = \Phi_r \int dt \left(-\dot{T}_R^2 + \left(\frac{\ddot{T}_R}{\dot{T}_R} \right)^2 \right) + \Phi_r \int dt \left(-\dot{T}_L^2 + \left(\frac{\ddot{T}_L}{\dot{T}_L} \right)^2 \right) = \Phi_r \int dt T_R, t + \Phi_r \int dt T_L, t.$$

On the other hand, using the above action and the corresponding generalized momenta, one can show that the Hamiltonian on each side of the boundary takes the form

$$\begin{aligned} H_R &= \frac{1}{2\Phi_r} \left[\frac{1}{2} p_{\tilde{\sigma}_R}^2 + p_{T_R} \exp(-\tilde{\sigma}_R) + \frac{1}{2} \exp(-2\tilde{\sigma} * R) \right] \\ H_L &= \frac{1}{2\Phi_r} \left[\frac{1}{2} p_{\tilde{\sigma}_L}^2 + p_{T_L} \exp(+\tilde{\sigma}_L) + \frac{1}{2} \exp(+2\tilde{\sigma}_L) \right]. \end{aligned} \tag{9.88}$$

Here,

$$p_{\tilde{\sigma}_R} = 2\Phi_r \dot{\sigma}_R \quad (9.89)$$

and

$$p_{T_R} = 2\Phi_r \left(-\cosh^2 \sigma_R \dot{T}_R + \frac{1}{\epsilon} \sinh \sigma_R \right). \quad (9.90)$$

Then, using Eq. 9.87, we can identify

$$H \equiv H_R = H_L. \quad (9.91)$$

Consequently, the classical phase space is only two-dimensional. This phase space can be parametrized in two equivalent ways. One can use the pair $(H, \delta t)$, where $\Delta t = T_R - T_L$ is the time difference between the left and right boundary trajectories. Alternatively, one can use the pair (ℓ, p_ℓ) , where ℓ is the renormalized geodesic distance between the left and right boundaries and p_ℓ is its conjugate generalized momentum.

The theory can therefore be canonically quantized. In this case, the Hilbert space is

$$\mathcal{H} = L^2(\mathbb{R}). \quad (9.92)$$

Using the Euclidean path integral, one can also construct a basis

$$\{ |\beta\rangle, \beta \in \mathbb{R}_+ \}$$

for this Hilbert space, as illustrated in Fig. 9.12.

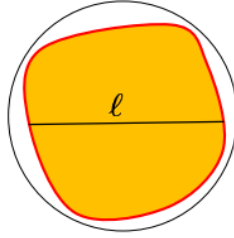


Figure 9.12: In Euclidean signature, the geometry is represented by a hyperbolic disk with a boundary (shown as a thick red line) having the topology of a circle. The path integral over the lower half-disk with boundary length $\frac{\beta}{2}$ prepares the wavefunction $\psi_\beta(\ell)$ for the state $|\beta\rangle$.

We now introduce matter fields. For example, consider a scalar field ϕ , which may have self-interactions. Since the matter fields are not coupled to the dilaton Φ , the spacetime geometry remains AdS_2 . We denote the boundary limit of the field ϕ by O , which defines a single-trace operator in the corresponding boundary quantum theory, as indicated in Eq. 6.2.

In the presence of matter fields, H_R and H_L are no longer independent, since each now contains a contribution from the matter fields. After quantization, the full Hilbert space

takes the form

$$\mathcal{H} = L^2(\mathbb{R}) \otimes \mathcal{H}_{\text{matt}}, \quad (9.93)$$

where $\mathcal{H}_{\text{matt}}$ is the Hilbert space associated with the matter fields. Once again, using the Euclidean path integral with insertions of the matter operator O on the Euclidean boundary, one can construct a basis for $\mathcal{H}$ [94, 95], as shown in Fig. 9.13.

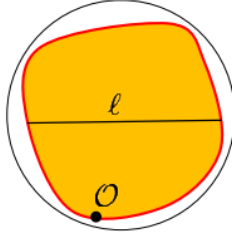


Figure 9.13: In the presence of matter, a basis for the bulk Hilbert space $\mathcal{H}$ can be constructed from a path integral over the lower half-disk with matter operators inserted along the lower boundary.

An important point is that even for finite G_N , there is only a single Hilbert space $\mathcal{H}$. In particular, this Hilbert space cannot be factorized into two independent Hilbert spaces associated with the R and L systems. We now denote the algebras generated by the single-trace operator O on the right and left boundaries by $\mathcal{A}_R^0$ and $\mathcal{A}_L^0$, respectively. These are type-III von Neumann algebras, analogous to those of the eternal AdS black hole at temperatures above T_{HP} . It is worth noting that the algebras $\mathcal{A}_R^0$ and $\mathcal{A}_L^0$ are equivalent to the algebras associated with the bulk regions R and L , respectively, where the causal wedge and entanglement wedge coincide.

We can also introduce boundary observers, as in the previous sections, with Hamiltonians $\hat{q}_R$ and $\hat{q}_L$. As discussed previously, the algebra of gravitationally dressed operators relative to these observers is obtained through the crossed-product construction, and the resulting algebra is of type-II.

In this case, even though we are in the semiclassical regime, $G_N \rightarrow 0$, the time difference between the two boundaries exhibits fluctuations of order $O(1)$. Consequently, the spacetime acquires a quantum and unstable character.

We now consider the regime of finite G_N (or finite Φ_r). In this regime, the boundary algebra $\mathcal{A}_R^0$ must be enlarged by including H_R , giving rise to a larger algebra denoted by $\mathcal{A}_R$. Similarly, $\mathcal{A}_L^0$ is enlarged to $\mathcal{A}_L$.

In this regime, spacetime becomes fully quantum. In particular, the R and L boundaries are fully dynamical, and the horizons are no longer sharply defined. Likewise, bulk subregions such as the causal and entanglement wedges are no longer sharply defined. It is therefore unclear how the spacetime can be decomposed into independent R and L regions. Interestingly, however, the algebra of operators of the quantum gravitational bulk system, $B(\mathcal{H})$, can still be factorized algebraically into right and left parts [94, 95],

$$B(\mathcal{H}) = \mathcal{A}_R \vee \mathcal{A}_L \quad , \quad \mathcal{A}'_R = \mathcal{A}_L. \quad (9.94)$$

An interesting feature is that $\mathcal{A}_R$ and $\mathcal{A}_L$ are type-II von Neumann algebras. Consequently, we can define a trace map tr as

$$\forall a \in \mathcal{A}_R \quad , \quad \text{tr} a = \lim_{\beta \rightarrow 0} \langle \beta | a | \beta \rangle \quad (9.95)$$

where $|\beta\rangle$ is the state prepared as shown in Fig. 9.13. We also know that in the semiclassical regime, $G_N \rightarrow 0$, or equivalently $\Phi_r \rightarrow \infty$, the state $|\beta\rangle$ is non-normalizable, and consequently the trace map tr defined in Eq. 9.95 ceases to exist. This signals the emergence of a type-III von Neumann algebra structure.

Chapter 10

Answers to Selected Questions

10.1 Question 2

As we know, the algebra $\mathcal{M}$ is closed. Consequently, for any $C, D \in \mathcal{M}$ one may set

$$A = C + D, \quad B = C - D. \quad (10.1)$$

From the properties of the state $\omega : \mathcal{M} \rightarrow \mathbb{C}$ we have

$$\begin{aligned} \omega(\alpha A + \beta B) &= \alpha\omega(A) + \beta\omega(B) \quad \text{for all } \alpha, \beta \in \mathbb{C} \text{ and } A, B \in \mathcal{M}, \\ \omega(A^\dagger) &= (\omega(A))^*. \end{aligned}$$

It follows that

$$\omega(A^\dagger) = \omega(C^\dagger + D^\dagger) = \omega(C^\dagger) + \omega(D^\dagger), \quad \omega(B) = \omega(C - D) = \omega(C) - \omega(D).$$

Therefore

$$\begin{aligned} \omega(A^\dagger B) &= \omega((C^\dagger + D^\dagger)(C - D)) = \omega(C^\dagger C + D^\dagger D + C^\dagger D + CD^\dagger) \\ &= \omega(C^\dagger C) + \omega(D^\dagger D) + \omega(C^\dagger D) + \omega(CD^\dagger). \end{aligned} \quad (10.2)$$

On the other hand,

$$\begin{aligned} \omega(A^\dagger A) &= \omega(C^\dagger C) + \omega(D^\dagger D) + \omega(C^\dagger D) + \omega(DC^\dagger), \\ \omega(B^\dagger B) &= \omega(C^\dagger C) + \omega(D^\dagger D) - \omega(C^\dagger D) - \omega(DC^\dagger). \end{aligned} \quad (10.3)$$

Combining (10.2) and (10.3) immediately yields the Cauchy–Schwarz inequality

$$|\omega(A^\dagger B)|^2 \leq \omega(A^\dagger A) \omega(B^\dagger B). \quad (10.4)$$

10.2 Question 3

In the GNS construction one must quotient out the null operators with respect to the state ω . The set of null operators is

$$\mathcal{T} = \{X \mid \omega(X^\dagger X) = 0\}. \quad (10.5)$$

From Question 1 we already know that ω satisfies the Cauchy–Schwarz inequality. Consequently, for every $A \in \mathcal{A}$ and every $X \in \mathcal{T}$,

$$0 \leq |\omega((AX)^\dagger(AX))|^2 = |\omega(X^\dagger X A^\dagger A)|^2. \quad (10.6)$$

Since $\omega(X^\dagger X) = 0$, it follows that

$$\langle AX | AX \rangle = \omega((AX)^\dagger(AX)) = 0 \quad \text{for all } A \in \mathcal{M} \text{ and } X \in \mathcal{T}. \quad (10.7)$$

Thus $\mathcal{T}$ is a left ideal, as required for the GNS construction.

10.3 Question 8

As indicated in Chapter 4, the modular conjugation operators satisfy

$$u_{\Psi\Omega}(s) = (\Delta_\Psi \Delta_\Omega)^{+is}, \quad (10.8)$$

where $|\Psi\rangle$ and $|\Omega\rangle$ are cyclic and separating vectors for the algebra $\mathcal{M}$. If a third cyclic and separating vector $|\Phi\rangle$ is available, one obtains the cocycle relation

$$\begin{aligned} u_{\Psi\Omega} &= (\Delta_\Psi \Delta_\Omega)^{+is} = (\Delta_\Psi \Delta_\Phi \Delta_\Phi^{-1} \Delta_\Omega)^{+is} \\ &= (\Delta_\Psi \Delta_\Phi)^{+is} (\Delta_\Omega \Delta_\Phi)^{+is} = u_{\Psi\Phi} u_{\Phi\Omega}. \end{aligned} \quad (10.9)$$

10.4 Question 10

Assume $\mathcal{N} \subseteq \mathcal{M}$ and that both algebras share a common cyclic and separating vector $|\Omega\rangle$. Then each algebra possesses a modular structure. Define the unitary

$$D(t) = \Delta_{\mathcal{M}}^{-it} \Delta_{\mathcal{N}}^{+it}. \quad (10.10)$$

Because modular flow of $\mathcal{M}$ leaves $\mathcal{M}$ invariant,

$$\Delta_{\mathcal{M}}^{-it} A \Delta_{\mathcal{M}}^{+it} \in \mathcal{M} \quad \text{for every } A \in \mathcal{M}, \quad (10.11)$$

it follows that

$$D(t) \mathcal{N} D^\dagger(t) = \Delta_{\mathcal{M}}^{-it} \mathcal{N} \Delta_{\mathcal{M}}^{+it} \subseteq \mathcal{M}. \quad (10.12)$$

Hence, for every $t \leq 0$, the conjugated algebra

$$\mathcal{N}_t := D(t)\mathcal{N}D^\dagger(t) \quad (10.13)$$

is a subalgebra of $\mathcal{M}$. This is precisely the half-sided modular inclusion.

10.5 Question 12

From the bulk reconstruction formula (6.71) one recovers the desired operator by means of the boundary representation π . At leading order one has the familiar HKLL expression

$$\phi^{(0)}(X) = \int d^d x K(X; x) \pi_\Omega(O(x)). \quad (10.14)$$

Inserting the higher-order terms of the bulk equation of motion (Eq. (6.2)) immediately yields the corrected bulk operator (6.72).

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